Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8 and read off the Standard Model: status assessment and reading guide

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In one sentence

Topological Fixed-Point Theory (TFPT) constructs, from exactly **two inputs** — a boundary constant $c_3 = \frac{1}{8\pi}$ (P1) and a five-slot carrier $g_{\rm car} = 5$ (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with E_8 ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ($\alpha^{-1} = 137.0359992$); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* ($v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$), not as free outputs. The honest one-line claim is on the status card below.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\rm geo}, G_{\rm net}, F_{\rm transfer}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading the introduction — the entry point and reading guide.

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1,1,2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}})$ are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

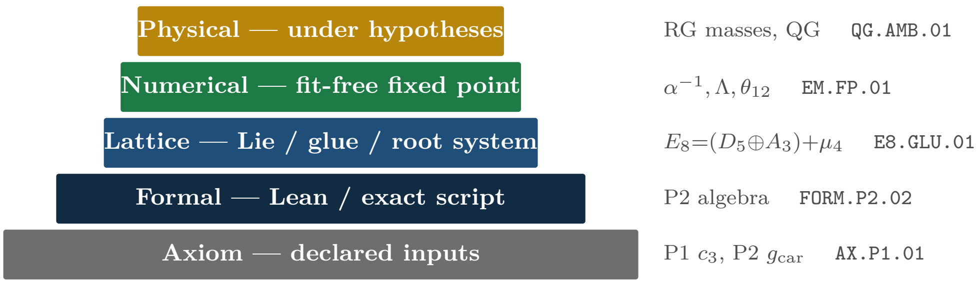
SM skeleton: three families, hypercharges, flavor matrix R *([E])*

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Reading the markers. Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: [E] *exact / machine-proven* — an exact identity (e.g. $240 = 16 \cdot 5 \cdot 3$), a Lie/lattice theorem (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$), a Lean/script formalisation, or a reproduced numerical fixed point; [C] *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); [O] *open / axiom* — a declared input or a genuine gap; [X] *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.



The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L); (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support):

1. the architecture: the two axioms $\{c_3, g_{\mathrm{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the α^{-1} fixed point plus ablation (`v3_em_alpha.py`);

4. the **single status ledger** `verification/status_ledger.csv` — the source of truth for every claim's grade, location, dependency and script;
5. the **open gates** as numbered research contracts (`tfpt_research_contracts`: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. `E8.GLU.01`, `EM.FP.01`) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers $(2, 3, 5, 16, \dots)$ reappears in every sector: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8, and E_8 is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less.* Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune.* The only genuinely free, continuous ingredient that remains is the number π.

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\text{car}}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n{+}1^n{+}2^n = 2{+}2^n$ generate the big Lie data directly: $p_1{=}4{=}|\mu_4|$, $p_2{=}6{=}|R^+(A_3)|$, $p_3{=}10{=}\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1p_2p_3 = 240, \quad \dim E_8 = p_1p_2p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1}{-}p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$.
[`verification/v23_anchor_generator.py`]

And a is itself half-forced ([E]/[C]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not "why $a = (1, 1, 2)$" but "why the carrier $g_{\text{car}} = 5$ / $(|\mu_4|, N_{\text{fam}}) = (4, 3)$" — that interface (with c_3) is the [O] input layer.

Two readings, one source (no contradiction). P1 and P2 are the *operational compiler inputs*; their anchor provenance is $a = (1, 1, 2)$ plus π. The *single structural bedrock postulate* is `QGEO.SYM.01` (the carrier μ_4 clock is the seam's conformal deck, see `tfpt_research_contracts`): it forces the μ_4 deck, hence $N_{\text{fam}} = 3$, $\deg E = -4$ and the tripartition $4 = 2{+}1{+}1$, of which c_3 and g_{car} are the two elementary readouts. "Two inputs" (the working axioms) and "one postulate" (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local$\Rightarrow \omega \circ \rho = \omega$ implication are now Lean-formalised (`FORM.QGEO.02`/`FORM.QGEO.01`); the postulate itself stays [O].

One source, four readings (the cleanest origin). The four-point seam divisor μ_4 on $\mathbb{P}^1$ generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \tfrac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\operatorname{rank} H_1(\mathbb{P}^1 \backslash \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles}\to A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions}\to D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates D_5 itself ([`verification/v197_rr_carrier_clifford_d5.py`]): $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor. So μ_4 is not merely glue: it is the *common source* of A_3, D_5, c_3 and E_8 ([`verification/v189_riemann_roch_carrier.py`]) — the arithmetic is exact [E], the physical reading of the carrier as the seam mode space stays [C].

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor* $a = (1, 1, 2)$ *into the* E_8 *audit hull.*

In one line:

> The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0, α^{-1}, the exponential scales) flows from π; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad ([`verification/v119_review_validation_2.py`]). The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions [E]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio)}, \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain)}, \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2 \operatorname{rank} E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\operatorname{rank} E_8 = p_0 + e_2$.

$a = (1, 1, 2)$ *[E]*

elementary $e_k(a)$	**power sums** $p_n = 2 + 2^n$	**derived motifs**
$e_1 = 4 \to \lvert\mu_4\rvert$ $e_2 = 5 \to g_{\text{car}}$ $e_3 = 2 \to \lvert\mathbb{Z}_2\rvert$ $c_3 = \frac{1}{2e_1\pi} = \frac{1}{8\pi}$	$p_0, p_1, p_2, p_3 = 3, 4, 6, 10$ $p_1p_2p_3 = 240 = \lvert R(E_8)\rvert$ $p_4 - p_3 = 8 = \operatorname{rank} E_8$ $\Rightarrow \dim E_8 = 248$	$4, 5, 8, 16, 25,$ $30, 41, 120, 240$ *one anchor,* *many projections*

The companion documents

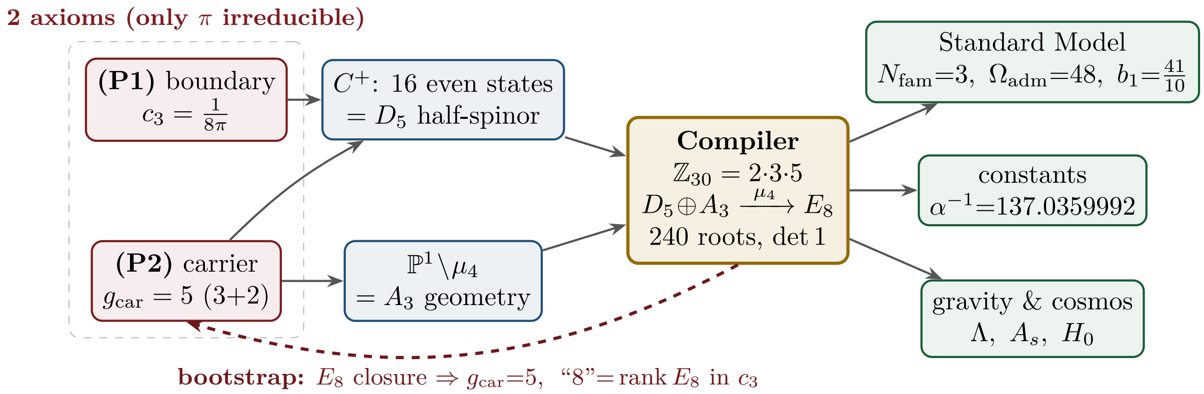
The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

Document	What it contains (former notes now as parts)
`tfpt_1_architecture_e8`	**Core.** The two axioms $\{c_3, g_{\rm car}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}{=}2{\cdot}3{\cdot}5$, the explicit $D_5{\oplus}A_3{+}\mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\mathrm{Spin}(10){\times}SU(4))/\Delta\mathbb{Z}_4$ (Part II).
`tfpt_2_standard_model`	**Standard Model.** The full SM in one $\varphi_0^{\rm ret}$-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/n_s, r, Starobinsky M, derived θ_{12}, quark c, the H2 splitting type, Λ-typing (Part III).
`tfpt_3_e8_audit_bootstrap`	E_8 **audit & bootstrap.** The seven E_8 slices as an audit raster ($E_6{\times}A_2$ reads the flavor matrix) (Part I); the old cascade $D{=}60{-}2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\rm car}{=}5$ and the "8" in c_3 as overdetermined fixed points, only π irreducible (Part III).
`tfpt_4_frontier`	**Frontier.** The honest status of η_B, m_p/m_e, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder.
`tfpt_5_redteam`	**Red Team.** The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.
`tfpt_research_contracts`	**Research contracts.** The open gates as numbered contracts with lemma chains and closing theorems: $(U_{\rm wall})$ (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = *two* line pairings, the third integrality-forced) and $G_{\rm metric}/G_{\rm net}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).
`tfpt_horizon_readouts` **(App. H)**	**Appendix H — horizon unit system (reframe, not new physics).** One seam constant $c_3{=}\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\rm GW}{=}c$ — standard physics in seam units, two compiler fingerprints ($1920{=}\lvert W(D_5)\rvert$, $\lvert\mu_4\rvert{=}4$); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
`origin_theory`	**Origin Theory.** Why no free *dimensionless compiler dial* remains beyond the anchor and π (the dimensionful $v_{\rm geo}$ and transfer physics stay typed): the $(g_{\rm car}, N_{\rm fam})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; **[E]** core plus one honestly-typed **[C]** cyclic interpretation.

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\mathrm{Spin}(10){\times}SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a $U_{\rm point}$ anchor; θ_{12} the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

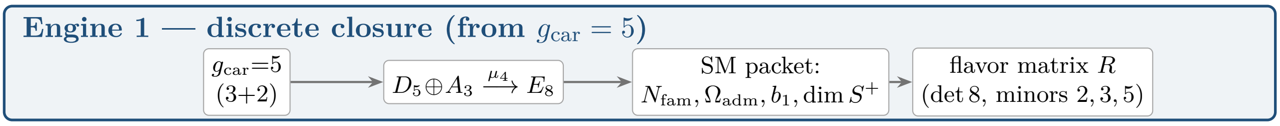
2 The architecture at a glance

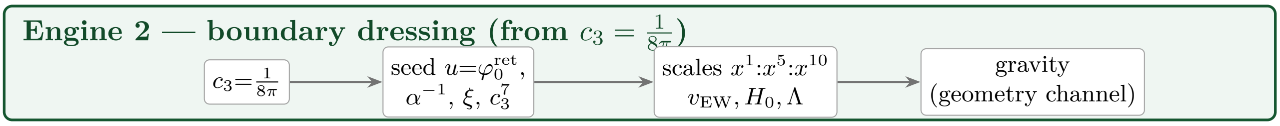
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.



Key statement (the loop closes): previously D_5, A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs.* $g_{\rm car} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0}{+}\binom{g}{1}{+}\binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not "prove itself"; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

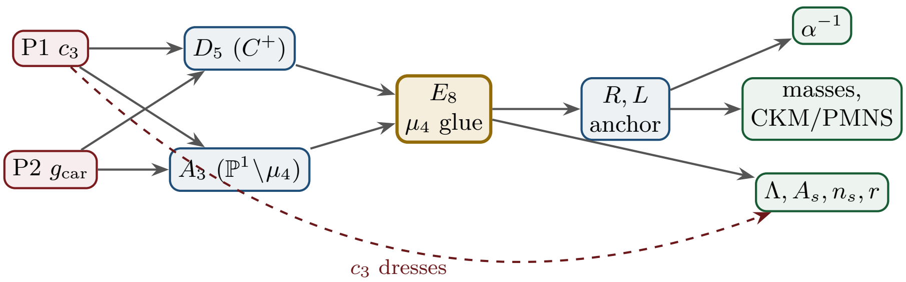
The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.





3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, $c_3{=}1/(8\pi)$	— (axiom)	[O]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\mathrm{car}}{=}5$	— (axiom)	[O]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[E]	$\dim S^+{=}16$, Y	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[E]	$N_{\mathrm{fam}}{=}3$, $\mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc$=\mathbb{Z}_4$, q-sum$=2$	D_5, A_3	[E]	$240, 248$	not even-unimodular
R, L	residue + winding matrix, anchor $(1, 1, 2)$	A_3	[E]	$\det 8$, minors $(2, 3, 5)$	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha){=}0$	c_3, $M{=}41$	[E]	137.0359992	no/second root (excl.)
masses	φ_0^{ret}-ladder $\lambda_Y^L \Lambda$	R, L, c_3	[E]/[C]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam $\varepsilon{=}c_3$	R, L	[E]/[C]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[C]	Λ, A_s, n_s, r	ambient RP fails

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R-bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6\log\frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([E] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [C]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ, so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([E]/[C], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just π, the discrete E_8-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} \quad = \quad (U_{\text{wall}}) \quad \oplus \quad (G_{\text{metric}}) \quad \oplus \quad (F_{\text{frontier}})$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank $= c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [E]; $(F_{\text{frontier}}) \to F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers; the four are one typed functor with a four-axiom contract (`CONTRACT.F.01`, v213). *Latest reductions (v134–v148):*

the flavor selector residue R4′ is now "derive *one map*" — $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data, (d, n) forces R *columnwise* (v136/v139), the third pairing is *integrality-forced* mod 11 (v142), and the pairing *values* are atom identities given the exponent-duality lift, including the new closure $|\mu_4| + g_{\rm car} = N_{\rm fam}^2$ (v145); the sheet/parity question `GATE.QGEO` has *no discrete freedom left* — the $\mathbb{Z}_3$ choice is *derived* (v141) and the *full* Möbius D_4 of the seam curve matches the integer model parity by parity (v146); R2's Q-system identification holds at the finite Lie level (v143), with the remaining net statement *scoped*: the odd glue sectors are Ramond modules, invisible to the untwisted NS module, and the glue choice is the R-projection (v148); and the seam quantum clock R1 is *typed closed* — edge-sector budget matches Volkov–Wipf exactly (v138), the det-ratio cancellation is derived in-family (v144), the ring sum *is* the Born-squared Gaussian zero-mode integral, and the quantum bend is det′-clean (v147). Beyond the five classes: *both* dual normals are now anchor data ($n = (p_2, p_0, e_2)(a)$ in the cusp frame, v149), R3's mechanism is exhibited at the gapped-model level (v150), the Calderón transfer answered (the DtN kernel is conically *clean*, v151), and its normalisation shown to be the dimensionful anchor, not a separate gap (v152). Two closing theorems then re-type the survivors: the **No-Unit Theorem** (v153) makes $v_{\rm geo}$ an *irreducible metrology primitive* — a dimensionless compiler cannot make an absolute scale, and $U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ are one unit in three dimensions — and the **Simple-Current Extension Theorem** (v154) states the central $G_{\rm net}$ closing as $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$ (index 4, c=8, μ=1 ⇒ holomorphic ⇒ E_8), exact, with only the seam-side identification left. *That seam-side identification is then driven to bedrock (v175–v181):* net existence and full-cone reflection positivity are discharged to **[E]** (v175), the realisation is assembled as one central theorem (v176) and split into the marks + kernel obligations whose finite cores close (v177/v178); the two then unify into one conformal-realisation premise (v179), which reduces — via uniformisation, Kerékjártó and the order-4 Möbius classification — first to a seam-*isometry* premise (v180) and finally, by equivariant uniformisation, to the strictly weaker conformal-*symmetry* / deck premise `QGEO.SYM.01` (v181): the carrier μ_4 clock is the conformal deck of the seam. Below that the residual is *definitional* (the seam's conformal structure *is* the μ_4-deck structure of the carrier), not a missing theorem — the single, irreducible structural residual of the whole theory. It is since sharpened to its *non-circular* form (v194–v198): the bedrock is cracked to the *state-invariance* $\omega \circ \rho = \omega$ (the DtN principal symbol commutes *exactly* with the clock; Tomita–Takesaki gives $[\rho, \Lambda_\Sigma] = 0$ with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity) — the maximally-operational form of the same postulate. *Its geometric normal form* (the Uniform node: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit→ μ_4) *and the conditional implication* mark-local ⇒ $\omega \circ \rho = \omega$ are now **Lean-formalised** (`FORM.QGEO.02`/`FORM.QGEO.01`, audit: pass, only the three standard kernel axioms); the premise `QGEO.SYM.01` *itself* stays **[O]**. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[E]	gapped transport ⇒ *unique* Perron–Frobenius attractor (v56); bootstrap web g_{car}=5 forced (v6/v14)
Newton constant G_N	[E]/[C]	not *independent*: branch pins $\bar{M}_{\text{Pl}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[E]	$g_{\text{car}}\lVert \text{Pl}_{1,a}(K)\rVert_1/(N_{\text{fam}}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
(U) flavor gate	[O]$\to v_{\text{geo}}$	**Gate 1 closed**: $U_{\text{point}}\to v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$
$R+R^2$ gravity	[E]/[C]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[E]/[C]	gap-decoupled admissible measure (Δ_{eff}=1.648>0)
ambient QG measure	[O](reduced)	**Gate 2 tier B**: holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$) and factors into the already-open A_2 net-existence + `GATE.QGEO` — zero new gates (v160–v165)
v_{geo} metrology unit	[O](primitive)	No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, m/μ a ratio — one unit, irreducible
Koide	[C]	$Q_{\text{src}} + \varphi_0/24$ as source→pole transfer conjecture
axion DM	[C]/[O]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[C]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[O]	cross-sector QCD/EW ratio
$N_\star$	[C]	reheating input; scalaron-reheating point $N_\star$=51.4 (v86); frozen band $[50, 60]$
$a = (1, 1, 2)$	[E]/[C]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection g=5 (QBL)	[E]/[C]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank $= c$); residue = the G_{net} premise itself — gate closes ⇒ carrier [E]
selector establishment R4′	[E]/[C]	$(d, n) \Rightarrow R$ columnwise (v136); both normals *anchor data* — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = *one* assignment
sheet/parity `GATE.QGEO`	[E]/[C]	H^1 characters = Spec$(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice *derived* (v141); full Möbius D_4 matched parity by parity (v146); residue = the module identification only
seam quantum clock R1	[E]/[C]	typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born^2 Gaussian zero-mode integral, bend det′-clean (v147); residue = measure + reference
E_8 Lean	[E]	hardening, not a blocker (script-certified)

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Only *two* research gates remain — (U_{wall}) and G_{metric}; the frontier items are typed interfaces, not structural holes. Of these, (U_{wall}) is the only genuine *physics* gate left: G_{metric} is now heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: Δ_{eff}=1.648>0 isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion **`tfpt_research_contracts`**, with priority on the selector-triangle pairings (the v_{geo} scale anchor) and the G_{net} inclusion theorem. [`verification/v36_spectral_action_g2.py`]

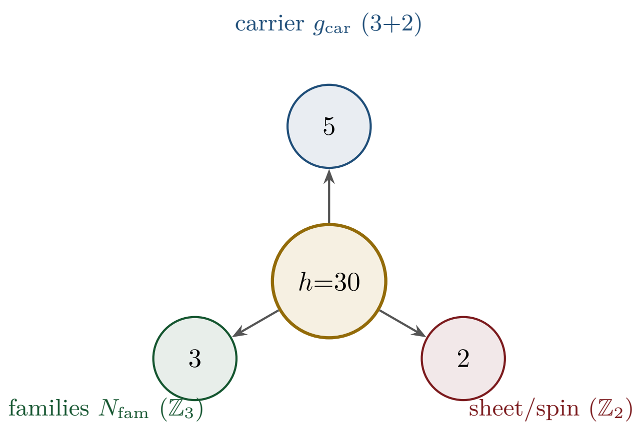
Sharper still: the structural residual is *one* condition. G_{metric}, the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition *"the seam carries no nontrivial abelian sector"* — holomorphy (μ-index 1), a homology-sphere seam link (Γ perfect $\Leftrightarrow 2I$, [`verification/v232_e8_kleinian_seam.py`]), exactly one 1-dim irrep ([`verification/v219_icosahedral_mckay.py`]) — all of which force E_8 ($\#(\text{mark-1}) = |H_1| = 1$ only for E_8; holomorphic $c = 8 = g_{\text{car}} + N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic $\Leftrightarrow \det K = 1$, the extension tower $D_5 \oplus A_3(16) \to D_8(4) \to E_8(1)$ being anyon condensation (the Kitaev E_8 state); the one open analytic step is then "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" = `QGEO.SYM.01`. So beneath the two gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ($f_2/f_0 = 1$, v259), the seam, the carrier-16 and E_8 live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo `QGEO.SYM.01` and adds *no new open item*, while the ambient quantum-gravity measure stays separate by design. [`verification/v234_seam_holomorphy_selection.py`] [`verification/v235_seam_chern_simons.py`]

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence. And even those two tighten:* the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined E_8-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. *A fourth, deeper forcing* comes from the McKay correspondence: E_8 is the exceptional top of the tower of finite $SU(2)$ subgroups, so its branch data is the icosahedron and the three atoms $(2, 3, 5)$ are its rotation-axis orders — the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees equal to the affine-E_8 Kac marks, $\sum = 30 = h(E_8) = 2{\cdot}3{\cdot}5$ ([`verification/v219_icosahedral_mckay.py`]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [`verification/v6_bootstrap.py`] [`verification/v14_carrier_uniqueness.py`] [`verification/v23_anchor_generator.py`] [`verification/v219_icosahedral_mckay.py`]

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

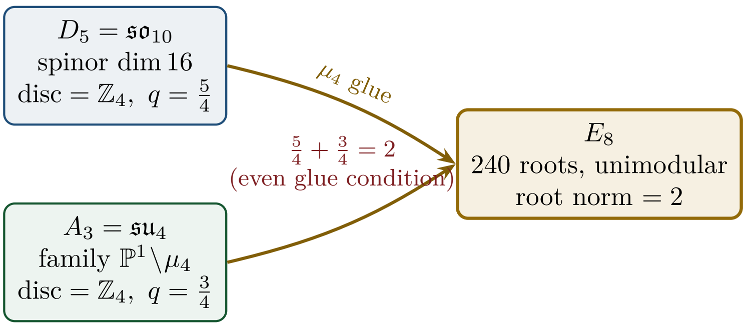


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. **[E]**
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. **[E]**
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30}; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \backslash \mu_4$ is the A_3 Coxeter element. **[E]**
- **why these three primes:** E_8 is the icosahedral top of the McKay tower ($2I$, order 120, irrep degrees = affine-E_8 marks summing to 30), so $2, 3, 5$ are the icosahedron's rotation orders; and the E_8 exponents $(\mathbb{Z}/30)^\times$ carry an order-4 totative character $\langle 7 \rangle$ — the same μ_4 seam clock. **[E]** [`verification/v219_icosahedral_mckay.py`] [`verification/v223_coxeter_totative_clock.py`]
- **a network reading (computational substrate).** On the same McKay $(2, 3, 5)$ graph the affine-E_8 marks are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4 \cdot m$ (it converges from *any* positive start to $(1, 2, 3, 4, 5, 6, 4, 2, 3)$); the spherical tower terminates at E_8 and the cascade $E_8 \supset E_7 \supset E_6 \ldots$ is nested-subgraph coarse-graining. So E_8 reads as the *universal attractor of a minimal* $(2, 3, 5)$ *network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. **[C]** A minimal hypergraph *rewrite* reproducing the *full* structure (lattice + recovery + readouts) stays open, and φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction. **[O]** [`verification/v298_e8_network_attractor.py`]

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\operatorname{disc}(D_5) = \operatorname{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind "positing of 248". **[E]** [`verification/v1_e8_glue.py`]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\text{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^{+}{=}16$, $N_{\text{fam}}{=}3$, $\Omega_{\text{adm}}{=}48$	[E]
hypercharge	roots of the charge polynomial $bsY^2 + (s{-}b)Y - 1$	[E]
$U(1)$ index b_1	$b_1 = \dfrac{g_{\text{car}}\, 2^{g_{\text{car}}-2} + 1}{10} = \dfrac{41}{10}$	[E]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[E]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[E]/[C]
electroweak scale	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[E]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[E]/[C]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[C]
inflation amplitude	$A_s = \dfrac{N_\star^2}{24\pi^2} c_3^7$ in the R^2 branch	[C]
E_8	$D_5 \oplus A_3 + \mu_4$-glue	[E]

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by $5 =$ carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

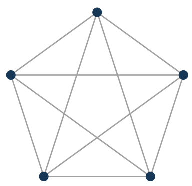
Typing note on Λ (a deliberate danger zone). "Λ" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [E] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier

With $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5:

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \qquad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda, \qquad (1,5,10) = \binom{5}{0},\binom{5}{1},\binom{5}{2}$$

EW is the unit x^1, Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. "Hubble = slot product, Λ = pair product." [E]



$\binom{5}{1} = 5$ vertices — action x^1 (Hubble slot)
$\binom{5}{2} = 10$ edges — Λ density pair x^{10}
$\binom{5}{3} = 10$ triangles — dual sector
$\binom{5}{4} = 5$ complements — inverse sector
$\binom{5}{5} = 1$ determinant — closure

K_5

K_5 on the five carrier slots; the action powers x^1, x^5, x^{10} are its $\binom{5}{k}$ faces (mnemonic; the physical scale map stays [C]).

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}}\,\lambda_Y^{L_{f,j}}\,\Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1-\varphi_0^{\text{ret}})}, \qquad \varphi_0^{\text{ret}} = \tfrac{1}{6\pi} + \tfrac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers (tr $= 9 = N_{\text{fam}}^2$, det $= 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret}-ladder $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c\,(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). *(Full derivation: document 2,* `tfpt_2_standard_model`*.)* [`verification/v18_quark_yukawa.py`] [`verification/v20_lepton_c_derivation.py`] [`verification/v46_grand_mass_volume.py`]

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1 \backslash \mu_4$ produce D_5 and A_3, glued to E_8; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30}/Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions $\Rightarrow$ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers $240, 248$ are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms 5/4 and 3/4 (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8.

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that $E_6 \times A_2$ reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2\cdot 27\cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic ↔ unitary representation); the algebraic selector R, the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from D_4, v69/v70; since v136/v139 the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\mathrm{Spec}(Q_+) = (1, 2, 3)$ is the A_3-exponent grading on the seam curve's cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point}, an anchor; Research Contract 1). [E]/[O]
2. **Gravity — exact in the R^2 regime.** Spectral action → $R{+}R^2$, BRST/Hodge kills the gauge directions, FRW reduction → Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n\geq 3} \to R^2$, valid for curvature $\ll M^2$). [E] in the regime
3. **Strong CP — decoupled.** $\theta_{\text{eff}}{=}0$ follows from three structural facts (γ_5-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/E_8 live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single premise `QGEO.SYM.01`. *What this does* not *close:* the full measured pole masses (the absolute scale v_{geo} + standard RG), the absolute QCD scales (m_p/m_e, [O]), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT / ambient measure ([O]). [E] (transfer model + boundary relative object) / [C] (continuum) / [O] (real 4D)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`1886 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.)*

The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, N_{fam}, hypercharge, b_1	[E]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[E]	— complete
bootstrap $(g, \mu) = (5, 4)$, "8" in c_3	[E]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[E]	— **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin [C]
fermion masses (φ_0^{ret}-ladder)	[E]/[O]	leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)	[E]/[C]	θ_{12} cond. derived; θ_{23} maximal; *full* PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]
flavour matrix / H2	[E]	dictionary *exact* (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d, n) (v136/v139); pairing values = atom identities (v142/v145); residue = *one* lift map
seam quantum clock (R1)	[E]/[C]	typed closed: reduced edge-sector clock (v124–v133); external VW edge = 2× reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)
inflation A_s, n_s, r, scalaron M	[C]	scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; n_s, r are bands over $N_\star \in [50, 60]$ (reheating point 51.4, v86)
boundary QFT (Modular Spectral Closure)	[E]/[C]	**one relative object** $\mathsf{TFPT}_{\text{QFT}}$: D_F = covariance induction (v258), cutoff = KMS weight $f_2/f_0 = 1$ (v259), seam/carrier/E_8 on one K3 (v260), certified (v261) — closed modulo `QGEO.SYM.01`
perturbative 4D S-matrix (S_{pert})	[E]/[C]	**Epstein–Glaser-constructible** (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ with one-loop unitarity (optical theorem, v278) — matter+gauge; gravity R^2/Weyl^2 non-unitary (Stelle ghost)
genuinely open — the actual TOE remainder (what the closure does not *cover)*		
QFT measure (admissible sector)	[E]	**closed: RP from cusp spectrum, OS reconstruction**
full measured pole masses	[C]	ladder $\times v_{\text{geo}}$, dressed by standard RG (no new physics); v_{geo} over-determined (v274)
absolute QCD scales (m_p/m_e, Λ_{QCD})	[O]	QCD matching contract (`FR.MPME.01`); not a compiler number
full PMNS dynamics (mass scale, CP)	[C]/[O]	angles typed; CP strength $J_{\text{PMNS}} = -0.0297$ derived (v270); absolute ν-scale = one seesaw ratio (v272)
(1) `SEAM.EQUIV.01` — one named theorem (*raw seam* $= (E_8)_1$ *at* $\tau = i$)	[O]	the single arrow unifying the flat-$\tau = i$ geometry and the det $K = 1$ holomorphy (v282/v286, import firewall); Route A is a citable theorem stack (v297), Route B *proves* the full-L^2 $\mathbb{Z}_4$ lift (v288); residual = *Flat-Away* (v289), its heat positive-definiteness proved+Lean (v295)
(2) `QG.AMB.01` — nonperturbative ambient measure	[O]	the genuinely separate strict-TOE summit: the perturbative R^2/Weyl^2 gravity ghost (Stelle) is the frontier; Tier-B's holomorphy bit (det $K = 1$, v277) is absorbed into `SEAM.EQUIV.01`
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[C]	**hardened:** $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8" = rank E_8; π stays the one primitive
η_B, m_p/m_e, Koide, dark matter	[O]	frontier handles only (see document 4)

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret}-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object* $\mathsf{TFPT}_{\text{QFT}}$ (the *Modular Spectral Closure*, v258–v261: D_F a covariance induction, the cutoff the seam KMS weight, seam/carrier/E_8 on one Kummer/K3) — closed modulo the *single* premise `QGEO.SYM.01`. *What that closure does* not *reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder $\times\, v_{\rm geo}$ + standard RG, [C]), the absolute QCD scales (m_p/m_e, [O]), the complete PMNS dynamics ([C]/[O]), and the nonperturbative *ambient/metric-sector* measure, i.e. *full quantum gravity*. The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278); only the nonperturbative ambient measure stays [O] — its R^2/Weyl^2 gravity sector is renormalizable but non-unitary (the Stelle ghost), the lone perturbatively-irreducible piece, with Tier-B reduced to the single holomorphy bit $\det K{=}1$ (v277) and the whole bedrock reduced to one geometric postulate, *the raw seam is the flat $\tau{=}i$ pillowcase* (v276). Sharper still (v282): the flat $\tau{=}i$ geometry and the E_8 holomorphy are *two faces of one object* — the $(E_8)_1$ character is $\chi_{E_8}{=}j^{1/3}$ so $\chi_{E_8}(i){=}1728^{1/3}{=}12$, and $\tau{=}i$ is at once the order-4 CM point (the flat geometry) and the $(E_8)_1$ partition-function modulus (the holomorphy), collapsing the two obligations to one: *the raw seam is $(E_8)_1$ at $\tau{=}i$*, with a direct numerical experiment on the pillowcase realising the chain (v280). That single open lemma is attacked from two decomposed routes — RP-state uniqueness (5/6 lemmas discharged; Troyanov reduces the rest to "the raw seam is the constant-curvature state", v284) and seam-net condensation (3/4 discharged; the open bit is $\det K{=}1$, v285) — whose open lemmas coincide, so the one statement that closes both is a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$. *Closing sprint* (v286–v288): that statement is now the single *named* theorem `SEAM.EQUIV.01` — the contract types four objects and six arrows and enforces an *import firewall* so the two routes never feed each other (no "E_8 smuggled into the geometry and pulled back out"); Route A (AQFT) reduces it to *one* standard import, "invertible Gaussian bulk ⇒ single-sector boundary" (4/5 lemmas discharged, v287); and Route B (DtN) *proves* the full-L^2 lift of the subprincipal $\mathbb{Z}_4$ block-diagonality (v288), shrinking the residual to the single sharper question *why is the raw seam subprincipal term mark-local?*. *Flat-Away round* (v289–v297): that residual is decomposed to one open analytic lemma, *Flat-Away* (RP+gap+4 marks ⇒ the curvature vanishes away from the marks, v289); a red-team shows $\mathbb{Z}_4$-invariance is *not* mark-locality (v290), so Flat-Away is attacked on three independent routes — the heat route, whose positive-definiteness is *proved* by convexity with the exact a_2 kernel in closed form and Lean-formalised (v295/v296, `FORM.FLATAWAY.01`); the spectral-rigidity route (positive-definite Hessian over the full smooth-$\mathbb{Z}_4$ space, v293); and the RP-energy/Troyanov route (v294) — while Route A's import is a citable theorem stack (v297) modulo the *same* geometric input, so both routes reduce to one shared external fact. The seam seed is *hardened*: $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K\, dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive. *4D-QFT fork policy (frozen, v265):* the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* (E_8 is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

10 Does this make the theory more probable?

Plausibility balance. *Pro:* (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors ⇒ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	**TFPT value & derivation**	**Experiment (running / upcoming)**	**St.**
solar angle $\sin^2\theta_{12}$	prediction of record $\frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2} = 0.306747$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36c_3^4$ (cond.)	**JUNO**: first 59.1 d run 0.3092±0.0087 (compatible), precision phase pending	[E]/[C]
reactor angle $\sin^2\theta_{13}$	$\varphi_0^{\mathrm{ret}} e^{-5/6} = 0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma{=}\frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[E]
atmospheric $\sin^2\theta_{23}$	$\approx \frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected)	NOvA / T2K / **DUNE** (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)	[C]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha){=}0$ (no fit, no second branch)	CODATA-2022 (data through 2022) 137.035999177(21): dev. 2.9×10^{-10} (1.9σ of its uncertainty)	[E]
tensor ratio r	$r = \frac{12}{N_\star^2} \in [0.0033, 0.0048]$ (frozen band, $N_\star \in [50, 60]$); scalaron-reheating point 0.0045 ($N_\star{=}51.4$, v86)	now $r{<}0.036$ (BICEP/Keck BK18); **CMB-S4** ($\sigma_r{\le}5{\times}10^{-4}$, obs. $\sim$2033)	[C]
tilt n_s	$n_s = 1 - \frac{2}{N_\star} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ($N_\star{=}51.4$, v86)	Planck (0.9649±0.0042); CMB-S4 sharpens	[C]
amplitude A_s	$\frac{N_\star^2}{24\pi^2} c_3^7$; 2.0×10^{-9} at $N_\star{=}55$ (the measured A_s prefers fast reheating, v86)	Planck (2.1×10^{-9})	[C]
neutron EDM	$\theta_{\mathrm{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)	**PSI nEDM** / SNS (running)	[E]
CP phase δ_{CKM}	$\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs γ_{PDG}	**LHCb** / Belle II γ combination; decision at $\sigma_\gamma \le 0.96^\circ$	[E]
cosmic birefringence β	$\beta = \frac{\varphi_0^{\mathrm{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)	ACT DR6 hint 0.215°±0.074° (systematics not yet understood — not a validation target); **Simons Obs.** / LiteBIRD	[C]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	**LEGEND / nEXO**, DUNE, KATRIN	[C]
flavor invariants	$\det R{=}h(D_5){=}8$, principal minors $(2, 3, 5)$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$ (exact)	any future CKM/PMNS global fit must satisfy these	[E]
H_0 vs. Λ	$v_{\mathrm{EW}} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine	SH0ES / **DESI** / Planck (Hubble tension)	[C]

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`*.*

The two decisive near-term tests

(1) JUNO and θ_{12}. JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the single prediction of record $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.306747$ (misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret} = c_3 + 36c_3^4$). The first 59.1-day JUNO run reports $\sin^2\theta_{12} = 0.3092 \pm 0.0087$ — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from ~ 0.307 at high significance falsifies the seam mechanism.

(2) r. The R^2 scalaron with M fixed by c_3^7 predicts a small $r = 12/N_\star^2$: the frozen band $r \in [0.0033, 0.0048]$ over $N_\star \in [50, 60]$, with the scalaron-reheating point $r = 0.0045$ ($N_\star$=51.4, v86) — already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \le 5\times10^{-4}$, first observations ∼2033–34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which $M_{\rm Pl}$ and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8; and the discipline rule "*every load-bearing number appears in at least one E_8 projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [`verification/v100_numerology_null_mc.py`].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying $M_{\rm Pl}, A_s$;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\rm eff} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor ($U_{\rm point}$) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+;
- a sharper geometry↔gravity bridge once the Hodge/R^2 branch is lifted from [C] to [E].

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\mathrm{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable.* Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\mathrm{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

Script	What it machine-checks
`v1_e8_glue.py`	glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$
`v2_carrier_pascal.py`	$g_{\mathrm{car}}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\mathrm{fam}}{=}3$, rank $E_8{=}8$, $\Omega_{\mathrm{adm}}{=}48$, $b_1{=}\frac{41}{10}$
`v3_em_alpha.py`	EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation
`v4_flavor_matrix.py`	residue matrix R: det $=8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$
`v5_e8_cascade.py`	cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to$ 240, IR tail 46080, variance 6552
`v6_bootstrap.py`	reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\mathrm{car}}{=}5$ three ways; $8{=}$ rank $E_8{=}h(D_5){=}\varphi(30)$
`v7_gravity_cosmo.py`	scalaron exponent 7, $M^2_{\mathrm{scal}}/\bar{M}^2_{\mathrm{Pl}}{=}c_3^7$, A_s, n_s, r, Ω_b, $\sin^2\theta_{12}$
`v8_horizon.py`	horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, S_{dS}, Page, $\beta_{\mathrm{rad}}{=}0.2424°$
`v9_neutrino_texture.py`	solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$, $\theta_{23}{=}45°$, $\theta_{13}{=}0$
`v10_projection_involution.py`	Q, Σ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; χ_Q, χ_K; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$
`v11_unique_KQ.py`	K, Q are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
`v12_mass_generation_polynomials.py`	sector/generation polynomials of K and the anchor-block det ladder $(9, 10, 16, 40)$
`v13_open_gates.py`	gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\mathrm{fam}}$
`v14_carrier_uniqueness.py`	$g_{\mathrm{car}}{=}5$ unique ($N_{\mathrm{fam}} \in \mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\mathrm{Tr}\,Y{=}\mathrm{Tr}\,Y^3{=}0$
`v15_bootstrap_classification.py`	$D_5{\oplus}A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
`v16_solar_dual_anchor.py`	$a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2}, -\frac{1}{2}, 1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$; full PMNS

continued on next page

Script	What it machine-checks *(continued)*
`v17_hexagonal_resolvent.py`	finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark c backbone)
`v18_quark_yukawa.py`	quark source ratios $\frac{m_u}{m_d}$=0.470085, $\frac{m_c}{m_s}$=13.61, $\frac{m_t}{m_b}$=40.80 exact; lepton $c=(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy
`v19_monodromy_moduli.py`	exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4-fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
`v20_lepton_c_derivation.py`	lepton c's derived: $\delta=\frac{1}{2}$ resolvent $c=\lvert\mu_4\rvert^w/(\frac{5}{4}-\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7},\frac{4}{3}$; product $\frac{2g_{\rm car}}{N_{\rm fam}^2}=\frac{32}{9}$ $\Rightarrow\frac{7}{6}$
`v21_solar_product_quark.py`	solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$; product $\frac{2g_{\rm car}}{N_{\rm fam}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
`v22_open_gates_audit.py`	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
`v23_anchor_generator.py`	anchor $a=(1,1,2)$: $e_k=(4,5,2)$, $c_3=\frac{1}{2e_1\pi}$; $p_n=2+2^n\Rightarrow 240,248$, binary ladder; inputs $\to\{a,\pi\}$
`v24_quark_ratio_closure.py`	quark ratios $\frac{55}{117},\frac{34}{47},\frac{3}{26}$ from TFPT blocks; mass ratios match $<0.03\%$; rational c gauge, Λ_q in $O(1)$
`v25_frontier_conjectures.py`	conjectures: Koide $Q_{\rm src}+\frac{\varphi_0^{\rm ret}}{24}\approx\frac{2}{3}$; axion $f_a=M_{\rm scal}/128\to m_a\approx 23.8\,\mu$eV
`v26_flavor_frontier_unification.py`	the "11" is not uniquely forced ($\Rightarrow$ [C]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
`v27_wall_representative.py`	explicit balanced wall rep $W_{\rm wall}$ (cols = perm$(0,1,2)$, rows $w=(2,1,1)$, pardeg 0); selector det R=8, Spec$(Q_+)=\{1,2,3\}$
`v28_gravity_fR.py`	$R+R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\rm Pl}$ scalaron, $N_\star=57\to n_s,r,A_s$; full metric measure open; $248c_3^2<6\log\frac{3}{2}$
`v29_research_contract_certs.py`	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick $W_{\rm wall}$) and $G5$ gap-dominance $2\lVert V\rVert<\Delta$
`v30_d4_character_variety.py`	$C_U^{(2)}$: full D_4-fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the R4′ selector establishment)
`v31_R_dictionary.py`	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ-invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
`v32_rh_splitting.py`	RH route: $\sum_k U^kA_0U^{-k}=4\,{\rm diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2\Leftrightarrow{\rm diag}(A_0)=(\frac{1}{2},\frac{1}{4},\frac{1}{4})$ (algebraic); $\prod M_k=I$ + R-bridge stay open (c_u/c_d not forced)
`v33_explicit_flat_bundle.py`	explicit valid $(U_{\rm wall})$ flat bundle (RH solve): cusp + $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ + $\lVert M_\infty-I\rVert\sim10^{-9}$ $(\prod M_k=I)$ + irreducible (case A)
`v34_h2_bridge_attempt.py`	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k=I$); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\rm min}$ dictionary is the remaining input (c_u/c_d not obtained)
`v35_quark_qcd_boundary.py`	the open gates are the discrete↔continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
`v36_spectral_action_g2.py`	G2: spectral action $\to R+R^2$ $(a_2\sim-R/3,\ a_4\vert_{R^2}\sim R^2/72)$; $M^2/\bar{M}_{\rm Pl}^2=6(4\pi)^2/f_0$, c_3^7 fixes f_0; gap-decoupling $\Delta_{\rm eff}$=1.648>0 $\Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
`v37_plucker_anchor.py`	anchor-plane Plücker apparatus: $\lVert{\rm Pl}(K)\rVert_1=11$, $\lVert{\rm Pl}_R(K)\rVert_1=26$, $\sum{\rm Pl}_R(L)=60$; pencil $\det(K+xQ)=3x^3+7x^2+6x+4$; $K_e-K_u=a$, $L_e-L_u={\rm Exp}(A_3)$; lepton ring $c_ec_\tau=\lvert\mathbb{Z}_2\rvert c_\mu$; Koide $u\to\frac{53}{54}u$ ([C])

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Script	What it machine-checks *(continued)*
`v38_uwall_killswitch.py`	($U_{\rm wall}$) U2 kill-switch hardened: D_4-cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq$0.35, reducible locus measure-zero $\Rightarrow$ case A generic, ($U_{\rm wall}$) survives; amplitude dictionary still research-level
`v39_uwall_selectors.py`	($U_{\rm wall}$) selector side closed on the explicit point: splitting type = anchor $a{=}(1,1,2)$, pardeg $=0$; $\det R{=}8{=}n{\cdot}a$, $\mathrm{Spec}(Q_+){=}\{1,2,3\}{=}3\alpha{+}1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\min}$ Hitchin) remain
`v40_harmonic_metric.py`	($U_{\rm wall}$) harmonic metric = FINITE linear algebra (polystable$\Rightarrow$unitary$\Rightarrow\Phi{=}0$): unique invariant form $H{>}0$, unitarised $\widetilde{M}_k$ cusp-class with $\lvert\widetilde{M}_k\rvert{\in}\{0,\frac{1}{2},\frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R
`v41_leg_assignment.py`	($U_{\rm wall}$) final leg test: amplitudes = $\delta{=}\frac{1}{2}$ resolvent (closed); $\Lambda_\mu{>}1$ rules out holonomy moduli; harmonic holonomy $\lvert\mathrm{diag}\rvert{=}(0,\frac{1}{2},\frac{1}{2})$ supplies $\delta{=}\frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
`v42_exterior_leg.py`	($U_{\rm wall}$) Exterior Leg Lemma: u,d share $(r,w){=}(1,1)$ so any scalar leg gives $c_u/c_d{\equiv}1$; the u/d split is the Λ^2F area $\mathrm{Pl}_{\mathbf{1},a}(K)$, $\lVert\cdot\rVert_1{=}11$ generated $\Rightarrow \frac{5\cdot 11}{9\cdot 13}{=}\frac{55}{117}$ [E]; phys. identification [C]
`v43_exterior_bridge.py`	($U_{\rm wall}$) Λ^2F bridge: $\Lambda^2(\widetilde{M}){=}\overline{\widetilde{M}}$ (exterior leg $=\bar{\mathbf{3}}$) confirmed; continuous action $\neq$ integer $\mathrm{Pl}(K)$ $\Rightarrow$ bridge is the discrete H2 invariant ($\det R{=}8$, $\mathrm{Spec}(Q_+)$ confirmed), not continuous
`v44_carrier_exterior.py`	($U_{\rm wall}$) Lie-level grounding: $16{=}\Lambda^{\rm even}(5){=}1{+}10{+}5$; $u^c{\in}\Lambda^2(5)$, $d^c{\in}\Lambda^4(5)$ $\Rightarrow$ exterior degrees $(2,4)$, sum$=6=\lvert R^+(A_3)\rvert$, diff$=2=\lvert\mathbb{Z}_2\rvert$; the exterior leg is the carrier's own Λ^2 grading (no special math)
`v45_family_exterior.py`	($U_{\rm wall}$) the "11" is Pascal: $\lvert\mathrm{Pl}(K)\rvert{=}(1,4,6){=}\{\Lambda^0,\Lambda^1,\Lambda^2\}(\mu_4)$; $\lVert\mathrm{Pl}(K)\rVert_1{=}11{=}\sum_{k\leq 2}\binom{4}{k}{=}16{-}g_{\rm car}$ (cumulative ext of μ_4 up to $\deg u^c{=}2$); $15{=}\dim\mathfrak{su}(4){=}10{+}5{=}N_{\rm fam}g_{\rm car}$
`v46_grand_mass_volume.py`	absolute mass-scaling is [E], not a product rule: sector dets $(\varphi_0^{\rm ret})^6,(\varphi_0^{\rm ret})^9,(\varphi_0^{\rm ret})^{10}$ (exponents $\lvert R^+A_3\rvert, N_{\rm fam}^2,\mathcal{A}_\Lambda$) $\Rightarrow$ $\det M_{\rm SM}{\sim}(\varphi_0^{\rm ret})^{25}{=}(\varphi_0^{\rm ret})^{g_{\rm car}^2}$; Q-rows $(4,5,6)$ sum $15{=}\dim A_3$; $25{+}15{=}40{=}\sum L$
`v47_selection_theorem.py`	Boundary Carrier Selection (Thm A): $\mu_4{\to}A_3, N_{\rm fam}{=}3$; 16-spinor$\to D_5$; $q(D_5){+}q(A_3){=}2$, glue $4{=}\lvert\mu_4\rvert$; only $D_5{\oplus}A_3$ meets all four conditions ($D_8,E_7{+}A_1,E_6{+}A_2,A_4{+}A_4,A_8$ excluded) [E]/[C]
`v48_em_ward.py`	EM boundary Ward (Thm C): $F_{U(1)}{=}$ Maxwell α^3 − Calderón $2c_3^3\alpha^2$ − transport $8b_1c_3^6\log\frac{1}{\varphi_{\rm seam}}$; $8b_1{=}\frac{4}{5}M{=}\frac{164}{5}$, $-\frac{5}{4}{=}q(D_5)$ [E]; Ward origin [C]
`v49_readout_rigidity.py`	Readout Rigidity (Thm U2): $c_u/c_d{=}\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q}$ $\Rightarrow$ no point-uniqueness needed; $(U_{\rm wall}){=}U_{\rm unitary}{+}U_{\rm H2}{+}U_{\Lambda^2}{+}U_{\rm point}$, only $U_{\rm point}$ open
`v50_q_geometry.py`	Q geometry (Thm Q): $Q_+{=}$ diag$(3,2,1)$-block $\chi{=}(t{-}1)(t{-}2)(t{-}3)$, Q_- $\chi{=}t(t^2{-}3)$; Q unique under rows$(4,5,6)$/cols$(9,5,1)$/spectra; geom. origin [C]
`v51_boundary_half_step.py`	glue norms $=(g_{\rm car},N_{\rm fam})/\lvert\mu_4\rvert$: $q(D_5){=}\frac{5}{4}$, $q(A_3){=}\frac{3}{4}$; four ops forced $\to$ {sum= 2 root norm, diff= $\frac{1}{2}{=}\lvert\mathbb{Z}_2\rvert/\lvert\mu_4\rvert{=}\delta$, prod= $\frac{15}{16}$, ratio= $\frac{5}{3}$}. So $\delta{=}\frac{1}{2}$ is the carrier−family count over the glue index, not chosen [E]
`v52_pencil_endpoints.py`	$K{+}xQ$ pencil endpoints: $P(-1){=}2{=}\lvert\mathbb{Z}_2\rvert$, $P(0){=}4{=}\lvert\mu_4\rvert$, $P(1){=}20{=}\det L$, $P(2){=}68{=}2p_5(a)$; $\det(K{-}Q){=}2$, tr $=N_{\rm fam}$ (audit ext. of v37) [E]
`v53_compiler_core.py`	Big picture: the whole integer skeleton from $(g_{\rm car},N_{\rm fam}){=}(5,3)$ — rank $E_8{=}8$, $\lvert\mathbb{Z}_2\rvert{=}2$, Pythagorean $\Delta_Y{=}g_{\rm car}^2{=}25{=}N_{\rm fam}^2{+}\lvert\mathbb{Z}_2\rvert{\cdot}$rank $E_8{=}9{+}16{=}\dim S^+$ (diff. of squares); anchor char-poly $(t{-}1)^2(t{-}2)$ unique [E]
`v54_seam_horizon_keystones.py`	The "8" in $c_3{=}\frac{1}{8\pi}$ overdetermined and gravitationally aligned ($2\lvert\mu_4\rvert{=}$rank $E_8{=}h(D_5){=}\varphi(30){=}8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1,(2/3)^6,(1/3)^6\}$, $\lambda_2{=}(2/3)^6$) governs *both* SM flavor gap and horizon Page recovery [E]

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Script	What it machine-checks *(continued)*
`v55_coxeter_cycle.py`	Computed E_8 Coxeter element: order exactly $30{=}\lvert\mathbb{Z}_2\rvert N_{\rm fam} g_{\rm car}$, eigenphases = exponents = totatives of 30, so rank $E_8{=}8{=}\varphi(30){=}$ live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda{=}32\pi^4$) **[E]**
`v56_unique_attractor.py`	Gapped boundary transport (gap $6\log\frac{3}{2}{>}0$) $\Rightarrow$ *unique* attracting fixed point, geometric rate $(2/3)^6$ (two starts $\to$ same direction); Coxeter in $\lvert\mu_4\rvert{=}4$ planes; entropy reset = rank-1 projector **[E]**
`v57_horizon_crosslinks.py`	Horizon cross-links: $c_3{=}\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4{=}1/\lvert\mu_4\rvert$ entropy density); Hod QNM $\omega_R/T_H{=}\ln 3{=}\ln N_{\rm fam}$; Hawking power denom $1920{=}\lvert W(D_5)\rvert$; one α^{-1} for EM/S_{dS}/Λ/Hubble **[E]** (+ **[C]** identification)
`v58_seam_horizon_chain.py`	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3{=}1/(2{\cdot}4\pi){=}\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}{=}M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) **[O]** open
`v59_area_law_evidence.py`	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim\frac{c}{3}\log L$), by the *same* gap that fixes the attractor; $8{=}\lvert\mathbb{Z}_2\rvert\lvert\mu_4\rvert$. The c_3 coefficient stays **[O]** open
`v60_lambda_metrology_branch.py`	Λ branch selection: $(8\pi)^2\delta_{\rm top}{=}\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\rm top}{=}\frac{64\pi^3}{3}{=}661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215°{\pm}0.074°$ vs $0.2424°$ (0.37σ) **[E]**
`v61_cft_bridge.py`	Boundary-CFT mirror: level-1 WZW central charges $c(E_8){=}8$, $c(D_5){=}5$, $c(A_3){=}3$ = (rank $E_8, g_{\rm car}, N_{\rm fam}$); conformal embedding $c_{\rm coset}{=}0$; $SU(3){\leftrightarrow}SU(4)$ reconciliation $N_{\rm fam}{=}\lvert\mu_4\rvert{-}1$, 4→3+1 (one geometry $\mathbb{P}^1\setminus\mu_4$, two sides) **[E]**
`v62_data_scorecard.py`	TFPT vs 2024/25 data: α^{-1}, $\sin^2\theta_{12}$, $\beta_{\rm rad}$ match; $\sin^2\theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r{\approx}0.004$ future falsifier **[E]**
`v63_seam_engineering_index.py`	Exact Seam-Engineering Index $\Xi{=}2\lVert V\rVert/\Delta{=}\frac{31}{24\pi^2\log(3/2)}{\approx}0.323$ ($2\lVert V\rVert{=}\frac{31}{4\pi^2}$, $248{=}\dim E_8$, $31{=}1{+}h^\vee$), $\Delta_{\rm eff}{\approx}1.648$; four-class possibility taxonomy **[E]**/**[O]**
`v64_causal_boundary_nogo.py`	Causal Boundary Engineering *conditional no-go*: gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives **[E]** (core)/**[C]** (chain)
`v65_falsification_layer.py`	Prediction layer with explicit failure thresholds: 3 core ($v_{\rm GW}{=}c$, no 4th gen, seam=8), 4 observational (β, r, n_s, θ_{12}), no-go + unitarity; $N_\star$ demoted to **[C]** input; none violated **[E]**
`v66_e8_casimir_degrees.py`	Organizing simplification: the compiler atoms *are* the E_8 Casimir degrees $\{2,8,12,14,18,20,24,30\}$ ($2{=}\lvert\mathbb{Z}_2\rvert$, $8{=}$rank, $14{=}\dim G_2$, $20{=}\det L$, $24{=}\lvert W(A_3)\rvert$, $30{=}h(E_8)$); $\sum{=}128{=}2^7$ (dS prefactor), $\sum\exp{=}120{=}\lvert R^+(E_8)\rvert$ **[E]**
`v67_area_law_coefficient.py`	Central-theorem *structural closure*: Fursaev–Solodukhin $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow S{=}2\pi c_3 A{=}\frac{1}{4}A$ ($2\pi c_3{=}\frac{1}{4}$); $c_3{=}\frac{1}{8\pi}$ is the *unique* value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k{=}c_3/2$ (then *forced*, v73) **[E]**
`v68_seeley_dewitt_residual.py`	Residual *resolved as an anchor*: $a_2{=}-\frac{d}{192\pi^2}R \Rightarrow$ the *absolute* $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the *scale* $1/G$ is the one dimensionful anchor (the dimensionless $k{=}c_3/2$ is forced, v73); cutoff-indep. R^2/scalaron derived **[E]**
`v69_d4_q_geometry.py`	D_4-equivariant Q-geometry (gate **[C]**/**[E]**): on $H_1(\mathbb{P}^1\setminus\mu_4){=}B_1{\oplus}E$, $Q_+{=}3w{+}1{=}\mathrm{diag}(1,2,3)$ (cusp weights on $\mu_4{=}\mathbb{Z}_4$ eigenspaces), $Q_-{=}E$-coupling $\sqrt{N_{\rm fam}} \Rightarrow t(t^2{-}3)$; Σ-split $=D_4{=}\mathbb{Z}_4{\rtimes}\mathbb{Z}_2$ **[E]**
`v70_q_integer_lift.py`	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1,i,-i\}$); $\det Q{=}3{=}N_{\rm fam}$, SNF diag$(1,1,3) \Rightarrow$ residual = one lattice invariant $\det Q{=}N_{\rm fam}$, not closable by D_4 alone **[E]**
`v71_simple_r_bridge.py`	*Simple* R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R{=}8{=}$rank E_8 lattice + Spec$(Q_+){=}\{1,2,3\}$ from v69) $\Rightarrow$ quark *ratios* are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual $U_{\rm point}{=}$ absolute norm. = an anchor **[E]**/**[O]**

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Script	What it machine-checks *(continued)*
`v72_q_det_from_cusp.py`	$\det Q{=}N_{\rm fam}$ derived from the cusp class: $\det Q{=}\lvert\operatorname{coker} Q\rvert{=}$ cusp-class order = denominator of the weights $\{0,\frac{1}{3},\frac{2}{3}\}$ (same data as Spec Q_+); coker $Q{=}\mathbb{Z}/N_{\rm fam}{=}$ triality deck group ⇒ integer-lift residual closed [E]
`v73_k_c3_half.py`	Central coefficient $k{=}c_3/2$ *forced*: (1/2) variational × $1/(\lvert\mathbb{Z}_2\rvert 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2){=}2$), both cutoff-indep ⇒ Fursaev–Solodukhin $S{=}A/4$; only the absolute 4D Newton scale stays the anchor (v68) [E]
`v74_compiler_micro_lemmas.py`	Spine quotient ladder $(\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{4},\frac{4}{3}$ = adjacent quotients of $(2,3,4,5)$); pencil differences $(2,16,48){=}(\lvert\mathbb{Z}_2\rvert,\dim S^+,\Omega_{\rm adm})$ sheet→1gen→3gens; anchor QF $a^\top Ka{-}\mathbf{1}^\top K\mathbf{1}{=}41{-}25{=}16{=}\dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L{=}R{+}61e_1^\top$, $a^\top R^{-1}\mathbf{1}{=}0$ (Sherman–Morrison rank-one invariance) [E]
`v75_upoint_to_vgeo.py`	**Gate 1 complete:** $U_{\rm point}{\to}v_{\rm geo}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = $\varphi_0^{\rm ret}$-powers (v46); (ratios, product)⇔(individuals) bijection ⇒ absolute amplitudes fixed up to *one* scale $v_{\rm geo}$ = the same anchor as $1/G$ (v68); two $[A]$ → one [E]/[O]
`v76_gmetric_reduction.py`	**Gate 2:** Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\rm eff}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{=}1.648{>}0$, $\lVert V\rVert{\le}248c_3^2{=}\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-*boundary* measure (conditional: RP+tightness ⇒ closes) [E]/[C]
`v77_e8_conformal_net.py`	**G6 route:** the seam boundary measure = the $(E_8)_1$ *lattice net* (rigorous, holomorphic $c{=}8$); level-1 $c(E_8){=}\frac{248}{31}{=}8$, $c(D_5){=}5$, $c(A_3){=}3$, embedding 5+3=8 (coset $c{=}0$); imports $G6$ into RCFT/conformal-net rigor [E]/[C]
`v78_vgeo_floor.py`	$v_{\rm geo}$ **floor:** no pure number is dimensionful ⇒ $v_{\rm geo}$ irreducible by logic; all scales are ratios × one $\bar{M}_{\rm Pl}$; $S_{dS}\rho_\Lambda{=}32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) [E]/[O]
`v79_review_identities.py`	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n{=}(3^n{-}(-2)^n)/5{=}1,1,7,13,55,133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}{=}1/$atom, $a^\top M^{-1}a{=}1$ for R,K,L); MacWilliams $(1,10,5)$; $6Y^2{-}Y{-}1{=}(2Y{-}1)(3Y{+}1)$. Rejects m_p/m_e, η_B near-formulas (firewall) [E]
`v80_operator_pencil_geometry.py`	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}{=}(N_{\rm fam}x{+}\lvert\mathbb{Z}_2\rvert)(N_{\rm fam}x{+}g_{\rm car}){=}(3x{+}2)(3x{+}5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3,10,16,20)$; type checker $\det B_{Q,K,R,L}{=}(9,10,16,40)$; audit: differential spectrum, $F_4{\times}G_2$ shadow $248{=}52{+}14{+}26{\cdot}7$ [E]/[C]
`v81_singular_pencil_matrices.py`	Double cover $y^2{=}\det B_{K+xQ}{=}(3x{+}2)(3x{+}5)$: Koide $x{=}{-}\frac{2}{3}$ and carrier $x{=}{-}\frac{5}{3}$ are the two branch points (deck $2{=}\lvert\mathbb{Z}_2\rvert$; disc$=81{=}N_{\rm fam}^4$; separation=1 transport period). Clearing matrices $C_{2/3}{=}3K{-}2Q$ ($D_5{\oplus}A_3$ glue, $B{=}(5,7)^\top(9,11)$, $\sum{=}240$), $C_{5/3}{=}3K{-}5Q$ (neutral, $\chi{=}(\lambda{+}2)(\lambda^2{+}\lambda{+}6)$); scalaron 7= mixed anchor disc$/N_{\rm fam}$ [E]
`v82_koide_attractor_splitting.py`	Koide attractor *forced*: the branch-divisor-preserving Möbius map fixing $q{=}2,5$ is unique, multiplier $(2/3)^6{=}\lambda_2$ of the established transfer operator ⇒ $F'(2){<}1$ attractor, $F'(5){>}1$ repeller; three postulates → one [C]. Splitting trichotomy disc 81/49/40: the clean split is non-generic (anchor-first) [E]/[C]
`v83_e8net_holomorphic_uniqueness.py`	Holomorphy pins $(E_8)_1$: #primaries = det Cartan (D_8:4 vs E_8:1), and a holomorphic $c{=}8$ chiral CFT is the lattice theory of the *unique* even unimodular rank-8 lattice (Minkowski–Siegel mass $1/\lvert W(E_8)\rvert$); carrier Pascal truncation $K{=}(g{-}1)/2$ = half-spinor split [E]
`v84_frozen_registry.py`	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula↔value): exactly *one* θ_{12} prediction of record; r, n_s as $N_\star$ *bands*; conditional entries carry their hypothesis by name [E]
`v85_master_cover.py`	Master-cover theorem: $\det B$ is GL(2)-covariant on span$\{K,Q\}$ ⇒ exactly *one* double cover up to Möbius (disc = $N_{\rm fam}^4\det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3}{=}{-}\mathrm{rank}E_8/N_{\rm fam}$ = carrier − one period); μ_4 is *not* a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron *scale* [E]

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Script	What it machine-checks *(continued)*
`v86_nstar_reheating.py`	$N_\star$ band → point [C]: $\Gamma{=}4M^3/48\pi\bar{M}^2{=}128\,\mathrm{GeV}$, $T_{\mathrm{reh}}{=}9.6{\times}10^9\,\mathrm{GeV}$, $N_\star(0.05/\mathrm{Mpc}){=}51.4 \Rightarrow n_s{=}0.9611$, $r{=}0.0045$ (inside the frozen band); honest tensions: n_s $+0.9\sigma$, A_s dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]
`v87_bulk_uniqueness_reduction.py`	Red-team Target A residual (ii) conditionally closed: holomorphic ⇒ Rep = Vect ⇒ 2D bulk *unique* (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits *six* modular invariants (incl. both E_8 pairings) ⇒ Target A = one residual [E]/[C]/[O]
`v88_cp_phase_audit.py`	Target-D CP residual quantified: frozen $\delta{=}\pi/3{+}3\lambda^2$ survives at $+0.98\sigma$ vs γ_{PDG}; the $0.07°$-near alternative $\pi/3{+}2\lambda^2$ recorded as look-elsewhere trap, *not* adopted; decision at $\sigma_\gamma{\le}0.96°$ [E]
`v89_carrier_index_lemma.py`	Carrier Index Lemma: KLM $\mu_A{=}[B{:}A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1{:}(D_5)_1{\times}(A_3)_1]{=}4{=}\lvert\mu_4\rvert$ (glue order *is* the index); μ-additivity $16/4^2{=}1 \Rightarrow$ holomorphy *follows*; Gate A (H)⇔(H′) index-4 extension [E]
`v90_conical_defect_chain.py`	Fursaev–Solodukhin factor 4π *derived* (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow S{=}A/4 \Leftrightarrow c_3{=}\frac{1}{8\pi}$ (sympy-solved); residual isolated to "seam determinant ⇒ EH form" [E]/[O]
`v91_spine_tetrahedron.py`	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120{=}\lvert R^+(E_8)\rvert$; $240{=}\lvert\mu_4\rvert\lvert E(K_4)\rvert\lvert E(K_5)\rvert$ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]
`v92_glue_uniqueness.py`	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two *chiralities*, swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier$\to SO(16)_1 \to (E_8)_1$, nothing else [E]
`v93_koide_relaxation_toy.py`	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho{=}{-}\varphi_0^{\mathrm{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t{=}2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a *continuous* generator [E]/[C]
`v94_sheet_diamond.py`	Sheet diamond: all four flavor operators on *one* surface $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$ (pencil = the cut $(1{+}x,x{-}1)$; diagonal-cut disc 153 negative control); winding line $\det B{=}2\det$ for all s; cofactor seam normal $(5,-9,6)$; Plücker canonicalisation: 11 *and* 26 both live in K; spine-quotient firewall *rejected* [E]
`v95_centered_diamond.py`	Centered cross: $Q{=}U{+}V$, $R/L{=}C{\mp}U$, $K/F{=}C{\mp}V$ (five objects → three); $\mathrm{Spec}\,V{=}\{0,1,2\}{=}N_{\mathrm{fam}}{\cdot}\mathrm{cusp}$; $\det C{=}14$, $\sum C{=}31{=}2^{g_{\mathrm{car}}}{-}1$; $\mathrm{Pl}_R(C){=}7(2,3,1) \parallel \mathrm{Pl}_R(L){=}10(2,3,1)$; G_2 reading audit-typed [E]
`v96_branch_kernel_selection.py`	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have *integer* kernels (carrier kernel = $\mathbf{1}$); collapse lemma ⇒ $(-1,1,0)$: up = −down, *lepton pairing* = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E]
`v97_sheet_conjugation_bridge.py`	Sheet-conjugation bridge (P1 → one [C]): the anchor forces the cusp conjugation uniquely ⇒ integer T_A, $\det{=}{-}1$; $a{=}e_2{+}e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\,\mathrm{span}\{\mathbf{1},a\}$; $\langle T_A,\Sigma\rangle{=}D_4$; sheet-index lemma (intertwiner dets even, min $2{=}\lvert\mathbb{Z}_2\rvert$); remaining [C]: Q_+ grading = A_3 discriminant grading [E]/[C]
`v98_discriminant_dictionary.py`	Dictionary *derived* (P1 closed modulo GATE.QGEO): $G{=}T_A\Sigma$ acts integrally on the cusp basis ($Ge_1{=}{-}e_1$, $Ge_3{=}e_2$, $Ge_2{=}{-}e_3$ = $B_1{\oplus}E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$-class-2 line; $T_AGT_A^{-1}{=}G^{-1}$ ($k{\mapsto}{-}k$); $q_{A_3}(1){=}q_{A_3}(3){=}\frac{3}{8}$, $q_{A_3}(2){=}\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities ⇒ P1 carries no separate [C]; residual = the one existing Q-geometry gate [E]

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Script	**What it machine-checks** *(continued)*
`v99_koide_flow_time.py`	Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\rm fam})(q-2)(q-5)=(\Delta/N_{\rm fam})\det B(q)$, time-1 map $=F$ [E]; non-integrality of t=2.84 is *data-fragile* (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [E]; conditional steps: n=2 excluded (-2.9σ), n=3=$N_{\rm fam}$ $\Rightarrow$ m_τ=1776.9427 MeV ($+0.14\sigma$), decision at $\sigma(m_\tau)\sim$0.01 MeV [C]; ln(46080) scale reading destroyed by controls (firewall)
`v100_numerology_null_mc.py`	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2{\times}2{\cdot}10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly *one* hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, *conditional on the declared grammar* [E]
`v101_horizon_anchor.py`	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1,1,-2)$ = traceless anchor; $S_{\rm Nariai}/S_{dS}=\frac{2}{3}=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ (per horizon $\frac{1}{3}$); interpolation $(x^2+1)/\Phi_3(x)$; three-sheet total $\lvert\mathbb{Z}_2\rvert S_{dS}$ conserved; $Q_{\rm geom}\in[\frac{3}{8},\frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$, deck = horizon swap; τ_{BH}=128 $g_{\rm car}M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading)
`v102_seam_orientation.py`	One orientation: the anchor is the *stationary repeller* in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its *unique* stationary point, curvature $\frac{2}{9}=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}^2$; evaporation = entropy ascent away from the anchor [E]; "one variational principle of the seam" stays a reading [C]
`v103_trisection_normal_form.py`	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow$ $\cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos\psi/N_{\rm fam}$ and $S_{\rm tot}/S_{dS}=\frac{4}{3}-\frac{2}{3}\cos\frac{2\psi}{3}$ — *one* cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=(\lvert\mathbb{Z}_2\rvert/N_{\rm fam})^{N_{\rm fam}}$ (Koide constant to the family power); invariant slope $d\sigma/dm\rvert_N=-\frac{8}{9}=-\operatorname{rank}E_8/N_{\rm fam}^2$; bridge: flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, missing = one *clock* [E]/[C]
`v104_nariai_clock.py`	The *classical* Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the dS_2 static-patch equation with $m^2=-2\Lambda=-\lvert\mathbb{Z}_2\rvert\Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\rm clock}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1,-2\}$; Nariai cubic = $(t-1)\cdot\chi_{\rm clock}$; entropy-deviation rate $2H=\lvert\mathbb{Z}_2\rvert H$; honest (2/3)-test *negative* for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]
`v105_residual_inventory.py`	The residual inventory: *one constant* ($2/3=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ exact in seven places across both sectors), *one anchor* (three dynamical appearances: configuration, Nariai roots, clock spectrum), *relocation theorem* (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10$ $\Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the *complete machine-pinned gap list*: exactly five structural objects + one scale + π; a sixth gap would fail the script [E]/[C]
`v106_review_validation.py`	External-review validation: seed normal form $\varphi_0^{\rm ret}=\frac{\lvert\mu_4\rvert}{N_{\rm fam}}c_3+\Omega_{\rm adm}c_3^{\lvert\mu_4\rvert}$ exact [E]; hypercharge moment $\operatorname{Tr}X^2$=120=5! computed from the charge table; factorial spine $5!=\lvert R^+(E_8)\rvert=\sum$exponents, $240=2{\cdot}120$, $1920=2^4{\cdot}5!$; degree-2 inventory (Pascal K=2 unique at g=5, glue norms sum 2, $c_3=\frac{1}{2}\times\frac{1}{4\pi}$, $\binom{5}{2}$=10); named [C] hypothesis *Quadratic Boundary Locality* ($\Rightarrow$ K=2 forced; would upgrade the Pascal selection to [E]); audit $240=16{\times}15$ non-unique [E]/[C]

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Script	**What it machine-checks *(continued)***
`v107_quantum_clock_target.py`	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0,-1,-2\} = -\operatorname{Spec}(V) = -N_{\mathrm{fam}}\times$cusp (the transfer operator's own grading) [E]; exact target identity $(1/3)^6 = \big((2/3)^6\big)^{\log_{3/2}3}$ — required bend $\log_{9/4}3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi)\cdot\Lambda/\bar{M}_{\mathrm{Pl}}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]
`v108_pascal_ladder.py`	Pascal Ladder Theorem: $2^{g-1}{=}\sum_{k\le K}\binom{g}{k} \Leftrightarrow g{=}2K{+}1$ (odd-g midpoint identity + even-g straddle) $\Rightarrow$ carrier selection $g{=}5$ *exactly equivalent* to degree bound $K{=}2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K{=}1$ one-family, $K{=}3$ nine-family, $K{=}4$ inconsistent; only $K{=}2$ also satisfies $g{+}N_{\mathrm{fam}}{=}8$ (v14 overdetermination) [E]/[C]
`v109_sheet_pairing.py`	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm\frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+\otimes S^+ = 0$; odd g flips parity); $S^+\otimes S^-{=}\Lambda^0{\oplus}\Lambda^2{\oplus}\Lambda^4$ exactly ($256{=}1{+}45{+}210$, zero-modes $(1,5,10)$ = the code); sheet-diagonal = odd forms ($2\Lambda^1{+}2\Lambda^3{+}\Lambda^5$); tower tops at Λ^{2K}, $K{=}2 \Rightarrow$ the seam scalar kernel is necessarily a *sheet pairing* [E]/[C]
`v110_calderon_sheet.py`	Calderón-sheet selection theorem: a Calderón involution ε on $S^+{\oplus}S^-$ certifies a scalar two-point datum *iff* it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2{=}\lvert\mathbb{Z}_2\rvert$ invariant kernels = the two Lagrangian glues; ladder genericity ($g{=}3,5,7$): sheet-oddness forces $K{=}(g{-}1)/2$, *not* $g{=}5$ (anti-overclaim); QBL residue split into *interface* ("ε sheet-odd", natural from the one-sided collar) + *analytic core* (slot-degrees ≤ 2) [E]/[C]
`v111_quadratic_transport.py`	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics ($=\mathfrak{so}(10)$, the Λ^2 term of $1{+}45{+}210$) sheet-even, generate the code from the vacuum *and* span the full $\operatorname{End}(S^+){=}256$ at word length $2 \Rightarrow$ *transport degree exactly 2* (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g{=}3$ control: degree selected, not rank [E]/[C]
`v112_selfcounting_channel.py`	Self-Counting Channel: $w \mapsto (w,-w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels $= 2^{g-1}$ *exactly*; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1}{=}\sum_{m\le K}\binom{g}{m}$ is *two countings of one set* (an identity, not a requirement); Hodge fold $\binom{g}{2j}{=}\binom{g}{\min(2j,g-2j)}$; QBL residue = one input ("a single scalar 2-point kernel"), selection carried by rank-8+integrality [E]/[C]
`v113_quasifree_kernel.py`	One kernel is the whole net: carrier $(D_5)_1{\times}(A_3)_1 = SO(10)_1{\times}SO(6)_1 = 16$ free Majoranas, $c{=}5{+}3{=}8$ at every tower level (only the glue grows, index $2{\times}2{=}4{=}\lvert\mu_4\rvert$); Wick/Pfaffian exact (all 210+210 higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P{=}5{=}g_{\mathrm{car}}$, seam level $8{=}\operatorname{rank}E_8$ (c = rank of the kernel); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]
`v114_torsion_delta.py`	Torsion normal form + $\delta{=}\frac{1}{2}$ theorem: flatness of the (U_{wall}) $\mathbb{Z}_4$-family $\Leftrightarrow$ $(MU)^4{=}\mathbf{1}$ (μ_4 = order of the twisted cusp generator); on the involutive branch ($T^2{=}\mathbf{1}$) the cusp trace forces $\operatorname{diag}M{=}(0,\frac{i}{2},-\frac{i}{2})$ with spec $\{1,\omega,\omega^2\}$ automatic $\Rightarrow$ $\delta{=}\frac{1}{2}$ exact and constant (v20's hand value = torsion identity); census: $\{1,i,-i\}$ branch varies, v33 point in its $d_1{=}0$ slice; residue = bundle-side branch selection [E]/[C]
`v115_anchor_residue.py`	The anchor pins the residue: μ_4-average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow$ diag $A_0{=}a/\lvert\mu_4\rvert$; off-weights uniquely $(8,0,5)/144$ $(=(\operatorname{rank}E_8,0,g_{\mathrm{car}})/(\lvert\mu_4\rvert N_{\mathrm{fam}})^2$, total $13{=}\Delta_Q)$; *exact* residue $A_0^\star$ with $\chi{=}\lambda(\lambda{-}\frac{1}{3})(\lambda{-}\frac{2}{3})$ is the RH solution ($\lVert M_\infty{-}\mathbf{1}\rVert{\sim}10^{-10}$); anchor forces $d_1{=}0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]

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Script	What it machine-checks *(continued)*
`v116_resonance_uniqueness.py`	Resonance theorem (v115 [N]→[E], linear — no Gröbner): twisted μ_4 averages collapse to $B_1{=}4a_{12}E_{12}{+}4\overline{a_{13}}E_{31}$; anchor exponents $\mathrm{diag}(2,1,1)$ give resonance gap 1 with obstruction $4\overline{a_{13}} \Rightarrow M_\infty{=}\mathbf{1} \Leftrightarrow a_{13}{=}0$; + $(8,0,5)/144$ + Diagonaleichung ⇒ flat anchor locus = *one* gauge orbit = exact $A_0^\star$ — U_{wall} bundle datum algebraically unique; sharpness control $\lVert M_\infty{-}\mathbf{1}\rVert \sim 8.5\lvert a_{13}\rvert$ [E]
`v117_monodromy_weyl_a3.py`	The monodromy is $W(A_3)$: the unitarised holonomy of $A_0^\star$ is *exact* (entries in $\frac{1}{2}\mathbb{Z}[i]$), diag $\widetilde{M}_0{=}(0,-\frac{i}{2},\frac{i}{2}) \Rightarrow d_1{=}0$, $\delta{=}\frac{1}{2}$ are matrix entries; $\langle U,\widetilde{M}_0\rangle$ enumerated exactly: order 24, statistics $(1,9,8,6)$, characters $(3,-1,0,1) = S_4{=}W(A_3)$ (standard⊗sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match $3.5{\times}10^{-10}$ [E]
`v118_hexagon_family_dictionary.py`	First exact H2 piece: $\mathrm{spec}(\widetilde{M}_0)\cup\mathrm{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6{=}\mathbb{Z}_2{\times}\mathbb{Z}_3$); $\det(\mathbf{1}{-}t\widetilde{M}_0){=}1{-}t^3 \Rightarrow \det_\mp{=}(\frac{7}{8},\frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e{=}\lvert\mu_4\rvert/(2\det_-)$, $c_\mu{=}N_{\mathrm{fam}}/(2\det_+)$, $c_\tau{=}\lvert\mu_4\rvert\det_-/N_{\mathrm{fam}}$, product $= \lvert\mu_4\rvert/\det_+ = \frac{32}{9}$); $\langle U,\widetilde{M}_0,-\mathbf{1}\rangle$ has order $48{=}\Omega_{\mathrm{adm}}$; the address table stays open [E]/[C]
`v119_review_validation_2.py`	Second review validation: anchor ratio triad $(e_3,e_1,e_2)/p_0{=}(\frac{2}{3},\frac{4}{3},\frac{5}{3})$ = (Koide branch, seed gain, carrier branch; flow critical points $(2,5){=}(e_3,e_2)$); the 121 audit lemma $(\mathbf{1}{+}a)^T R(\mathbf{1}{+}a){=}121{=}11^2{=}\lVert\,\mathrm{Pl}(K)\rVert_1^2$ (orderings $\{105,121,135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\mathrm{adm}}{=}2p_1p_2$, $10b_1{=}1{+}p_1p_3$, $\Delta_Y{=}e_2^2{=}p_0^2{+}2\,\mathrm{rank}$; flavor surface = v94/v95 (presentational) [E]
`v120_address_table.py`	Address table (frozen v18/v20 integers): lepton words $(8,5,3){=}(\mathrm{rank}\,E_8, g_{\mathrm{car}}, N_{\mathrm{fam}})$; addresses = divmod by the hexagon ($p_2{=}6$): $e{\to}(2,1)$, $\mu{\to}(5,0)$, $\tau{\to}(3,0)$, $\sum r{=}p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3, down = $p_1{+}p_3$, all quarks = $24{=}\lvert W(A_3)\rvert$, all nine = $40{=}p_1p_3{=}a^T Ra$; top at the vacuum site $(0,0)$; the assignment derivation stays open [E]/[C]
`v121_address_pinning.py`	Address pinning: $L{=}R{+}2U$ (v95) ⇒ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a; margins = atoms (rows (e_1,rank,p_3), columns $(e_1,\Delta_Q,g_{\mathrm{car}})$); *census theorem*: among all hexagon matrices with atom margins (17 of them) exactly *one* has $\det{=}8$ — and it is R (also unique with SNF $(1,1,8)$; all 17 determinants distinct) ⇒ the address table carries *zero* residual information; the H2 residue = the six atom margins [E]/[C]
`v122_margin_theorem.py`	Margin theorem: the *established* selectors — the D_4 annihilator $n{=}(5,-9,6)$ with $n^T R{=}(8,0,0)$ (v94), $\det R{=}8$, $\det K{=}4$ (the v71 quartet) — pin R *uniquely* among hexagon matrices ($64 \to 12 \to 1$) ⇒ the atom margins, the anchor column and all sum rules are *consequences* (their input status is revoked); bonus: $\det L{=}20$ holds automatically on the locus (the quartet is redundant); H2 introduces *no* new residual class [E]/[C]
`v123_inventory_update.py`	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries *three* loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to *five* classes — R1 clock, R2 index-4 net, R3 EH, R4′ selector establishment (replacing the *discharged* H2 dictionary), R5 Q realisation — + v_{geo}, π; a sixth class fails the script [E]
`v124_resummed_clock.py`	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1{-}n/N_{\mathrm{fam}})^{p_2} \Rightarrow \mathrm{rate}(n){=}{-}p_2\ln(1{-}n/N_{\mathrm{fam}})$ reproduces $\{0,\Delta,6\ln 3\}$, the bend $\log_{3/2}3$ in one line; the linearisation carries slope $\lvert\mathbb{Z}_2\rvert{=}2$ relative to the classical GP clock $\{0,-1,-2\}$; the pole at $n{=}N_{\mathrm{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]
`v125_glue_qsystem.py`	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*{=}\lvert\mathbb{Z}_4\rvert\mathbf{1}$) ⇒ Jones index $4{=}\lvert\mu_4\rvert$ (the v89 value); locality = isotropy ($q(k(1,1)){=}k^2 \in \mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, $4{=}2{\times}2$; R2 sharpens from "find" to "identify" [E]/[C]

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Script	**What it machine-checks** *(continued)*
`v126_clock_wall_bridge.py`	Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0,\frac{1}{3},\frac{2}{3}\}$ = exactly spec $A_0^\star$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$; checkpoint 1 passed *classically* ($p_2/N_{\text{fam}}=2=\lvert\mathbb{Z}_2\rvert$ = the v104 entropy rate $2H$); $\det(\mathbf{1}-A_0^\star)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]
`v127_ring_resummation.py`	The clock is a log determinant: rate $=-p_2\operatorname{Tr}\ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, *one per hexagon site*, coupling = the parabolic weight; the k=1 ring = the classical term, the k=2 ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha\to1$; v107's "O(1) resummation" is thereby *typed* (ring/log-det), its derivation open [E]/[C]
`v128_graded_hull.py`	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5\oplus A_3$+glue coordinates, coset decomposition 240=52+64+60+64; the grading is exactly additive on *all* 6720 root-pair sums $\Rightarrow$ E_8 is a $\mathbb{Z}_4$-*graded* Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1)\vert_{l=1}$ = the sheet-doubled GP zero modes [E]/[C]
`v129_entropy_power_law.py`	The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS}\in\{1,\frac{2}{3},\frac{1}{3}\}$ (v101); rate $=p_2\ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\text{fam}}$= MS weight = entropy-deficit share $\Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma\propto(S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]
`v130_born_square.py`	Born square: amplitude $\sim(S/S_{dS})^{n_{\text{zero}}/2}=(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $=\lvert A\rvert^2=(S/S_{dS})^6$ $\Rightarrow$ $h=n_{\text{zero}}/2=3=N_{\text{fam}}$, $p_2=2h$ (the Born rule) — the v129 weight audit is *derived*; the 3 = the dimension of the l=1 space = the number of proper conformal Killing vectors of S^2; the $\times2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E]/[C]
`v131_measure_is_area.py`	The measure is the area: $\lVert Y_{1m}\rVert^2=r^2=A/(4\pi)$ *exactly* (all three l=1 modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2}$ $\Rightarrow$ amplitude S^3, Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3, cancelling in all ratios; the R1 residue = *one* det-ratio cancellation $\det'_n/\det'_{dS}=1+O(\text{deficit})$ [E]/[C] *(cancellation since derived within the SdS family, v144)*
`v132_detratio_anomaly.py`	Det-ratio anomaly = the Koide constant: $\zeta(0)\vert_{\det'}$ of the GP sphere operator $-\Delta-2$ (the 3 zero modes removed) $=a_1-3=\frac{7}{3}-3=-\frac{2}{3}=-\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ exactly ($a_1=\frac{1}{N_{\text{fam}}}+\lvert\mathbb{Z}_2\rvert$; numerical continuation $\sim10^{-7}$) — the eighth 2/3 appearance, the first as a *spectral* anomaly; consequence $\det'(cL)=c^{\zeta(0)}\det'(L)$; the R1 residue = *one* $\zeta(0)$ budget statement (the 4d sum) [E]/[C]
`v133_zeta_budget.py`	The ζ budget, both routes exact (Euclidean Nariai $=S^2\times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}=-e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2=\frac{161}{45}$, $\zeta(0)=-\frac{109}{45}$ (*no* atom; continuation $\sim10^{-8}$); zero modes 3+3 = the horizon pair $\Rightarrow$ the budget structurally favours the *reduced* seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E]/[C]
`v134_dual_anchor.py`	Dual anchor lemma: $d{:=}a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2},-\frac{1}{2},1)$, $d\cdot\mathbf{1}=0$, $d\cdot a=1$, $(1,1,-2)=-2d$ — the inverse flavor response *is* the Nariai root (v101); the invariance is exact via Sherman–Morrison ($\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1}=\frac{1}{4}(1,1,-1)$; controls: $\mathbf{1},e_1,n$ do *not* lie on the plane); (d,n) = the dual normal pair of the flavor boundary; a third purely *algebraic* leg of the flavor$\leftrightarrow$horizon bridge [E]/[C]
`v135_det_surface.py`	Determinant surface: $\det M(s,t)=3s(t+1)(t+2)+t^2+5t+8$ — two s-independent iso-volume walls $t=-2\Rightarrow2=\lvert\mathbb{Z}_2\rvert$, $t=-1\Rightarrow4=\lvert\mu_4\rvert$ ($\det K=4$ is a *layer*); winding line $6s+8\Rightarrow(8,14,20)$, $\det F=32=2^{g_{\text{car}}}$; $\det B(s,0)=2\det M(s,0)$ exactly, B-wall only at $t=-2$; row-sum axes $C=(7,11,13)$, $U=(3,3,3)$, $V=(1,2,3)=\operatorname{Spec}(Q_+)$; micro-identities $e_1^2+e_2^2=41=10b_1$, $p_1^2+p_2^2=52$= the carrier coset root count (v128) [E]/[C]

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Script	What it machine-checks *(continued)*
`v136_dual_normal_selector.py`	Dual-normal selector: (d,n) pins R *columnwise* — $d{\cdot}c_j{=}a_j$, $n{\cdot}c_j{=}8\delta_{1j}$, kernel = primitive $(6,8,7)$, census: each column unique (1 of 216) ⇒ R4′ shrinks from 6^9 to three column problems; the $A_0^\star$ zero mode carries $\sigma{=}(2,-9,5){=}(\lvert\mathbb{Z}_2\rvert,-N_{\mathrm{fam}}^2,g_{\mathrm{car}})$ ($\lVert v\rVert^2{=}\frac{16}{9}$, $16{=}\dim S^+$; adj $=\frac{2}{9}P_v$ = the SdS curvature); $n{\cdot}\sigma{=}121{=}11^2$; $W(A_3)$ orbit control: the naive lift $\sigma{\to}n$ fails — a genuine mechanism, not a group element [E]/[C]
`v137_qplus_cohomology.py`	The Q_+ grading is cohomology: $H^1(\mathbb{P}\setminus\mu_4){=}\chi_1{\oplus}\chi_2{\oplus}\chi_3$ ($\omega_k{=}z^{k-1}dz/(z^4{-}1)$, character i^k, residues $\zeta^k/4$); degrees $(1,2,3){=}\mathrm{Spec}(Q_+){=}$ the A_3 exponents ($h{=}4{=}\lvert\mu_4\rvert$); cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{=}\mathrm{spec}(A_0^\star)$ ⇒ MS weights = cusp classes = characters = Q_+: *one* spectrum, four readouts; the QGEO residue = the naturality of the canonical map [E]/[C]
`v138_vw_firewall.py`	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\mathrm{PI}}{=}-\frac{98}{45}{=}\frac{22}{45}{-}\frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); **edge match**: $\alpha_{\mathrm{edge}}{=}-\frac{8}{3}{=}2{\times}(-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom ⇒ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [E]/[C]
`v139_selector_triangle.py`	Selector triangle: $d{=}\frac{3}{2}a{-}2\cdot\mathbf{1}$ exactly (pure anchor data, no R) ⇒ the first selector is *derived*; the frame $(\mathbf{1},a,\sigma)$ has $\det{=}11{=}\lVert\,\mathrm{Pl}(K)\rVert_1$, and n is the *unique* covector with atom pairings $(2,8,121){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8,11^2)$; on the constrained line the pairing is $11(11{+}t)$, a square only at the true n; review lift exact (audit): $n{=}\mathrm{reverse}(\sigma){+}\lvert\mu_4\rvert e_3$; R4′ residue = three pairings [E]/[C] *(since v142: two — the third pairing is integrality-forced)*
`v140_canonical_map.py`	The canonical map exists and is rigid: the plain deck U (characters $(0,1,3)$) cannot match H^1; the *sheet-twisted* $-U$ ($(2,3,1)$) and the cusp exponential i^{Q_+} ($(1,2,3)$) both carry the H^1 character set $\{1,2,3\}$ (the v118 sign twist); multiplicity-freeness ⇒ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q_+} pulls the Euler degree back to Q_+, $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to *one* $\mathbb{Z}_3$ choice [E]/[C] *(the choice since derived, v141)*
`v141_deck_selection.py`	Deck Selection Theorem (R5): the established integer deck $G{=}T_A\Sigma$ (v97/v98) pairs the $Q_+{=}1$ line with the self-conjugate character 2 ⇒ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2,3,1)$ = chars$(-U)$ survives; i^{Q_+} excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ [E]/[C]
`v142_frame_integrality.py`	Frame Integrality (R4′): pairing triples of *integer* covectors against $(\mathbf{1},a,\sigma)$ form an index-11 sublattice ($x{-}3y{+}z\equiv 0$ mod 11, $11{=}\lVert\mathrm{Pl}(K)\rVert_1$); with $(x,y){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8)$ the third pairing is forced $\equiv 0$ mod 11, primitivity forces even t; Cramer reductions for the two line pairings; R4′: three pairings → two [E]/[O] *(since v145: the values are atom identities — residue = one lift map)*
`v143_graded_frobenius.py`	Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k{=}C_{-k}$ exact on all 240 roots ⇒ Killing pairing nondegenerate per $g_k\times g_{-k}$; glue average = carrier projector (index $\lvert\mathbb{Z}_4\rvert{=}4$); each sector one $W(D_5){\times}W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement [E]/[C]
`v144_detratio_family.py`	Det-ratio cancellation within the SdS family (R1, the v131 step): e_2-rigidity gives $r_br_c{=}1{-}\Delta^2/3$ exactly ⇒ det′-ratio = $(1{-}\Delta^2/3)^{4/3}$ (via $\zeta(0){=}{-}\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27}{=}\lvert\mu_4\rvert(\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6\to\frac{14}{3}$ obstruction stated sharply [E]/[C] *(the bend since shown det′-clean, v147)*
`v145_pairing_atoms.py`	R4′ values are *atom identities*: with the lift $n{=}w_0(\sigma){+}\lvert\mu_4\rvert e_3$ ($w_0{=}$ the A_3 exponent duality), $n\cdot\mathbf{1}{=}\lvert\mu_4\rvert{-}\lvert\mathbb{Z}_2\rvert{=}\lvert\mathbb{Z}_2\rvert$, $n\cdot a{=}\lvert\mu_4\rvert a_3{=}8$ via the new closure $\lvert\mu_4\rvert{+}g_{\mathrm{car}}{=}N_{\mathrm{fam}}^2$ ($4{+}5{=}9$, $\sigma\perp w_0(a)$), $n\cdot\sigma{=}\lVert\sigma\rVert^2{-}N_{\mathrm{fam}}^2{+}\lvert\mu_4\rvert g_{\mathrm{car}}{=}121$; residue = derive the *one* lift map [E]/[C]

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Script	What it machine-checks *(continued)*
`v146_moebius_d4.py`	Möbius D_4 realisation (R5 at its floor): the full dihedral $\langle z\mapsto iz,\ z\mapsto 1/z\rangle$ on $H^1(\mathbb{P}^1\backslash\mu_4)$ matches the integer model $\langle G, T_A, \Sigma\rangle$ exactly — ι^* swaps $\chi_1\leftrightarrow\chi_3$, fixes χ_2 with $+1$, det $=-1$ ($=T_A$ in the cusp basis); $\delta\iota=\Sigma$ class; parities match (v98) — the `GATE.QGEO` premise reduces to the module identification **[E]**/**[C]**
`v147_clock_gaussian.py`	The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born^2 Γ-ratio $(1-\alpha)^{p_2}$, rate $=-p_2\ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2}3$ lives between the two Nariai levels of *one* geometry $\Rightarrow$ *det′-clean*; the $6\to\frac{14}{3}$ obstruction is localised to the α=0 reference **[E]**/**[C]**
`v148_fock_census.py`	NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the *even* classes $\{0,2\}\times\{0,2\}$ — the odd glue sectors $k(1,1)$ are *Ramond* modules; the R zero-mode module (2^8=256) splits 128+128 into the odd sectors of the *two* Lagrangian glues (v92): choosing the glue *is* choosing the R-projection (128= one $SO(16)$ chirality), $248=120_{\mathrm{NS}}+128_{\mathrm{R}}$; R2 is irreducibly a twisted-sector net statement **[E]**/**[C]**
`v149_cusp_normal.py`	Both dual normals are *anchor data* (R4′): in the cusp frame n pairs to $(6,3,5)=(p_2,p_0,e_2)(a)$ — one anchor invariant per Q_+ line ($d=\frac{3}{2}a-2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2,8,121)$ frame data and the w_0-lift (v145); audit: $\sigma\to(5,1,2)$, sum $14=\dim G_2$; residue = *one* assignment **[E]**/**[C]**
`v150_replica_eh_model.py`	The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and t-independent (image sum $(N^2-1)/(12N)=C(2\pi/N)$); the gap makes the ζ-variation *finite and cutoff-independent*, $\Delta\log\det'=2C(\gamma)\ln m$; linearisation $=(\ln m/12\pi)\int\sqrt{g}R$ — the EH form; target equation k=$c_3/2\Leftrightarrow\ln m=\frac{3}{4}=q(A_3)$ (audit); Calderón transfer + normalisation stay open — `SEAM.THEOREM.01` narrows, does *not* close **[E]**/**[O]** *(Calderón transfer since answered — the DtN kernel is conically clean, v151)*
`v151_bfk_split.py`	BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition *independent* (C_N=C_D, derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant *vanishes identically*, C_{DtN}=$C_{\mathrm{cone}}(\gamma)-2C_D(\gamma/2)$=0; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the *local* half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise **[E]**/**[O]**
`v152_norm_is_anchor.py`	R3's normalisation *is* the dimensionful anchor: the EH coefficient is $k=\ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio m/μ is physical; k=$c_3/2\Leftrightarrow m/\mu$=$e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), *not* a separate gap; the irreducibles stay {one scale, π}; audit: the log-ratio $=\frac{3}{4}=q(A_3)$ is the target value, NOT a derivation **[E]**/**[O]**
`v153_no_unit_theorem.py`	No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L\mapsto\lambda L$, a mass / $1/G$ is not, so a dimensionless compiler *cannot* select an absolute scale; the three residuals collapse — $U_{\mathrm{point}}\sim v_{\mathrm{geo}}$, $1/G\sim v_{\mathrm{geo}}^2$, $m/\mu=e^{3/4}$ (a ratio); v_{geo} is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, π} **[E]**/**[O]**
`v154_simple_current_theorem.py`	Simple-Current Extension Theorem (the central G_{net} theorem, assembled): A=$(D_5)_1\otimes(A_3)_1$ extended by the isotropic glue L=$\langle(1,1)\rangle$ has index $\lvert L\rvert$=4, c=5+3=8, $\mu(B)$=$\mu(A)/\lvert L\rvert^2$=16/16=1 $\Rightarrow$ holomorphic $\Rightarrow B$=$(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification **[E]**/**[C]**
`v155_quasifree_boundary.py`	Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$ (a contraction, hence a valid covariance), so the G_{net} analytic premise (v110(b)/v113) *follows* from 'the seam bulk is a reflection-positive free field' — the *same* premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise **[E]**/**[O]**

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Script	What it machine-checks *(continued)*
`v156_seam_net_construction.py`	Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k{=}\lvert k\rvert$ (free chiral dispersion); 16 free Majoranas give $c{=}8{=}5{+}3$; the E_8 currents are $248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1{+}248q{+}4124q^2{+}\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a *checked* identity, not an assertion; residual = the free-bulk premise (v155) + net existence **[E]**/**[C]**
`v157_rigid_fixed_point.py`	Freeness as a *rigid fixed point* (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $\lvert k\rvert$ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c{=}8$ theory has *no* marginal $(1,1)$ field (all $\bar{h}{=}0$) ⇒ rigid, isolated, no room for an interaction; (iii) the same $\lvert\mathbb{Z}_2\rvert$ that halves 8π in c_3 IS the Ramond projection giving $\mu{=}1$; (iv) holomorphic $c{=}8$ unique = $(E_8)_1$ = free. Premise (A) shrinks from 'prove Gaussianity' to 'the gapped flow reaches the holomorphic fixed point' **[E]**/**[C]**
`v158_fixed_point_stable.py`	The free chiral $c{=}8$ fixed point is *stable* (exact dimension count, the finite core of (A)): Majorana $h{=}\frac{1}{2}$, bilinear current $h{=}1$, quartic $h{=}2$; the relevant window $(0,1)$ for bosonic operators is *empty*, the $h{=}1$ currents are chiral (not $(1,1)$-marginal, v157), the lowest interaction (quartic) is $h{=}2{>}1$ irrelevant ⇒ isolated stable fixed point; (A) reduces to a basin-of-attraction statement **[E]**/**[C]**
`v159_pyrate_gauge_crosscheck.py`	PyR@TE cross-check of the F_{transfer} gauge inputs: the carrier/SM content gives $(b_1,b_2,b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1{=}\frac{41}{10}$ holds *three* independent ways — carrier algebra $10b_1{=}g_{\text{car}}2^{g_{\text{car}}-2}{+}1{=}41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and the external RGE generator PyR@TE 3 (`arXiv:2007.12700`) which writes $\beta_{g_1}{=}(\frac{41}{10})g_1^3$ verbatim; the $M{=}41$ split $10b_1{=}40_{(\text{ferm})}{+}1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in `verification/pyrate`. The transfer functor's gauge inputs are not free knobs **[E]**/**[C]**
`v160_seam_gaussianity_from_pf.py`	Seam Gaussianity as a *conditional* fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free ⇒ all higher connected cumulants $\kappa_{2n}{=}0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality ⇒ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a SECOND fixed ray, breaking uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0 — the value $(2/3)^6$ only sets the rate. **The load [O]**: v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}{=}1820$, $\binom{16}{6}{=}8008$ cumulant directions ⇒ (A) reduces to the single internal statement 'the admissible TFPT transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum' **[E]**/**[O]**
`v161_one_particle_reduction.py`	The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T{=}\Gamma(t)$, the second quantisation of a one-particle contraction t; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\operatorname{spec}(\Gamma(t)\vert_{\deg m})$ = products of m one-particle eigenvalues **[E]**; (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly **[E]**; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization ⇔ one state) **[E]**; (4) quasi-freeness *forced* not assumed — the $120{=}\dim\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant **[C]**. **Residual [O]**: v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open **[E]**/**[O]**

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Script	**What it machine-checks** *(continued)*
`v162_seam_transport_identification.py`	The final identification, forced as far as it goes — the v161 one-particle statement carries *no* new open content: (1) the gap $(2/3)^6$ is *forced* — cusp weights $= \mathrm{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$, $p_2{=}6{=}\lvert R_+(A_3)\rvert$, all carrier data **[E]**; (2) μ_4-equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} = \lvert\mu_4\rvert\,\mathrm{diag}(X)$, commutant of U = the diagonal algebra — so the gapped μ_4-equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ **[E]**; (3) the cusp grading *is* the A_3 grading on the 16 Majoranas (v137) **[E]**; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) **[E]**. **Closure** [C]/**[O]**: with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the *already-open* A2 (net existence) and R1 (seam clock, `HOR.CLOCK`) — so (A) factors exactly into A2 + R1, adds *no* new open item, and is closed conditionally on them (a unification, not an unconditional proof) **[E]**/**[O]**
`v163_rg_stability_flavor.py`	F_{transfer} scale-tagging: the lepton-mixing seeds ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0}{2}{=}0.3067$, $\sin^2\theta_{13}{=}\varphi_0 e^{-5/6}{=}0.0231$) are *RG-stable* — PMNS running is y_τ^2-suppressed ($\sim 1.8{\times}10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b{\sim}41$) run through $y_t^2 \sim O(10\text{–}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2}YY^\dagger Y$ carries $y^2_{\max}$ (PyR@TE SM RGEs) **[E]**/[C]
`v164_qcd_scale.py`	the QCD confinement scale is parameter-free from the carrier $b_3{=}{-}7$: running $\alpha_s(M_Z)$ down (two-loop, n_f thresholds) gives $\alpha_s(m_b){\sim}0.22$, $\alpha_s(m_c){\sim}0.38$ and confinement ($\alpha_s{\sim}1$) at $\sim$0.5 GeV — so the QCD-scale *numerator* of m_p/m_e is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest 'Open / not forced' frontier non-claim ($m_p{=}O(1)\Lambda$ is lattice) **[E]**/**[O]**
`v165_premise_a_closure.py`	Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries *zero* independent open content: (1) A2 (net existence) closes by *assembly* of established mathematics [C]— 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c{=}8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B){=}16/16{=}1$, $248{=}120{+}128$, character $E_4/\eta^8{=}j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN $= \lvert k\rvert$ (v156/v157); (2) R1 (seam clock) reduces to `GATE.QGEO` [C]— μ_4-equivariance forces diagonal-in-cusp, the flat μ_4-equivariant generator is *unique* $= A_0^\star$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = 'seam boundary = the μ_4-punctured sphere $\mathbb{P}^1\backslash\mu_4$' = `GATE.QGEO` (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\mathrm{geo}}\}$, a *theorem* by the No-Unit Theorem (v153). **Final accounting**: (A) = A2 assembly [C] + R1=`GATE.QGEO` + the $\{\pi, v_{\mathrm{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof **[E]**/**[O]**
`v166_higgs_free_seam.py`	the Higgs quartic from the *free seam* (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\mathrm{seam}}){=}0$ *and* $\beta_\lambda(M_{\mathrm{seam}}){=}0$ (Shaposhnikov–Wetterich double criticality, here *motivated* not assumed). Two-loop PyR@TE-sourced SM running with the measured (m_H, m_t) gives $\lambda(\bar M_{\mathrm{Pl}}){\sim}0.002$, $\beta_\lambda(\bar M_{\mathrm{Pl}}){\sim}0$ — the famous SM near-criticality, now *explained* by the free seam (2-loop sharpens $\lambda(\bar M_{\mathrm{Pl}})$ from $\sim$0.022 to $\sim$0.002). The double condition predicts $m_H{\sim}$129–134 GeV; at the scalaron scale it gives $\sim$107 GeV (too low) $\Rightarrow$ the BC lives at $\bar M_{\mathrm{Pl}}$ (seam=Planck) **[E]**/[C]

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Script	**What it machine-checks *(continued)***
`v167_inventory_post_closure.py`	The terminal residual inventory (post premise-(A) closure; line v105 → v123 → v167): after the (A)-chain (v160–v165), premise (A) is *no longer* an independent structural class (`GATE.METRIC.16-19`). The residual collapses from v123's FIVE classes to: **Tier 0** — the irreducible core $\{\pi, v_{\text{geo}}\}$, a *theorem* (No-Unit, v153), nothing to close; **Tier 1** — ONE hinge (seam = flavour = horizon): A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary $= \mathbb{P} \setminus \mu_4$, cohomology $b_1 {=} N_{\text{fam}} {=} 3$); **Tier 2** — folded ($\to v_{\text{geo}}$, $\to G_{\text{net}}$); **Tier 3** — F_{transfer}, one functor with four typed conditional interfaces (η_B carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]
`v168_qgeo_rigidity.py`	The QGEO *Rigidity Theorem* — the finite half of the Tier-1 hinge `GATE.QGEO`, hardened. Exact: $\mu_4 {=} \{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z{\to}iz, z{\to}1/z\rangle$ of order $8{=}\lvert D_4\rvert$ (faithful) ⇒ four marks with faithful D_4 are the μ_4 square up to Möbius; $H^1(\mathbb{P} \setminus \mu_4)$ has rank $4{-}1{=}3{=}N_{\text{fam}}$; the forms $\omega_k {=} z^{k-1}dz/(z^4{-}1)$ ($k{=}1, 2, 3$) are μ_4-eigenforms with $z{\to}iz$ eigenvalue i^k = the three nontrivial characters χ_1, χ_2, χ_3, weights $(1, 2, 3)$ = A_3 exponents $= \text{Spec}(Q_+)$. `QGEO.RIGID.01` [E] (math half closed); `QGEO.REALIZE.01` [C] (seam-collar $= \mathbb{P} \setminus \mu_4$ stays the realisation premise) [E]/[C]
`v169_etaB_boltzmann_interface.py`	The η_B leptogenesis transfer *operationalised* (Tier-3 F_{transfer}, not a new class): type-I thermal leptogenesis $\eta_B {\sim} 0.96{\times}10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}{=}240°$. Davidson–Ibarra $\varepsilon_1 {=} \frac{3}{16\pi} M_1 m_3/v^2 {\sim} 8.5{\times}10^{-7}$ for $M_1{=}10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1) {\sim} 0.03$–$0.19$. **Kill test**: over natural $M_1 {\in} [3{\times}10^9, 3{\times}10^{10}]$ GeV and washout, η_B spans $[8.4{\times}10^{-11}, 4.7{\times}10^{-9}]$ which *brackets* the observed $6.1{\times}10^{-10}$ (canonical point gives $6.0{\times}10^{-10}$, not tuned) ⇒ the route is viable; M_1/washout are scenario inputs, so η_B stays [C]. If excluded, the route falls, not the theory [C]
`v170_diamond_slice_compression.py`	E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P{=}(3, 4, 6, 10)$ ($p_n{=}2{+}2^n$) and Sheet-Diamond determinants $D{=}(3, 4, 8, 14, 20, 32)$. The K_4 edge products $(12, 18, 30, 24, 40, 60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$; all seven 248-slices follow from $P, D, \Delta{=}13$ and $(e_1, e_2, \dim S^+)$; one affine map $\text{row}(C{+}sU{+}tV) = (7, 11, 13){+}s(3, 3, 3){+}t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]
`v171_os_moment_cluster.py`	Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T){=}\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n){=}A{+}Br^n{+}Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram ⇒ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, $\xi{=}0.41105$, Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$; gap safety $\Delta_{\text{eff}}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{>}0$ with $31{=}1{+}h^\vee(E_8)$, $c((E_8)_1){=}248/31{=}8$ [E]/[C]
`v172_trace_anomaly_seed.py`	Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam S^2 ($\chi{=}2$) the $(E_8)_1$ ($c{=}8$) Weyl anomaly is $c\chi/6{=}\frac{8}{3}$, halved by the one-sided $\lvert\mathbb{Z}_2\rvert$ to $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3{=}\frac{1}{6\pi}$. Plus QCD $b_3{=}11{-}\frac{2}{3}N_f{=}7$ = scalaron exponent $\Omega_{\text{adm}}{-}10b_1{=}48{-}41$ (confinement↔inflation), and the Volkov–Wipf factorisation $-98/45{=}-2\cdot 7^2/\dim(D_5)$ [E]
`v173_pfaffian_cp_car.py`	Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}{=}0$ — the sheet-odd Calderón involution $\{D, \Sigma\}{=}0$ pairs the spectrum $(\pm\lambda)$, γ_5-Hermiticity $D^\dagger{=}\gamma_5 D\gamma_5$ makes the Pfaffian real ($\arg \in \{0, \pi\}$), and reflection positivity ($Z{>}0$) selects the positive branch ⇒ $\theta_{\text{eff}}{=}0$ [E]

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Script	What it machine-checks *(continued)*
`v174_seam_fock_readings.py`	Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})$=$2^{15}$=32768 reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16$=2^{11}), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k\le 2}\binom{4}{k}$, with $11 = \dim S^+ - g_{\text{car}}$ (full 2^4=16) and c_u/c_d=55/117 reading the QBL-protected boundary state space; (3) Affleck–Ludwig g-theorem c_{UV}=5+3=8=$c_{IR}((E_8)_1)$ so Δc=0 forces an exact isometry (boundary-entropy form of v157/v158) **[E]**
`v175_net_existence_full_cone.py`	The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to **[E]**, isolating `QGEO.REALIZE.01` as the single open premise (nothing fabricated): (1) **full-cone RP for all** m — the CAR second-quantisation functor $\Gamma(t)$=$\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction t acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}$=65536); the exterior-power spectrum theorem makes every eigenvalue a subset product of spec(t), verified on all 2^{16} products to lie in $[0,1]$ (contraction), with eigenvalue-1 multiplicity 2^8=256 (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces *for all* m to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) **A2 net existence** assembled + verified — $(E_8)_1$ character E_4/η^8=1+$248q$+$4124q^2$+$34752q^3$+$213126q^4$+$1057504q^5$ (level-1=248), embedding 248=120+128, E_8 Cartan even unimodular (det 1); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = `QGEO.REALIZE.01`, finite half v168, realisation half left **[O][E][O]**
`v176_seam_collar_realisation.py`	The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — *not* a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks and admissible c=8 data, *then* its conformal boundary is Möbius-equivalent to $\mathbb{P}\setminus\mu_4$ with H^1 grading $(1,2,3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already **[E]**: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius D_4 order 8, b_1=N_{fam}=3), Calderón DtN symbol $\lvert k\rvert$ (v156), H^1 character = generation grading $(1,2,3)$=A_3 exponents=Spec(Q_+) (v137/v168), μ_4-equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4$=$(E_8)_1$ + full-cone RP all m (v154/v175); the whole structural residual reduces to the ONE open premise `QGEO.REALIZE.01` (the physical seam collar's boundary *is* this $\mathbb{P}\setminus\mu_4$), left honestly **[O][E][O]**
`v177_seam_marking_kernel.py`	The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed *constructively* **[E]** and the open residual sharpened into exactly TWO named obligations. **[E]** UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, the reflection $z \mapsto 1/z$ gives $\sigma\rho\sigma = \rho^{-1}$ (D_4, order 8) — a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P},\mu_4)$; **[E]** COHOM: $\rho^*\omega_k = i^k\omega_k$ so the H^1 grading is $(1,2,3)$=A_3 exponents=Spec(Q_+); **[E]** MODULE: the H^1 →generation iso is *unique* (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); **[E]** NET (v154/v175): index-4 μ_4 extension of $(D_5)_1 \otimes (A_3)_1$=$(E_8)_1$. The TWO genuine open obligations **[O]**: `QGEO.MARKS.01` (raw seam → genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (raw seam Calderón kernel = the μ_4-equivariant free gapped contraction as an *operator*); the whole structural residual is now exactly these two doors **[E][O]**

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Script	**What it machine-checks** ***(continued)***
`v178_marks_kernel_attempt.py`	An honest *attempt* at the two open obligations `QGEO.MARKS.01` + `QGEO.KERNEL.01` — *not* a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. *Marks core* [E]: a genus-0 double with an order-4 clock ρ ($\sim z \mapsto iz$, fixed points $\{0, \infty\}$) and reflection τ forces the marks to be a generic ρ-orbit $\{1, i, -1, -i\}=\mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$, $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful D_4 order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4-equivariant operator is forced diagonal and *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $\lvert k\rvert$, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O]
`v179_conformal_realisation.py`	The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise `QGEO.CONF.01` (honest unification; the conformal realisation stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 *is* P1 — $c_3=1/(\lvert\mathbb{Z}_2\rvert\, 2\pi\,\chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$, the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock *is* the carrier — Coxeter element of $W(A_3)=S_4$, order $h(A_3)=4=\lvert\mu_4\rvert$ (v117); (3) marks = one orbit is *forced* — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$, a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $\lvert k\rvert$ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise `QGEO.CONF.01`: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]
`v180_clock_is_mobius.py`	Cracking `QGEO.CONF.01` one level further — the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, non-circular premise `QGEO.ISO.01`: 'the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) *is* $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) isometry⇒conformal⇒Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P})=\mathrm{PSL}(2, \mathbb{C})$; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the milder isometry premise; the new residual `QGEO.ISO.01` (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O]

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Script	**What it machine-checks** *(continued)*
`v181_clock_is_conformal_symmetry.py`	The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise `QGEO.SYM.01` — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. The clock is the Coxeter element of $W(A_3)$=S_4 (v117), order $h(A_3)$=4=$\|\mu_4\|$, the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below `QGEO.SYM.01` the residual is *definitional*, not a missing theorem: 'the seam's conformal structure *is* the μ_4-deck structure of the carrier clock' (the carrier μ_4 glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain REALIZE(v175)→theorem(v176)→MARKS+KERNEL(v177)→finite cores(v178)→`CONF.01`(v179)→`ISO.01`(v180)→`SYM.01`(v181) has reached bedrock; we stop here honestly **[E][O]**
`v182_reviewer_residual_map.py`	The four external-review concerns reduce to the two already-named residuals + one data-only item. A (mapping $2D \to 4D$) **[E]**/**[C]**: the Plücker readout $11 = \sum_{k\leq 2} \binom{4}{k} = \dim S^+ - g_{\rm car} = 16-5$ (QBL count, v174) and the holographic reduction give A = `QGEO.SYM.01` + $F_{\rm transfer}$. B (UV gravity) **[E]**: 'gravity = geometric boundary readout' *is* `QGEO.SYM.01`. C (grammar tuning) **[E]**/**[C]**: frozen (v84) + minimal + over-determined (the anchor a=(1, 1, 2) in ≥ 6 independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). D (Koide Möbius flow) **[E]**/**[C]**: the Möbius nature is forced by Aut($\mathbb{P}$)=PSL(2, $\mathbb{C}$) on the deck, t∼2.84 only locates the point, and the missing RG generator *is* $F_{\rm transfer}$, so D = `QGEO.SYM.01` + $F_{\rm transfer}$. NET: three reduce onto {`QGEO.SYM.01`, $F_{\rm transfer}$}, one waits for data — a unification of the concern structure, not a closure (premises stay **[O]**/**[C]**) **[E][C]**
`v183_koide_f_corner_transfer.py`	The Koide source→pole factor $\frac{53}{54}$ has an *operator* origin (external-review proposal, validated): $u_{\rm pole}/u_{\rm source} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. (1) $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6\, 78 + \det R\, 8 + 2{\cdot}27{\cdot}N_{\rm fam}$); (2) F=R+Q is the *missing* Sheet-Diamond corner $M(1, 1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52{+}1$; (3) $\frac{53}{54} = 1-\frac{1}{2N_{\rm fam}^3}$ exactly (54=2·27); (4) negative control: only F gives 53 ($R/Q/K/L \to 32/21/35/56$). Upgrades `FR.KOIDE.03` from a bare guess to an operator identity; **[E]** the factor is exact, **[C]** the leptonic source→pole reading the F corner stays an $F_{\rm transfer}$ conjecture **[E][C]**
`v184_etaB_anchored_boltzmann.py`	Honest test of the external-review η_B proposal (M_1=$M_R\phi_0^4$, $\tilde m_1$=m_3/A_Λ) — firewall verdict: *sharper scenario*, not a closure. The washout anchors plausibly ($\tilde m_1$=m_3/A_Λ=$m_3/10 \approx 5\,$meV, A_Λ=10=$\|E(K_5)\|$, in the strong-washout window) **[C]**; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3 \approx 6{\times}10^{14}$ GeV and the nearest clean TFPT powers of $\bar M_{\rm Pl}$ both miss it (c_3-power off ∼75%, ϕ_0-power off ∼40%), so M_1=$M_R\phi_0^4$ merely relocates the free input $M_1 \to M_R$ (the reviewer's $6.0{\times}10^{-10}$ uses M_R=$1.3{\times}10^{15}$; the seesaw value gives $3.4{\times}10^{-10}$, the tell that M_1 is unpinned). η_B stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — η_B stays **[C]**, a sharper scenario **[C]**

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Script	What it machine-checks *(continued)*
`v185_axion_relic_solver.py`	Axion relic, honest misalignment estimate: the determinant-line axion $f_a{=}M_{\rm scal}/128 \approx 2.39{\times}10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) can provide all dark matter, but *only* near the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this f_a (prefactor $\sim$0.026/θ^2, $\theta{\sim}O(1)$ gives $\sim$21% of $\Omega_{\rm DM}$, need $\theta^2_{\rm eff}{\sim}4.7$), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\rm relic}(f_a,\theta_i,N_{\rm DW}){\overset{?}{=}}\Omega_{\rm DM}$ is a kill test for the *branch*, not the theory (no free singlet WIMP: 128=(16,4)+(16,4)) [C][O]
`v187_ftransfer_laws.py`	The $F_{\rm transfer}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, η_B, the axion relic, $m_p/m_e{+}\Lambda_{\rm QCD}$ — must *never* be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads `status_ledger.csv` and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3{=}{-}7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]
`v188_frontier_wording_guard.py`	The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It forbids stale sentences whose semantics an earlier round superseded — "leptogenesis interface ([E])" (v184 retyped Route B to [C]), "closed branch also fixes the leptogenesis inputs" (M_R is the seesaw scale, not a compiler power), "relic scale f_a, m_a open" (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), "computed $Q{=}0.664\ldots$ near, not exact" (v183 gave 53/54 an exact operator origin), "Suite now 187 modules" (the count is 187, highest ID `v188`; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]
`v189_riemann_roch_carrier.py`	The Riemann–Roch carrier reading: $(g_{\rm car}, N_{\rm fam}) = (5,3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg{+}1 = 5$ (function side) and rank $H_1(\mathbb{P}^1 \backslash \mu_4) = \deg{-}1 = 3$ (cycle side, already `QGEO.COHOM.01`/v177); $g_{\rm car} = 5 = {\rm rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, rank D_5+ rank $A_3 = 5{+}3 = 8$, v1), $N_{\rm fam} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by ${\rm rank}(D_5) = 5$, but the physical identification is conjectural and does *not* replace the over-determined $g_{\rm car} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C]
`v190_nariai_entropy_bound.py`	The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\rm tot}/S_{dS} = (x^2{+}1)/\Phi_3(x)$ ($x{=}r_b/r_c$; $\Phi_3{=}x^2{+}x{+}1$ the $N_{\rm fam}{=}3$ cyclotomic, already in `tfpt_horizon_readouts`) satisfies $S_{\rm tot} \geq \frac{2}{3} S_{dS}$ with equality exactly at the Nariai merge $x{=}1$, via the perfect square $3(x^2{+}1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x{>}0$ is $2/3 = \lvert Z_2\rvert/N_{\rm fam}$) [E]
`v191_universal_branch_line.py`	The universal branch line, honestly typed [C] — an exact affine *relabeling*, not a theorem. The map $q = \frac{7}{2} + \frac{N^2_{\rm fam}}{2} m$ carries the SdS branch points $m = \pm 1/N_{\rm fam}$ onto the flavor branch points $\{2, 5\} = \{\lvert Z_2\rvert, g_{\rm car}\}$ ($m{=}0{\to}7/2$). The arithmetic is exact, but the map exists for *any* two pairs of points (two intervals are always affinely equivalent), so the slope $N^2_{\rm fam}/2$ and midpoint 7/2 are *determined* by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = \lvert Z_2\rvert/N_{\rm fam}$ ramification of the mass↔transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]

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Script	**What it machine-checks *(continued)***
`v192_qgeo_energy_clock.py`	The energy-conserving-clock reformulation, honestly typed: `QGEO.ENERGY.01` ($\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) *relocates* the bedrock `QGEO.SYM.01` to a more checkable operator identity, but does *not* eliminate it. The downstream chain (conformality $\to$ Möbius $\to z \mapsto iz \to \mu_4 \to$ `QGEO.SYM.01`) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda = \|k\|$. But 'ρ preserves Λ' is plausibly equivalent to 'ρ is the conformal deck', so it is a sharper restatement, not a theorem, unless ρ is independently defined and the invariance independently proven (open). Bedrock stays **[O]**
`v193_qgeo_energy_commutator.py`	`QGEO.ENERGY.02`: the energy-*commutator* contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, ρ the carrier clock, Λ_Σ the DtN energy form of the *raw* RP seam. Three sub-claims: (1) **[E]** ρ is carrier-defined (Coxeter of $W(A_3){=}S_4$, order $4{=}\|\mu_4\|$, v117), not imported from the seam; (2) **[O]** the non-circularity hinge — Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) **[E]** $[\rho, \Lambda_\Sigma]{=}0$ forces the $(1,2,3)$ grading on H^1 (`QGEO.COHOM.01`). Exact **[E]** rider: the induced $k{=}(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m{=}q(A_3){=}\frac{3}{4}$ (FAMILY), not $q(D_5){=}\frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. Proof target **[O]**; if proven, `QGEO.SYM.01` becomes an energy-rigidity theorem **[O]**
`v194_raw_seam_dtn_rp.py`	`QGEO.DTN.01` — subclaim 2 of v193 ("the raw RP-seam DtN is RP-definable") tackled: a *reduction*, not a closure. (1) **[E]** finite core: $\rho{:}z{\mapsto}iz$ on $H^1{=}\langle w_1, w_2, w_3\rangle$, $w_k{=}z^{k-1}dz/(z^4{-}1)$, acts as $\rho^* w_k{=}i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma]{=}0$ on H^1 (v177). (2) **[E]** hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without assuming $\mathbb{P}^1 \backslash \mu_4$. (3) **[O]** residual relocates: full-L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). `QGEO.SYM.01` reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still **[O]**, not closed **[O]**
`v195_marks_lefschetz_character.py`	`QGEO.MARKS.02` — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing D_4 marks, they are *forced* to be a free μ_4 orbit. **[E]**: $\rho{:}z{\mapsto}iz$ (order 4) fixes only $\{0, \infty\}$; $H^1{=}\chi_1{+}\chi_2{+}\chi_3$ (the A_3 exponents $(1,2,3)$, v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; rank $H_1{=}n{-}1{=}3 \Rightarrow n{=}4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks $=\mu_4$ (cross-ratio 2); $\mathrm{Tr}(\rho\|H^1){=}i{+}i^2{+}i^3{=}{-}1$, $L(\rho){=}1{+}1{=}2{=}\#$fixed. **[O]** residual relocates from "the seam is $\mathbb{P}\backslash\mu_4$ (geometric)" to "H^1 carries the A_3-exponent rep" — less circular **[E][O]**
`v196_seam_energy_functional.py`	`QGEO.VARI.01` — the E_{fail} variational functional (review path 1), a numerically-testable sharpening of the v194/`ENERGY.02` target: $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4{-}1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, with $E_{\text{fail}}{=}0 \Leftrightarrow$ the `QGEO.SYM.01` conditions. **[E]** on the finite H^1 block ($\rho{=}\,\mathrm{diag}(i, -1, -i)$, Λ diagonal, $\Theta{=}$conj) all three vanish $\Rightarrow E_{\text{fail}}{=}0$; **[E]** a μ_4-breaking Λ or non-order-4 ρ gives $E_{\text{fail}}{>}0$ (μ_4 a true minimiser); **[E]** the zero locus is the seam-deck conditions; **[O]** the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure **[E][O]**
`v197_rr_carrier_clifford_d5.py`	`ARCH.RRCAR.02` — hardening the Riemann–Roch carrier from g_{car} to D_5 itself (review point 7): the 5-dim carrier mode space $C_{\text{car}}{:=}H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10){=}D_5$ half-spinor by its even Clifford algebra. **[E]** $\dim\Lambda^{\text{even}}(\mathbb{C}^5){=}\binom{5}{0}{+}\binom{5}{2}{+}\binom{5}{4}{=}1{+}10{+}5{=}16{=}2^{g_{\text{car}}-1}{=}\dim S^+$; the $\mathfrak{so}(10){=}D_5$ half-spinor $2^4{=}16{=}S^+{=}\Lambda^{\text{even}}(C_{\text{car}})$; closes the chain $\mu_4{\to}g_{\text{car}}{\to}D_5$ from one four-point divisor (cycle side rank $H_1{=}3{=}N_{\text{fam}}(A_3)$; function side $h^0{=}5{=}g_{\text{car}}$; Clifford spinor $16{=}D_5$ half-spinor $=$ carrier). **[C]** "carrier $=$ Clifford spinor of the seam mode space" stays physical (as v189), does not change the over-determined $g_{\text{car}}{=}5$ **[E][C]**

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Script	What it machine-checks *(continued)*
`v198_modular_commutator_reduction.py`	`QGEO.MODULAR.01` — cracking the bedrock to a state-invariance. The v194 residual ($[\rho,\Lambda_\Sigma]$=0 on full L^2, framed as Bisognano–Wichmann with a circularity worry) is (a) made *exact* at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $\lvert k\rvert=\sqrt{-d^2}$ is diag($\lvert n\rvert$) and the clock $\rho{:}z{\mapsto}iz$ is diag(i^n) in the Fourier basis — both diagonal $\Rightarrow$ $[\rho,\lvert k\rvert]$=0 exactly on *all* of L^2; [E] Tomita–Takesaki: $[\rho,\Lambda_\Sigma]$=0 follows from $\omega{\circ}\rho{=}\omega$ (the modular Δ is intrinsic to (M,Ω); a state-preserving symmetry commutes with it), needing *no* conformal covariance — removes the BW circularity; [E] the finite H^1 block is already μ_4-invariant (v177). [O] residual: the full raw seam state is μ_4-invariant ($\omega{\circ}\rho{=}\omega$) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational `QGEO.SYM.01`. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O]
`v199_seam_state_invariance.py`	`QGEO.STATE.01` (work package A) — attacking $\omega{\circ}\rho{=}\omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular (ρ = Coxeter of $W(A_3){=}S_4$, ρ^4=1, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega{\circ}\rho{=}\omega$ $\Leftrightarrow$ $[\rho,H_1]$=0 ($C_\Sigma{=}\frac{1}{2}(1{+}\operatorname{sgn}H_1)$); [E] character criterion: $[\rho,H_1]$=0 $\Leftrightarrow$ H_1 block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$; principal symbol $\lvert k\rvert$= diag($\lvert n\rvert$) passes automatically. [O] residual: the bounded sub-principal symbol has *no* off-character matrix elements — non-circular, sharply typed, holds on H^1 (v177); not a closure [E][O]
`v200_seam_variational_scan.py`	`QGEO.VARI.02` (work package B) — the discretised variational test: v196 had $E_{\rm fail}$=0 at the μ_4 config on the fixed block; here the clock angle is free and $E_{\rm fail}$ is minimised over it. [C] scanning $E_{\rm fail}(\theta){=}\lVert\rho(\theta)^4{-}1\rVert^2$+faithfulness over $\theta{\in}(0,\pi)$ gives the unique minimum at $\theta{=}\pi/2$ ($E_{\rm fail}$=0: order-4 *and* three distinct characters $i,-1,-i$ = the $(1,2,3)$ grading); decoys $(0,\pi,\pi/3,2\pi/3)$ all lose. So $\arg\min E_{\rm fail}$ = the μ_4 normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite H^1 grading, not the full operator-valued minimisation (the v199 residual) [C][O]
`v201_seam_subprincipal_marks.py`	`QGEO.SUBPRIN.01` — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is *not* an independent postulate. [E] the seam DtN is $\Lambda{=}\lvert k\rvert{+}M_f$, M_f = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n\lvert M_f\rvert n'\rangle{=}f_{n-n'}$. [E] M_f is μ_4-character-block-diagonal $\Leftrightarrow$ f has Fourier support only on modes $\equiv 0 \pmod 4$ $\Leftrightarrow$ f is $\mathbb{Z}_4$-invariant ($\lvert k\rvert$ already commutes, v198). [E] a μ_4-mark sum $f(\theta){=}\sum_{j=0}^{3} g(\theta{-}2\pi j/4)$ has $f_m{=}4g_m[m{\equiv}0 \bmod 4]$ for *any* profile g — automatically $\mathbb{Z}_4$-invariant, so (v195 marks=μ_4 orbit) *forces* the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. [O] new residual: $\omega{\circ}\rho{=}\omega$ reduces to 'the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck v177)', the narrowest, structurally-definitional form of `QGEO.SYM.01` [E][O]
`v202_rare_kaon.py`	The rare kaon sector $K\to\pi\nu\bar\nu$ from the TFPT CKM point ([C] downstream flavor readout, *not* a compiler power). The kernel fixes the full CKM matrix ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}\lvert V_{cb}\rvert$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\mathrm{BR}(K^+\to\pi^+\nu\bar\nu){=}9.45\times10^{-11}$ and $\mathrm{BR}(K_L\to\pi^0\nu\bar\nu){=}3.33\times10^{-11}$. [E] CKM point + exact apex $(\bar\rho,\bar\eta){=}(0.1374,0.3509)$, $\lambda_t{=}V_{ts}^*V_{td}$; [C] the branching ratios (*not* parameter-free — $X_t,P_c,\kappa_\pm$ are external EW/QCD input); [E] $\mathrm{BR}(K^+)$ $\sim 1.2\sigma$ below NA62 $(13.0^{+3.3}_{-3.0})\times10^{-11}$; [E] Grossman–Nir ratio $0.352\ll4.3$; [X] dated kill test: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7,12]\times10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X]

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Script	**What it machine-checks** *(continued)*
`v203_eht_achromatic.py`	The EHT achromatic polarization-rotation intercept $\beta_{\rm BH}$ around a horizon-scale black hole — the surviving core of the old UFE/BH notes (*not* the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). **[E]** the coupling is a compiler number: $\beta_{\rm BH}(r){=}Q_e^{\rm eff}Q_m^{\rm eff}/(256\pi^4r^2){=}16c_3^4(Q_eQ_m/r^2)$, $16c_3^4{=}1/(256\pi^4)$ exactly. **[E]** α-kernel cross-link: $16c_3^4{=}\delta_{\rm top}/3$, $\delta_{\rm top}{=}48c_3^4{=}3/(256\pi^4)$ — the *same* top-form coefficient that fixes the α-kernel precision-zone correction (one row from $\beta_{\rm rad}{=}\varphi_0/4\pi$). **[C]** TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E{\cdot}B$ reversal; the amplitude Q_eQ_m is an MHD/GR weight, not a compiler output. **[X]** dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) **[E][C][X]**
`v204_muon_seam_g2.py`	The muon anomalous magnetic moment $a_\mu{=}(g_\mu{-}2)/2$ as a seam vertex readout (**[C]** downstream, *not* a compiler power; archive integration of the old muon-$g{-}2$ note). The carrier fixes the second-order defect $\delta_2{=}B\gamma\,\delta_{\rm top}^2{=}\frac{5}{4}\delta_{\rm top}^2$ ($\delta_{\rm top}{=}\Omega_{\rm adm}c_3^4{=}48c_3^4{=}3/256\pi^4$, v2/v23; $B\gamma{=}\frac{3}{2}{\cdot}\frac{5}{6}{=}\frac{5}{4}$, v51); the seam-loop projection $a_\mu^{\rm seam}{=}\delta_2/(2\pi){=}45/(524288\pi^9)\approx 2.879\times10^{-9}$. **[E]** the value is an exact compiler number ($\delta_2{=}4!\,{\rm Tr}_{S^+}(X^2)c_3^8$, Tr=120=5!); **[C]** the identification of $\delta_2/(2\pi)$ as a magnetic vertex ($1/2\pi{=}4c_3$) is a physical bridge; **[E]** 0.81σ vs dispersive $\Delta a_\mu{=}(2.49{\pm}0.48)\times10^{-9}$; **[C]** HONEST: lattice/CMD-3 HVP shrinks it ($\sim1.5\times10^{-9}$), the fixed value then $\sim1.5\sigma$ high; **[X]** dated kill test: a converged Δa_μ outside $2.879\times10^{-9}{\pm}0.5\times10^{-9}$ excludes the mechanism **[E][C][X]**
`v205_xi_threequarter.py`	$\xi{=}c_3/\varphi_{\rm tree}{=}3/4$: an *independent* route to the gravitational 3/4 (archive integration of the old Eliminating-K note). The torsion-compression ansatz $\kappa^2{=}\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2{=}8\pi G$ fixes $\xi{=}8\pi c_3^2/\varphi_0{=}c_3/\varphi_0$ (since $8\pi c_3^2{=}c_3$), and at tree level $\varphi_{\rm tree}{=}1/(6\pi)$ this is exactly 3/4. **[E]** tree identity $\xi{=}c_3/\varphi_{\rm tree}{=}3/4$; **[E]** Einstein-limit reduction $8\pi c_3^2{=}c_3\Rightarrow G$ is an output; **[E]** over-determination $3/4{=}q(A_3){=}N_{\rm fam}/\lvert\mu_4\rvert{=}\ln(m/\mu)$ (v152 $k{=}c_3/2$) — the torsion-compression ξ and the gapped EH replica are two independent appearances of the same gravitational 3/4; **[E]** with the retained seed $\xi{=}c_3/\varphi_0{=}0.748303$ (-0.226%); **[C]** the $\kappa^2{=}\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) **[E][C]**
`v206_h0_lambda_branch.py`	The cosmological-constant Letter branch: the seam number $\delta_\Sigma{=}m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1{=}\Omega_{\rm adm}{=}48$, the Planck-convention split, and the quantitative $H_0{=}66.5$–67.1 (archive integration of the tfpt-31 CC note; sharpens the qualitative H_0-vs-Λ row). **[E]** $\delta_\Sigma^{\rm lead}{=}48\,e^{-2\pi\cdot137.036/3}\approx1.085\times10^{-123}$; **[E]** convention split $M_{\rm Pl}^4/\bar M_{\rm Pl}^4{=}(8\pi)^2{=}631.65$ (unreduced vs reduced — must be displayed); **[E]** $\rho_\Lambda{=}M_{\rm Pl}^4\delta_\Sigma\approx2.4\times10^{-47}\,{\rm GeV}^4$ (observed vacuum order); **[E]** 123-orders cross-link $\lvert\log_{10}\delta_\Sigma\rvert\approx122.96$, consistent with the v60 split 122.948; **[C]** under flatness closure $H_0^{\rm lead}{=}66.51$, $H_0^{\rm disc}{=}67.10$ — 1.6–0.5σ below Planck, $\sim6\sigma$ below SH0ES: the Hubble tension is *not* relieved by raising the vacuum scale; H_0 uses imported $\Omega_b,\omega_{\rm DM}$, a **[C]** readout **[E][C]**
`v207_asymptotic_safety.py`	Asymptotic Safety as a *second*, independent, **[O]** route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise `QGEO.REALIZE.01`); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, *not* load-bearing, does *not* close $G_{\rm metric}$. **[E]** the axion coupling is fixed and small $y^2{=}16c_3^2{=}1/(4\pi^2){=}0.025330$ exactly ($g_{a\gamma\gamma}{=}{-}4c_3$ from the cubic); **[E]** existence by IVT: $\partial_tg/g{=}2{+}\eta_R$ is $+2{>}0$ as $g{\to}0$ and <0 at large g (torsion K^2 damping) $\Rightarrow$ a root $g_*{>}0$; **[E]** $g_*{=}2/a_1{>}0$ finite, Reuter-type $O(1)$, no new free input; **[O]** HONEST: a continuum-FRG ansatz the discrete-compiler theory did *not* adopt, primary stays the seam-net (v175) **[E][O]**

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Script	What it machine-checks *(continued)*
`v208_bh_thermodynamics.py`	Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). **[E]** modular T_H: $\Delta^{it}{=}e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta{=}2\pi$ and $T_H{=}\kappa/(2\pi)$; the 2π is the seam unit $1/(4c_3)$ ($1/2\pi{=}4c_3$, v58), reproducing $T_H{=}c_3/M$ (v57); seam$\leftrightarrow$horizon identification **[C]**; **[E]** Wald entropy: with the induced $f(R){=}R{+}R^2/(6M_s^2)$ ($M_s{=}c_3^{7/2}\bar{M}_{\rm Pl}$, v28), $S_W{=}(f_R A)/(4G){=}(A/4G)(1{+}R_h/3M_s^2)$ — leading $A/4G$ is the c_3 area law ($1/4{=}1/\lvert\mu_4\rvert$), the correction an exact scalaron consequence; **[E]** SdS $T_H{=}(1/4\pi r_h)(1{-}\Lambda r_h^2)$, $1/4\pi{=}2c_3$; **[E]** area law $S_{BH}{=}M^2/(2c_3){=}4\pi M^2{=}A/4G$ **[E][C]**
`v209_bh_regular_core.py`	The black hole as a seam fixed point (topological defect), *not* a singularity: the honest **[O]** synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. **[E]** no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2{=}(2/3)^6{=}64/729$ with gap $\Delta{=}6\log(3/2){>}0$ (v56) — the core is a fixed point, $\rho{\to}\infty$ replaced by $\varphi{\to}\varphi_*$ **[C]**; **[E]** maximal BH = anchor: Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = the traceless projection of $a{=}(1,1,2)$ (v101/v190); **[E]** information stays in the boundary field: Page recovery decays at $\lambda_2{=}(2/3)^6$ per step (v54) **[C]**; **[E]** holographic remnant $S_{BH}{=}A/4G$, $1/4{=}1/\lvert\mu_4\rvert$ **[C]**; **[O]** HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), *not* load-bearing, *not* a metric **[E][O]**
`v210_mark_local_dtn.py`	A numerical *sharpening* of v201/`QGEO.SUBPRIN.01` on *realistic* Steklov curvature profiles, carried to the quasi-free *state* level (the actual $\omega{\circ}\rho{=}\omega$, not just $[\rho,\Lambda]{=}0$); *not* a closure of `QGEO.SYM.01`. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four μ_4 marks $f(\theta){=}\sum_{j=0}^{3} g(\theta{-}j\pi/2)$, builds the Steklov DtN $\Lambda{=}\lvert D_\theta\rvert{+}M_f$. **[E]** every off-(mod 4) Fourier coefficient of the real f vanishes to $< 10^{-16}$ (exact 4-fold cancellation), although g has all modes. **[C]** $\lVert[\rho,\Lambda]\rVert < 10^{-15}$. **[C]** *state invariance* (new vs v201): the quasi-free covariance $C{=}\frac{1}{2}(1{+}\operatorname{sgn} H_1)$ satisfies $\rho C\rho^\dagger{=}C$ to $< 10^{-13}$ — the state, not just the operator, is clock-invariant. **[C]** negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$; convergence stable for $N{\in}\{16,32,64\}$. **[O]** certifies 'mark-local DtN $\Rightarrow$ $\omega{\circ}\rho{=}\omega$' for concrete profiles *and* at the state level; the premise that the physical seam *is* mark-local stays **[O]** — the narrowest definitional form of `QGEO.SYM.01`, not its closure **[C][O]**
`v211_axion_spine_angle.py`	Axion DM, the spine-angle branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ (108°): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170°$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$). **[E]** rational motive $\theta_i = 3\pi/5$ exactly (spine quotient 3/5, no fit); **[E]** the v185 finite-T solver ($\theta{=}1 \to 0.03$) reaches $\Omega_{\rm DM}$ at $\theta \approx 106°$, so 108° is on target (harmonic $0.03\theta^2{=}0.107$); **[C]** *robust*: 108° is 62° below the 170° hilltop (mild-anharmonic, not exponentially sensitive); **[O]** an alternative ansatz to $\theta_i{=}\pi(1{-}\varphi_{\rm seam}){\approx}170°$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); **[X]** kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch **[E][C][O][X]**
`v212_leptogenesis_decuple.py`	Leptogenesis, the scalaron-decuple branch: both Boltzmann inputs share the decuple $A_\Lambda{=}10{=}\lvert E(K_5)\rvert$ — $M_1{=}M_{\rm scal}\varphi_0^2/A_\Lambda \approx 8.65{\times}10^9$ GeV and $\tilde{m}_1{=}m_3/A_\Lambda \approx 5$ meV. A cleaner **[C]** route than v184's $M_1{=}M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw M_R). **[E]** shared decuple $A_\Lambda{=}10$; **[E]** $M_1{=}8.65{\times}10^9$ GeV inside the thermal window $[3{\times}10^9, 3{\times}10^{10}]$ (v169); **[E]** $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; **[E]** Davidson–Ibarra $\varepsilon_1 \approx 8.5{\times}10^{-7}$, $\eta_B \approx 1.2{\times}10^{-9}$ brackets observed $6.1{\times}10^{-10}$; **[C]** HONEST: M_1 uses only $\{M_{\rm scal}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; **[X]** kill test falls the route, not the theory (`FR.ETAB.01` independent) **[E][C][X]**

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Script	What it machine-checks *(continued)*
`v213_ftransfer_functor.py`	The `F_transfer` *functor contract* (`CONTRACT.F.01`): the four frontier transfers are ONE typed functor $F_{\text{transfer}}{=}F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel **[E]** composed with an external continuous solver [C]; consolidates v183 (F_{pole}), v212 ($F_{\text{Boltzmann}}$), v211 (F_{relic}), v164/`FR.MPME` (F_{QCD}), the structural complement of the v187 typing guard. **[E]** axiom 1 μ_4-deck equivariance ($\lambda_2{=}(2/3)^6{=}64/729$ the deck transfer eigenvalue); **[E]** axiom 2 Plücker preservation ($53{=}a^T(R{+}Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1,22,30)$, $\\|\mathrm{Pl}(K)\\|_1{=}11$); **[E]** axiom 3 positivity/stochasticity ($\operatorname{spec} T{=}\{1,(2/3)^6,(1/3)^6\}$ positive, $\kappa_f{\in}(0,1)$); **[E]** axiom 4 external modules explicit ($b_3{=}{-}7$ a carrier output, $\lvert b_3\rvert{=}\Omega_{\text{adm}}{-}10b_1{=}7$); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure **[E]**[C]
`v214_seam_pillowcase.py`	The *pillowcase* reduction of `QGEO.SYM.01` (`QGEO.PILLOW.01`): the two separately-tracked open QGEO residuals — v180/`QGEO.ISO.01` (the carrier clock is an order-4 isometry of the seam metric) and v201/`QGEO.SUBPRIN.01` (the DtN sub-principal symbol is mark-local, the seam flat away from the μ_4 marks) — are ONE statement, 'the seam carries the uniformising flat orbifold (pillowcase) metric', and the order-4 clock is *derived* from cross-ratio 2 (v168), not assumed. **[E]** $\chi_{\text{orb}}(S^2(2,2,2,2)){=}2{-}4(1{-}\frac{1}{2}){=}0 \Rightarrow$ flat (Troyanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits π sum to $4\pi{=}2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}}{=}0$, flat away from the marks). **[E]** NEW LINK: cross-ratio$(\mu_4){=}2$ (v168) $\Rightarrow j(\lambda){=}256(\lambda^2{-}\lambda{+}1)^3/(\lambda^2(\lambda{-}1)^2)$ with $j(2){=}1728$ (all six harmonic cross-ratios $\{2,-1,\frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a *consequence* of cross-ratio 2. **[E]** $j{=}1728 \Leftrightarrow \tau{=}i$ (square torus) $\Leftrightarrow$ CM by $\mathbb{Z}[i] \Leftrightarrow \mathrm{Aut}{=}\mathbb{Z}/4$ ($j{=}0$ gives $\mathbb{Z}/6$; generic $\mathbb{Z}/2$). **[E]** explicit CM $(x,y) \mapsto (-x,iy)$ on $y^2{=}x^3{-}x$ has order $4{=}z \mapsto iz$. **[E]** negative control $\lambda{=}3 \Rightarrow j{=}21952/9 \notin \{0,1728\} \Rightarrow \mathrm{Aut}{=}\mathbb{Z}/2$, no clock. **[E]** UNIFICATION: 'seam = uniformising flat pillowcase metric' implies both v201 mark-locality and v180 `QGEO.ISO.01`. **[O]** does *not* close `QGEO.SYM.01` — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175's **[E]**
`v215_seam_deck_killtest.py`	The bedrock `QGEO.SYM.01` ($\omega{\circ}\rho{=}\omega$) recast as a *falsifiable* kill-test (`QGEO.KILL.01`), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega{\circ}\rho{=}\omega \Leftrightarrow [\rho,C]{=}0$; the principal symbol $\lvert k\rvert$ commutes exactly, so the residual is the sub-principal M_f with $[\rho,M_f]{=}0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant; measure form = the raw Gaussian bulk form A is μ_4-deck-invariant. Four levers (carrier-side **[E]**, seam-side **[O]**): **K1** exactly 4 marks (order-4 clock free orbit μ_4 + $b_1{=}\#\text{marks}{-}1{=}3{=}N_{\text{fam}} \Rightarrow \#\text{marks}{=}4{=}\lvert\mu_4\rvert$); **K2** square config (cross-ratio $2 \Rightarrow j{=}1728$, $\mathrm{Aut}{=}\mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j{=}0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support (μ_4-mark sum $\Rightarrow f_m{=}0$ for $m \not\equiv 0 \bmod 4 \Rightarrow [\rho,M_f]{=}0$); **K4** transfer gap $(2/3)^6{=}64/729$ (cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}$). Freeze-bound (`freeze_file.csv seam_deck_symmetry`). **[O]** the decisive kill is *deferred* (`QGEO.REALIZE.01`): the raw $\mathbb{P}\setminus\mu_4$ collar DtN must yield 4 square marks *without* inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure

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Script	**What it machine-checks** *(continued)*
`v216_marks_gauss_bonnet.py`	The four seam marks *emerge* from Gauss–Bonnet (`QGEO.MARKS.03`), executing the K1 lever of the v215 kill-test. [E] mark count $= 2\chi$: $\mathbb{Z}_2$ branch points (deficit π) on a flat sphere (χ=2) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = \lvert\mu_4\rvert = e_1(a) = N_{\mathrm{fam}}+1$ (the same χ=2 as the 8 in c_3=$1/(8\pi)$). [E] the Euclidean sphere 2-orbifolds (χ_{orb}=0) are exactly $(2,3,6),(2,4,4),(3,3,3),(2,2,2,2)$. [E] all-order-2 (the $\lvert\mathbb{Z}_2\rvert$ sheet branch) $\Rightarrow$ uniquely $(2,2,2,2)$; [E] N_{fam}=3 (rank H^1=#marks−1) $\Rightarrow$ uniquely $(2,2,2,2)$, picking the 4-mark square over the 3-mark hexagonal $(3,3,3)$; [E] a free order-4 deck $\Rightarrow$ uniquely $(2,2,2,2)$ — three independent selectors converge. [O] residual: the square modulus (cross-ratio 2 $\Rightarrow$ j=1728, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does *not* close `QGEO.REALIZE.01`
`v217_marks_emergence_scan.py`	The *numerical* emergence scan (`QGEO.EMERGE.01`, the sibling of v216 as v210 is to v201): with the number of marks n *and* their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$} on the actual seam DtN/state. [C] n=4 equally-spaced $\Rightarrow \lVert\rho_4 C\rho_4^\dagger - C\rVert \sim 10^{-14}$ (clock-invariant state); [C] family count selects n=4 (scan $n \in \{2..8\}$, only $n-1$=3=N_{fam}; n=3 $\to$ 2 families = hexagonal, excluded; n=5 $\to$ 4); [C] gap: μ_4-equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$), cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}$ $\Rightarrow$ transfer $(1-w)^6$=$\{1,(2/3)^6,(1/3)^6\}$, gap $(2/3)^6$=64/729; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square ~ 0.4); [C] negative controls n=3, 5, 6. [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does *not* close `QGEO.REALIZE.01`. Python-only
`v218_diamond_axis_geometry.py`	Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: C center, U winding, V sheet), F the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C{+}xU)$=$14{+}6x$ linear (slope 6=$\lvert R^+(A_3)\rvert$), $\det(C{+}yV)$=$14{+}14y{+}4y^2$ quadratic, 2nd difference 8=rank E_8; $\det B(C{+}xU)$=$2\det(C{+}xU)$, anchor-block sheet curvature 6=$\lvert R^+(A_3)\rvert$. [E] PLÜCKER LADDER: $\mathrm{Pl}(K)$=$(-1,6,4)$ $\to$ $\mathrm{Pl}(C)$=$(0,14,14)$ $\to$ $\mathrm{Pl}(F)$=$(1,22,30)$, steps $(1,8,10)$=$(N_\Phi, \mathrm{rank}\,E_8, A_\Lambda)$ and $(1,8,16)$=$(N_\Phi, \mathrm{rank}\,E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\mathrm{Disc}(\chi)$=$q(r)^2\mathrm{Disc}(q)$ for Q,K,C,F; squares $(1,3,4,6)$, kernels $(13,48,65,105)$ with 48=Ω_{adm}, 105=$N_{\mathrm{fam}}g_{\mathrm{car}}\cdot 7$. [C] heuristic: F the transfer-completion corner, a search principle, not a hard filter; the G_2/F_4 pair-sum labels stay audit-only (cf. v94/v95) [E][C]
`v219_icosahedral_mckay.py`	McKay bedrock: *why* the atoms are $\{2,3,5\}$. E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T{\to}\hat{E}_6$, $2O{\to}\hat{E}_7$, $2I{\to}\hat{E}_8$). The McKay graph is built *from the group*: the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1,2,3,4,5,6,10\}$, containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1,2,2,3,3,4,4,5,6\}$ (sum 30=$h(E_8)$, sum of squares 120=$\lvert R^+(E_8)\rvert$); the natural-rep McKay adjacency *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 [E], the reading that it explains $(2,3,5)$ [C], the raw-seam origin [O]
`v220_cp_hexagonal_modulus.py`	CP lives in the *hexagonal* phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)$=1728, Aut=$\mathbb{Z}/4$ (the μ_4 clock); its exceptional partner is the hexagonal modulus $j(\rho)$=0, ρ=$e^{i\pi/3}$, Aut=$\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$, $\arg\rho$=$\pi/3$. [E] the frozen leading CKM reading δ=$\pi/3{+}3\lambda_C^2$=68.654° (v88) has leading term $\pi/3$=$\arg\rho$; the $\lvert\mathbb{Z}_2\rvert$ alternative $\pi/3{+}2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]

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Script	**What it machine-checks** *(continued)*
`v221_seam_qecc.py`	The seam as a *finite recoverability* code (recovery rate $(2/3)^6{=}64/729$). [E] code dimension $\dim S^+{=}16{=}2^{g_{\rm car}-1}$; the gapped transport T on the cusp-weight 3-space (deviations $(1,-1,0)$ and the Nariai anchor $(1,1,-2)$) is symmetric, doubly stochastic (CPTP), spectrum $\{1,(2/3)^6,(1/3)^6\}$; on the trace-zero subspace $\|T\delta\|{\le}(2/3)^6\|\delta\|$ hence $\|T^n\delta\|{\le}(2/3)^{6n}\|\delta\|$ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]
`v222_cm_norm_duality.py`	CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$, $j{=}1728$): $N(g_{\rm car}{+}i\|\mu_4\|){=}5^2{+}4^2{=}41{=}10b_1$ — the EM index *is* the Gaussian norm of the carrier-glue vector; $N(3{+}2i){=}13{=}\Delta_Q$. [E] HEX (Eisenstein $\mathbb{Z}[\omega]$, $j{=}0$): $N(3{+}2\omega){=}3^2{-}3{\cdot}2{+}2^2{=}7{=}$ scalaron. The $(3,2){=}$(colour,weak) split generates $(5,6,7,13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5,4){=}21{\neq}41$). The architecture bridge square$\leftrightarrow$EM, hex$\leftrightarrow$scalaron is [C][E][C]
`v223_coxeter_totative_clock.py`	The μ_4 clock is the order-4 character of the E_8 Coxeter cycle. [E] $(\mathbb{Z}/30)^\times{=}\{1,7,11,13,17,19,23,29\}{=}$ the E_8 exponents = live Coxeter phases; $\varphi(30){=}8{=}\mathrm{rank}\,E_8$, $30{=}2{\cdot}3{\cdot}5$. [E] $(\mathbb{Z}/30)^\times{\cong}\mathbb{Z}/4{\times}\mathbb{Z}/2$ with order-4 generator 7 ($7^2{=}19$, $7^4{=}1$ mod 30): a literal μ_4 clock $\langle 7\rangle{=}\{1,7,13,19\}$ on the exponents; the conjugate pairs $m{+}(30{-}m){=}30$ give $\|\mu_4\|{=}4$ invariant planes the clock permutes. Only E_8 has $\varphi(h){=}$rank *and* $h{=}2{\cdot}3{\cdot}5$; the seam-clock identification is [C][E][C]
`v224_diamond_ftransfer_path.py`	$F_{\rm transfer}$ as *one* path on the Sheet Diamond, not four external interfaces. [E] family $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$: the corners $K{=}M(1,-1),C{=}M(1,0),F{=}M(1,1)$ lie on the sheet axis ($s{=}1$); the sheet axis is curved $\det M(1,t){=}4t^2{+}14t{+}14$ (2nd diff $8{=}\mathrm{rank}\,E_8$), the winding axis $\det M(s,0){=}6s{+}8$ flat (slope $6{=}\|R^+(A_3)\|$); Plücker steps $(1,8,10),(1,8,16)$. [C] $F_{\rm transfer}$ is the $K{\to}C{\to}F$ sheet geodesic, Koide source$\to$pole its 1-D Möbius projection (v37/v183) [E][C]
`v225_dual_normal_frame.py`	The dual normal frame (d,n): one boundary compass for flavor and horizon, with CP as orientation. [E] $d{=}a^\top R^{-1}{=}a^\top L^{-1}{=}(-\tfrac12,-\tfrac12,1){=}-\tfrac12(1,1,-2)$ (the Nariai/horizon normal; $d{\cdot}\mathbf{1}{=}0$, $d{\cdot}a{=}1$); $n{=}(5,-9,6)$ with $n^\top R{=}(\det R,0,0){=}(8,0,0)$, $n^\top L{=}(20,0,0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1},d,n){=}21{=}N_{\rm fam}\cdot 7$. [C] CP as orientation: $\mathrm{Im}\,\det(\mathbf{1},d,\rho n){=}21\sin(\pi/3)$ for $\rho{=}e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]
`v227_degree_exponent_channel_split.py`	248=120+128 as a magnitude/phase channel typing (replaces the over-strong S^--dark-matter reading). [E] E_8 exponents sum $120{=}\|R^+(E_8)\|$ (*magnitude* channel); degrees $\{2,8,12,14,18,20,24,30\}$ sum $128{=}2^{\mathrm{rank}-1}{=}\mathrm{rank}\cdot\dim S^+$ (*phase/glue* channel); $248{=}120{+}128{=}\mathrm{adj}(D_8){+}\mathrm{spinor}(D_8)$; $\sum\deg-\sum\exp{=}\mathrm{rank}{=}8$. [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]
`v228_rr_index_gate.py`	P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch *index gate*; bedrock language for P2, not a proof from nothing). [E] $\deg D{=}4{=}\|\mu_4\|$ on $\mathbb{P}^1\Rightarrow h^0(\mathcal{O}(D)){=}5{=}g_{\rm car}$ (Riemann–Roch, genus 0); $\mathrm{rank}\,H_1(\mathbb{P}^1\backslash 4){=}3{=}N_{\rm fam}$; $h^0{+}\mathrm{rank}\,H_1{=}8{=}\mathrm{rank}\,E_8$, $h^0{-}\mathrm{rank}\,H_1{=}2{=}\|\mathbb{Z}_2\|$; $\Lambda^{\rm even}(\mathbb{C}^5){=}1{+}10{+}5{=}16{=}\dim S^+$. The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]
`v229_lepton_frobenius_algebra.py`	The charged leptons as an étale Frobenius algebra over the μ_6 resolvent. [E] exact coefficients $(c_e,c_\mu,c_\tau){=}(\tfrac{16}{7},\tfrac43,\tfrac76)$, ring closure $c_ec_\tau{=}\tfrac83{=}\|\mathbb{Z}_2\|c_\mu$, product $\tfrac{32}{9}{=}2^{g_{\rm car}}/N_{\rm fam}^2$. [E] $A{=}\mathbb{Q}[t]/(m)$, $m{=}\prod(t{-}c)$: the trace-form Gram is nondegenerate ($\det{=}\mathrm{disc}(m){\neq}0$, A étale) — a commutative Frobenius algebra; the C_6 shift has characteristic polynomial $t^6{-}1$ (spectrum $\mu_6{=}\mu_3$ family $\times\,\mu_2$ sheet). [C] the PMNS extension via $\mathrm{Aut}(A)$ and the hexagonal CM point [E][C]

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Script	What it machine-checks *(continued)*
`v230_center_budget_norms.py`	The Sheet-Diamond center budget $(7,11,13)$ is the three *local norms* of the theory. **[E]** $C{=}R{+}Q\,\mathrm{diag}(1,0,0)$, row sums $(7,11,13)$: $7{=}N_{\mathbb{Z}[\omega]}(3{+}2\omega)$ (Eisenstein/hex = scalaron), $13{=}N_{\mathbb{Z}[i]}(3{+}2i)$ (Gauss/square = Δ_Q), $11{=}\binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2}{=}1{+}4{+}6$ (QBL boundary Fock count on the 4 μ_4 corners = $\dim S^+{-}g_{\mathrm{car}}{=}16{-}5$). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is **[C][E][C]**
`v231_cp_mu6_phases.py`	Both CP phases are μ_6 powers of *one* hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho{=}e^{i\pi/3}$ (the $j{=}0$ Eisenstein CM unit): **[E]** ρ is a primitive 6th root ($\rho^6{=}1$, $\rho^3{=}{-}1$ the $\mathbb{Z}_2$ sheet half-turn); **[E]** δ_{CKM} leading $=\arg(\rho^1){=}\pi/3$ (quark), $\delta_{\mathrm{PMNS}}=\arg(\rho^4){=}4\pi/3$ (lepton, the *phase lattice*); **[E]** $\rho^4{=}{-}\rho \Rightarrow \delta_{\mathrm{PMNS}}{=}\delta_{\mathrm{CKM,lead}}{+}\pi{=}(\mathrm{CM\ unit}){\times}(\mathbb{Z}_2\ \mathrm{sheet})$ — the two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; **[E]** the C_6/μ_6 monodromy has charpoly $t^6{-}1$ (spectrum μ_6 carries ρ); **[E]** orientation $\mathrm{Im}\det(\mathbf{1},d,\rho^1 n){=}{+}21\sin(\pi/3)$, $\mathrm{Im}\det(\mathbf{1},d,\rho^4 n){=}{-}21\sin(\pi/3)$ (sheet-flipped Jarlskog, v225); **[E]** NEG μ_3 has no -1 (no sheet; $\mu_6{=}\mu_3$ family $\times\,\mu_2$ sheet). REDUCTION: 2 CP phase inputs $\to$ 1 hexagonal unit + the sheet; the physical CP derivation stays **[C]**, the deck stays $\mathbb{Z}/4$ (v215) **[E][C]**
`v232_e8_kleinian_seam.py`	The seam as the E_8 *Kleinian singularity* — a canonical algebraic-geometric model for the open seam-realisation premise `QGEO.REALIZE.01` (the du Val side of the McKay correspondence, v219). **[E]** the McKay graph of $2I$ is affine E_8; dropping the unique trivial-rep node (the single Kac mark 1) leaves the *finite* E_8 Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms $1,2,4$). **[E]** eight exceptional $\mathbb{P}^1$'s $=\mathrm{rank}\,E_8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}} = \#$ nontrivial $2I$ irreps — a fourth reading of the 8 in $c_3{=}1/(8\pi)$. **[E]** the negated intersection form is the E_8 Cartan (even, $\det{=}1$, positive definite; each $\mathbb{P}^1$ a (-2)-curve). **[E]** one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I){=}0$: the link is the Poincaré homology 3-sphere, $\pi_1{=}2I$ of order $120{=}\lvert R^+(E_8)\rvert$. **[C]** seam reading: the raw RP seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for `QGEO.REALIZE.01`, not a P2 proof (presupposes E_8); bedrock **[O][E][C][O]**
`v233_cp_triality_phase.py`	The CP phase is the *universal* family/triality phase, only sheet-split — one step past v231 in $\mu_6{=}\mu_3$ (family/triality) $\times\,\mu_2$ (sheet). **[E]** $\mu_6{=}\mu_3{\times}\mu_2$: the $\mathbb{Z}_3$ triality centre $\{1,\omega,\omega^2\}$ = the cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{\times}2\pi$, the sheet $\mu_2{=}\{1,-1\}$; $\rho{=}\omega^2{\cdot}(-1)$. **[E]** SAME FAMILY CLASS: $\rho^4{=}\rho^1{\cdot}\rho^3$ with $\rho^3{\in}\mu_2$, so the quark (ρ^1) and lepton (ρ^4) CP phases have the *same* image in $\mu_6/\mu_2{\cong}\mu_3$ and *opposite* sheet — the CP phase *is* the universal triality phase ω^2 (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. **[E]** ω is the μ_3 factor of the order-30 $(2,3,5)$ Milnor/Coxeter monodromy that builds E_8 (v232). Reduces Target D from 'one hexagonal unit + sheet' (v231) to 'the universal triality phase + sheet'; the physical CP derivation stays **[C][E][C]**
`v234_seam_holomorphy_selection.py`	The Seam-Holomorphy selection certificate: the entire structural residual is *one* condition with *three* equivalent faces, all forcing E_8. The condition: *the seam carries no nontrivial abelian sector.* **[E]** (i) AQFT — the boundary net is holomorphic (μ-index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$; (iii) rep theory — exactly one 1-dim irrep. **[E]** $\#$(mark-1 affine nodes) $=\lvert\Gamma^{\mathrm{ab}}\rvert = \lvert H_1(S^3/\Gamma)\rvert = \#$(1-dim irreps): $A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$ — only E_8 gives 1 ($2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). **[E]** holomorphic $c{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice $=E_8$. **[E]** the three faces (v83/v154, v232, v219) are the SAME E_8-selector, so Target A, G_{net}, the carrier P2 and the Kleinian seam are *one* gate. **[O]** the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is 'free bulk $\Rightarrow$ holomorphic boundary' **[E][O]**

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Script	What it machine-checks *(continued)*
`v235_seam_chern_simons.py`	The closing step in abelian Chern–Simons language: holomorphic ⇔ $\det K{=}1$ — a genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics. **[E]** a free gapped bosonic 2+1d bulk = an even integer K-matrix with #anyons $= \lvert\det K\rvert$, edge $c = \text{signature}(K)$, holomorphic ⇔ $\lvert\det K\rvert{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8, $c = \text{sig}{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$. **[E]** holomorphic ⇔ det 1 ⇔ E_8 = the Kitaev E_8 quantum-Hall state. **[E]** condensation: a Lagrangian glue of order $\sqrt{16}{=}\lvert\mu_4\rvert{=}4$ takes det 16→1 (a partial order-2 isotropic only → 4=D_8). **[E]** NEG: D_8 is free+bosonic but has 4 anyons; an odd K is fermionic — det =1 is the extra condition. **[O]** residual: 'the free RP seam condenses the order-$\lvert\mu_4\rvert$ Lagrangian glue' = the sheet/QGEO selection, now with holomorphy⇔ det 1 a theorem **[E][O]**
`v236_brieskorn_capstone.py`	The $(2,3,5)$ Brieskorn singularity $x^2{+}y^3{+}z^5$ is the *one* generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents *are* the atoms $(\lvert\mathbb{Z}_2\rvert, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$. **[E]** Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}1{\cdot}2{\cdot}4{=}8{=}\text{rank}\,E_8$ = the 8 in $c_3{=}1/(8\pi)$ (a fifth independent origin). **[E]** the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$, m the E_8 exponents (charpoly Φ_{30}, $\deg\varphi(30){=}8{=}\mu$) — so it *is* the order-30 E_8 Coxeter element (v55). **[E]** both clocks: the monodromy group $\langle h\rangle{=}\mathbb{Z}/30{=}\mathbb{Z}/2{\times}\mathbb{Z}/3{\times}\mathbb{Z}/5$ carries the μ_3 triality (CP, v233) as h^{10}; the Galois group $(\mathbb{Z}/30)^{\times}{=}\mathbb{Z}/2{\times}\mathbb{Z}/4$ carries the μ_4 clock (×7, v223). **[E]** Milnor lattice $= E_8$, link = Poincaré sphere. **[C]** one singularity generates atoms + 8 + E_8 + order-30 + both clocks + seam; a unification, not a closure **[E][C]**
`v237_seam_sre_closure.py`	The closing step as a *physical* condition: no topological ground-state degeneracy ⇔ $\det K{=}1$ ⇔ the seam bulk is short-range-entangled (SRE/invertible) — sharpening `GATE.HOLO.02`'s residual from definitional to falsifiable. **[E]** the K-matrix ground-state degeneracy on a genus-g surface is $\lvert\det K\rvert^g$ (carrier → 16, $D_8 \to 4$, $E_8 \to 1$ on the torus), so 'no degeneracy on any closed surface' ⇔ $\det K{=}1$. **[E]** among rank-8 even positive-definite K ($c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$) only E_8 has det 1 — the unique nontrivial bosonic SRE 2+1d phase (Kitaev E_8), edge holomorphic; D_8/carrier are LRE. **[E]** a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$, the torus does), so the residual is 'the seam unique-vacuum extends to the torus' = SRE. **[E]** NEG: an LRE bulk (D_8) has 4-fold torus degeneracy + non-holomorphic boundary. **[O]** sharpened residual: 'the free RP gapped seam is short-range-entangled (no topological order)' = $\det K{=}1$ = the Kitaev E_8 phase — a yes/no statement, sharper than v235/`QGEO.SYM.01`; not a closure **[E][O]**
`v238_modular_lindblad_dynamics.py`	`DYN.SEMIGROUP.01` — the static→dynamic flip: the seam's modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure *statically*; this reads the same objects *dynamically*. **[E]** the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the μ_4 clock is conserved ⇔ $[\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. **[E]** $L = \log T$ is a doubly-stochastic generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time, cf. v64). **[E]** keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap Δ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. **[E]** GKSL split (imaginary + real≤ 0 spectrum). **[O]** geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = `QGEO.SYM.01`, v198) — a target, not a closure.

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Script	What it machine-checks *(continued)*
`v239_kms_thermal_time.py`	`DYN.KMS.01` — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$, and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$. **[E]** KMS at $\beta = 1$: $\omega(A\,\sigma_i(B)) = \omega(BA)$ ($\omega(X) = \mathrm{Tr}(\rho X)$, $\rho = e^{-\Lambda_\Sigma}/Z$), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$). **[E]** the seam unit $\beta_{\mathrm{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). **[E]** one object with v238. **[O]** the geometric-boost identification = `QGEO.SYM.01` — a target, not a closure.
`v240_gns_os_reconstruction.py`	`DYN.GNS.01` — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data (M, ω), not inputs. **[E]** GNS: $\langle A, B\rangle = \mathrm{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ($\rho > 0$), $\dim n^2$, identity cyclic+separating. **[E]** modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving $*$-automorphism, generator $i[\log\rho, \cdot] = \Lambda_\Sigma$ (v238). **[E]** OS Hamiltonian $H_{\mathrm{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$, gap $= 6\log(3/2)$ = the v64 mass gap. **[E]** $H_{\mathrm{OS}} = -L$: minus the v238 dissipative generator (recovery semigroup = Euclidean time). **[O]** the geometric/operator-level step = `QGEO.SYM.01`/BW + the standard OS theorem under RP/gap — a target, not a closure.
`v241_dhr_sectors.py`	`DHR.SECTORS.01` — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, `FORM.GLUE.01`): on $\mathbb{Z}_4 \times \mathbb{Z}_4$, $q(x, y) = (5x^2{+}3y^2)/8 \bmod 1$, $\theta(a) = e^{2\pi i q(a)}$. **[E]** 16 sectors $= \mathbb{Z}_4 \times \mathbb{Z}_4$ ($= \det K_{\mathrm{carrier}} = \dim S^+$), fusion = abelian group law. **[E]** condensable $\Leftrightarrow \theta = 1 \Leftrightarrow$ isotropic $\Leftrightarrow 5x^2{+}3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0,0), (1,1), (1,3), (2,2), (3,1), (3,3)\}$. **[E]** tower = v235/v237: Lagrangian $\langle(1,1)\rangle$ (order 4, self-dual) $\to$ 1 sector $= (E_8)_1$; order-2 $\langle(2,2)\rangle \to 4 = D_8$; carrier $16 \to D_8\ 4 \to E_8\ 1$. **[E]** NEG: a non-isotropic anyon $(1,0)$ is confined. **[C]** sector↔SM-multiplet map is the interpretation; the MTC is exact.
`v242_spin_statistics_modular.py`	`DHR.MODULAR.01` — spin-statistics and the modular (S, T) data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$. Same MTC as v241 ($\mathbb{Z}_4{\times}\mathbb{Z}_4$, $q = (5x^2{+}3y^2)/8$, $\theta = e^{2\pi i q}$). **[E]** spin-statistics: $T = \mathrm{diag}(\theta_a) \to 6$ bosons ($\theta = 1$), 2 fermions ($\theta = -1$) $= \{(0,2), (2,0)\}$, 8 anyons. **[E]** modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$. **[E]** Verlinde $N^c_{ab} = \sum_x S_{ax}S_{bx}S^*_{cx}/S_{0x} = \delta(c, a{+}b)$ — fusion recovered from modular data. **[E]** Gauss–Milgram $(1/\sqrt{16})\sum\theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$ = the $(E_8)_1$ central charge (v156/v175). **[C]** the boson/fermion split is the kinematic skeleton; the sector↔multiplet map is the interpretation.
`v243_haag_ruelle_braiding.py`	`DHR.SMATRIX.01` — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a{+}b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$. **[E]** mass gap $\Delta = 6\log(3/2) > 0 \to$ HR in/out states. **[E]** 2-particle S = monodromy: $\lvert M\rvert = 1$, diagonal in conserved labels $\to$ no particle production, factorised (integrable) scattering. **[E]** Yang–Baxter: M is a bicharacter $\to$ multi-particle factorisation. **[E]** crossing+symmetry: $M(a, b) = M(b, a)$, $M(a, -b) = M(a, b)^*$. **[E]** holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$, trivial braiding (det 1/SRE). **[O]** the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).

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Script	What it machine-checks *(continued)*
`v244_spectral_action_contract.py`	`CONTRACT.QFT4D.01` — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net $\to$ 4d physics). A single Dirac operator whose $\mathrm{Tr}\, f(D/\Lambda)$ yields gravity AND matter. [E] carrier spinor = one generation: $SO(10)$ half-spinor $= \dim S^+ = 2^{g_{\mathrm{car}}-1} = 16$ $(15+\nu_R)$, $N_{\mathrm{fam}} = 3$. [E] SM gauge group inside the carrier: $SU(3){\times}SU(2){\times}U(1)$ (dim 12) $\subset SU(5)$ (24) $\subset SO(10){=}D_5$ (45); colour $SU(3) \subset A_3{=}SU(4)$ (Pati–Salam); $\mathrm{rank}(D_5){+}\mathrm{rank}(A_3) = 8 = \mathrm{rank}\, E_8$. [E] one Dirac operator $\to$ gravity ($a_2{\sim}R$, $a_4{\sim}R^2$, v36) + matter ($a_4 \sim F^2$ Yang–Mills $+\lvert D\phi\rvert^2$ Higgs, Gilkey). [X] hooks: $g_3{=}g_2{=}\sqrt{5/3}\, g_1$ + a Higgs-quartic relation. [O] `CONTRACT.QFT4D.01`: obligations (seam$\to$Dirac D; finite geometry = carrier; spectral action reproduces SM couplings) $\to$ the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.
`v245_spectral_matter_content.py`	`QFT4D.MATTER.01` — the computable core of `CONTRACT.QFT4D.01` (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the $SO(10)$ **16**. [E] **16** = one generation: $SO(10)\,\mathbf{16} = SU(5)(\mathbf{10}{+}\bar{\mathbf{5}}{+}\mathbf{1})$; SM dims 6+3+1+2+3+1 = 16 = $\dim S^+$, $N_{\mathrm{fam}} = 3$ (15 Weyl fermions $+\,\nu_R$). [E] electric charges $Q = T_3 + Y$ are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: $\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2U(1) = 0$, $[SU(3)]^2U(1) = 0$ (exact). [E] unification: $\sin^2\theta_W = (\sum T_3^2)/(\sum Q^2) = 3/8$, i.e. $g_3 = g_2 = \sqrt{5/3}\, g_1$. [E] Higgs = the $(1,2)_{1/2}$ inner fluctuation (one doublet). [O] residual: the seam$\to$Dirac realisation + measured-coupling reproduction stay the open contract.
`v246_unification_data_crosscheck.py`	`QFT4D.RGTEST.01` — the honest data cross-check for the spectral-action prediction (v244/v245): does $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ at Λ survive the *measured* SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, M_Z): $\alpha_i^{-1} = (59.01, 29.59, 8.47)$, SM 1-loop $b = (41/10, -19/6, -7)$ (v159). [E] the prediction: $\alpha_1{=}\alpha_2{=}\alpha_3$, $\sin^2\theta_W{=}3/8$ at Λ. [X] tension: pairwise crossings spread ~ 4 decades ($\sim 10^{13}$–10^{17} GeV), residual spread at $2{\times}10^{16}$ $\Delta\alpha^{-1} \sim 8.8$ — $g_1{=}g_2{=}g_3$ not realised by the plain SM. [X] $\sin^2\theta_W$ 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread ~ 3.2 at $1.7{\times}10^{14}$ GeV). [O] no rescue: SM gauge content + no new states (the E_8 spine is arithmetic, not a threshold tower) $\Rightarrow$ no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.
`v247_e8_branching_no126.py`	`PS.E8BRANCH.01` — the E_8 audit hull forbids the **126** and supplies exactly $\{1, 10, 16, 45\}$ for $SO(10)$. Carrier $D_5{\oplus}A_3 = SO(10){\times}SU(4) \subset E_8$ (via $SO(16)$); $\mathbf{248} = (45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4}) = 248$. [E] ranks 5+3=8; [E] dims = 248 ($SO(16)$: 120 adj +128 spinor); [E] $SO(10)$ content $\{1, 10, 16, 45\}$, the **126** ABSENT $\Rightarrow$ 126_H algebraically forbidden, 10+16+45 E_8-supplied not fitted; [E] caveat only one 45/one 15 $\to$ proton-marginal; [O] the field-content principle + NCG complement (v250).
`v248_ps_algebra.py`	`PS.ALGEBRA.01` — the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ and its exact SM matching. [E] unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$ (rank 5, dim 21); [E] $\mathbf{16} = (4,2,1){+}(\bar{4},1,2) = 8{+}8$; [E] $Y = T_{3R} + (B{-}L)/2$ gives the exact SM hypercharges and $Q = T_{3L} + Y$ the exact charges; [E] matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$. [O] A_F uniquely from seam+sheet+μ_4; [O] caveat $SU(4)_c \subset SO(10){=}D_5$ is not the carrier $A_3{=}SU(4)$ (family).
`v249_ps_unification.py`	`PS.RGTEST.01` — carrier-native Pati–Salam/$SO(10)$ two-step unification kill-test + negative controls (suite check of `experiments/gauge-unification`). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread ~ 8.8 @ $2{\times}10^{16}$); [C] $M_{PS}/M_s = 1.0$–1.7 (1-loop; 2-loop ~ 1.0–1.2) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match ($b_{2R}{>}0$; also E_8-forbidden v247); [X] NEG proton decay selects content (minimal $M_{GUT}{\sim}2{\times}10^{15}$ excluded; $+(15,1,1)$ safer). [O] the gauging fork, exact κ, the one-45 marginality.

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Script	**What it machine-checks** *(continued)*
`v250_dirac_triple_contract.py`	`CONTRACT.QFT4D.DIRAC.01` — the spectral-triple contract that would make 'carrier gauged' a theorem. Target (A, H, D, J, γ) almost-commutative, A_F = PS (v248), $H_F = 3{\times}16$, D_F from $(R, L, \varphi_0, \text{Majorana})$. [E] $H_F = 3{\times}16 = 48$; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] σ = scalaron hypothesis (rescues first-order $+\,\nu_R$ Majorana; $M_{PS}{=}\kappa M_s$) — caveat $\sigma \leftrightarrow \nu_R\nu_R$ not automatic; [O] inner fluctuations → 10+16+45 not 126. Reduces 4d QFT to one object + finite NCG checks.
`v251_first_order_sigma.py`	`PS.DIRAC.01` — the first-order condition is violated *exactly* by the Majorana/σ term (Chamseddine–Connes–van Suijlekom 'beyond first order' → Pati–Salam $+\,\sigma$), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: σ does NOT 'rescue' first order — one DROPS it and the inner fluctuations generate PS $+\,\sigma$. [E] order-zero; [E] real-even ($J^2{=}{+}1$, $\gamma^2{=}1$, $D\gamma{=}{-}\gamma D$, KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$, supported exactly on the $\nu_R\nu_R^c$ blocks (σ the unique violator). [C] σ = scalaron (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).
`v252_full_finite_triple.py`	`PS.DIRAC.02` — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. H_F = particle⊕antiparticle doubling of three $SO(10)$ **16**-plets (dim 96). [E] dim 96; D_F Hermitian+odd; reality $J^2{=}{+}1$, $JD{=}DJ$; grading $\gamma^2{=}1$, $[\gamma, a]{=}0$, $J\gamma{=}{-}\gamma J$; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order → PS+σ); spectrum = fermion masses + seesaw $m_{\text{light}}{=}m_D^2/M_R$. [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion.
`v253_kappa_scalaron_ps.py`	`PS.KAPPA.01` — pinning κ in $M_{PS} = \kappa\, M_s$, discharging residual 6(b) of v249. $M_s = c_3^{7/2}\bar{M} \approx 3.06 \times 10^{13}\,\text{GeV}$ (exact). [E] scalaron scale exact; [E] κ pinned by RG (scan PDG errors × the two E_8-allowed contents) to a tight $O(1)$ interval ~ 1.3 (1-loop $[1.0, 1.7]$, 2-loop $[1.0, 1.2]$) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, a ratio of heat-kernel traces over the 96-dim H_F (v252) — Λ cancels, κ scale-free and $O(1)$. [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments f_2/f_0 fixed.
`v254_inner_fluctuations.py`	`PS.NCG.FLUCT.01` — the inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not **126**) + the no-junk obligation + the σ quantum numbers. [E] $\Omega^1_{D_F}$ purely chirality-odd → finite fluctuations are scalars; [E] no junk / no **126**: SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$); [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam — σ is the CCvS beyond-first-order scalar; [E] σ is an SM singlet $(1, 1, 0)$ with $B{-}L = 2$; [C] σ = scalaron consistency.
`v255_spectral_action_expansion.py`	`PS.SPECACT.01` — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N{=}96$); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2\theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \to {\sim}125\,\text{GeV}$ with σ; [E] scalaron = R^2 spin-0 mode (gravitational $a_4 \sim f_0 N$); [C] $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ explicit; [O] exact decimal needs the cutoff moments fixed.

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Script	What it machine-checks *(continued)*
`v256_orientability_poincare.py`	`PS.NCG.ORIENT.01` — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\mathrm{cap}(p,q) = \mathrm{Tr}(\gamma\,\pi(p)\,J\pi(q)J^{-1})$ to be antisymmetric. **[E]** skew form forced by $J\gamma{=}-\gamma J$ + order-zero + reality (verified); **[E]** on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1,1,-1)$ — the structural obstruction; **[E]** orientability local condition $[\gamma_F, \pi(a)] = 0$, full orientability is a degree-4 cycle on $M \times F$; **[C]** the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; **[O]** a self-contained machine proof needs the complete canonical bimodule.
`v257_quasiorient_modpd.py`	`PS.NCG.ORIENT.02` — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\,\nu_R$) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. **[E]** strict orientability fails ($\gamma \notin \mathrm{span}\{\lambda(a)\rho(b)\}$); **[E]** quasi-orientability holds (left/right A-irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\mathrm{LR}}(\mathrm{even}, \mathrm{odd}) = \{0\}$); **[E]** strict Poincaré fails (KO-6 skew form, corank 1, the equal-L/R-neutrino case); **[C]** modified Poincaré duality (CCM) / Stephan's PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.
`v258_dirac_covariance_induction.py`	`PS.DIRAC.03` — D_F is the modular/covariance *induction* of the boundary seam state, not an independent posit; closes the 'posited Yukawa vs derived operator' gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance C (contraction, spec $\subset (0,1)$) with one-particle modular Hamiltonian $H = \log((1{-}C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1{+}e^{H})^{-1}$ (β=1, v239). **[E]** inversion identity $\log((1{-}C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); **[E]** $C_F = (1{+}e^{D/\mu})^{-1}$ a positive contraction; **[E]** chirality-odd ($D_F^{\mathrm{ind}}\gamma = -\gamma D_F^{\mathrm{ind}}$); **[E]** unitarily equivalent to v252 (inducing C_F from the 9 Yukawas returns the same spectrum); **[E]** Majorana = off-diagonal covariance (seesaw $m_{\mathrm{light}}{=}m_D^2/M_R$); **[C]** $C_F = P_F C_\Sigma P_F$, so D_F is the KK shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ and the v18 Yukawas are the readout of C_Σ.
`v259_modular_cutoff_kappa.py`	`PS.SPECACT.02` — the spectral-action cutoff *is* the seam KMS weight, fixing $f_2/f_0{=}1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at β=1 (v239) and $2\pi{=}1/(4c_3)$ (v58) fixes β, so the heat-kernel cutoff $f(u){=}e^{-u}$ is *derived.* **[E]** KMS cutoff moments $f_0{=}f(0){=}1$, $f_2{=}\int f{=}1$, $f_4{=}\int uf{=}1$, so $f_2/f_0{=}f_4/f_2{=}1$; **[E]** seam-fixed β=1 (no new parameter beyond c_3); **[E]** non-vacuous neg control: a Gaussian cutoff e^{-u^2} gives $f_2/f_0{=}\sqrt{\pi}/2 \neq 1$; **[E]** $\kappa{=}\sqrt{(f_2/f_0)c_{PS}/c_{\mathrm{grav}}}{=}\sqrt{c_{PS}/c_{\mathrm{grav}}}$, no cutoff function left; **[C]** the RG $\kappa{\sim}1.3$ pins $c_{PS}/c_{\mathrm{grav}}{\sim}1.7$ ($O(1)$); the **[O]** free scheme becomes a **[C]** finite computation over the built triple; **[C]** the same logic fixes $\lambda(\Lambda)$.

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Script	**What it machine-checks** *(continued)*
`v260_k3_kummer_unification.py`	`ARCH.K3.01` — one Kummer/K3 surface carries the *seam*, the *carrier-16* and the E_8 lattice as facets of one smooth object, the geometric unification of the v1 'glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$' step (the K3/Kummer proposal). The seam square torus $\tau = i$ (j=1728, v214) under the elliptic involution $z \mapsto -z$ squares to the Kummer K3. [E] the E_8 Cartan is even, det =1, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice $L_{K3} = U^3 \oplus E_8(-1)^2$ has rank 22=b_2, det =−1 (unimodular), is even, signature $(3, 19)$, with $E_8(-1)^2$ an orthogonal summand; [E] seam = $T^2/(z \mapsto -z)$: 4=$\lvert(\mathbb{Z}/2)^2\rvert$ fixed points, pillowcase $S^2(2, 2, 2, 2)$, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes $\lvert A[2]\rvert = 2^4 = 16 \to 16$ exceptional $\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$ (v197), 16=4×4; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, *not* literally $E_8(-1)^2$ (distinct sublattices of $H^2(K3)$); [C] two faces of E_8 in K3 (local du Val $\mathbb{C}^2/2I$, v232/v236; global H^2 summand); [C] unification (lattice [E], carrier↔node identification [C]).
`v261_modular_spectral_closure.py`	`QFT.MSC.01` — the *Modular Spectral Closure*: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into *one* relative object $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to *one* open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones $[B{:}A]$=4 (v89) = $\lvert\mu_4\rvert$=4 (v92) = seam marks 2χ=4 (v216) = pillowcase/Kummer fixed points $\lvert(\mathbb{Z}/2)^2\rvert$=4 (v260); [E] one carrier-16: $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5)$= Kummer nodes 2^4=16, $H_F = 2N_{\text{fam}} \cdot 16 = 96$; [E] one gap $\Delta = -\log((2/3)^6) = 6\log(3/2) > 0$ = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov $\log((1-C)C^{-1}) = H$, v258; KMS cutoff f_2/f_0=1, v259; K3 lattice signature $(3, 19)$/det −1/even, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate; [C] the master statement (Modular Spectral Universality) holds modulo `QGEO.SYM.01` — the honest sense in which the QFT layer is complete.
`v262_fqcd_mp_me.py`	`FR.MPME.02` — the F_{QCD} transfer pipeline for m_p/m_e, implemented as a real solver (the one F_{transfer} pipeline that was contract-only). *Not* a compiler power — a [C] standard-physics transfer reproduction. [E] $b_3 = -7 = -(11 - 2n_f/3)$ at n_f=6 (the only TFPT number in the run); [E] a two-loop $\alpha_s(M_Z)$=0.1179 run with thresholds gives $\alpha_s(2\,\text{GeV}) \sim 0.30$ (real ODE solve); [C] $\Lambda^{(3)}_{\overline{\text{MS}}} \sim 0.33\,\text{GeV}$ × the lattice bridge $C_p = m_p/\Lambda$ (external $O(1)$) over the closed m_e gives $m_p/m_e \sim 1824$ vs the measured 1836.15 within the ∼ 7% external budget; [X] firewall: m_p/m_e is not an admissible compiler integer ($SU(9)$ absent from $D_5 \oplus A_3$). Advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]).
`v263_mnu_from_df.py`	`FLAV.PMNS.02` — the light neutrino matrix M_ν as the type-I seesaw of D_F, advancing the solar-angle texture (v9) from a posited matrix to a seesaw *object* built from D_F's blocks. [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ symmetric + seesaw-suppressed (the light masses an output of D_F); [E] a $\mu\tau$-symmetric (m_D, M_R) gives a $\mu\tau$-symmetric M_ν (the v9 texture is in the seesaw image); [C] diagonalising gives θ_{23}=45° and $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$=0.30675 under the seam misalignment; [O] θ_{13}=0 in the leading texture (physical $\sin^2\theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}$ a separate readout), the CP magnitude δ_{PMNS}=240° is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

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Script	**What it machine-checks** *(continued)*
v264_raw_seam_marklocal.py	QGEO.ENERGY.03 — the raw-seam ⇒ mark-locality *reduction* (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit π each, sum $4\pi{=}2\pi\chi(S^2)$), curvature *only* at the 4 marks; [E] the curvature sourced at the μ_4 orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support *only* on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] M_f then commutes with the clock $\rho{=}\mathrm{diag}(i^n)$; [E] with $\|k\|$ already commuting (v198), $[\rho, \Lambda]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ on this metric; [O] the residual *is* the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the μ_4 orbit) breaks it. A reduction/sharpening, not a closure.
v265_qft4d_fork_freeze.py	QFT4D.FORK.01 — freeze the 4D-QFT fork as a *branch policy* (not an open ambiguity); writes qft4d_fork_freeze.json + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread ∼8.8 at $2{\times}10^{16}$, v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch (E_8 content $\{1, 10, 16, 45\}$ no **126** v247, $\sin^2\theta_W{=}3/8$ v248, κ $O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake τ_p window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.
v266_ps_threshold_proton.py	PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes ps_threshold_proton.json. Two-step PS solve + $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the carrier-native PS UV branch survives proton decay *only* with the E_8-allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal 16H $\to M_{GUT}{\sim}2.4{\times}10^{15}$, $+(15, 1, 1) \to M_{GUT}{\sim}5.9{\times}10^{15}$; [X] minimal KILLED ($\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K $2.4{\times}10^{34}$, even high-band); [C] $+(15, 1, 1)$ survives now ($\tau_p{\sim}1.6{\times}10^{35}$ yr); [X] *Hyper-K decisive* (τ_p at the $\sim 1.4{\times}10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.
v267_qgeo_rigidity_minimal_axiom.py	QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier μ_4 clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$) *forces* everything downstream; it does *not* close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of a); [E] cross-ratio 2 ⇔ $j{=}1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) ⇔ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2{=}x^3{-}x$; [E] downstream forced: square ⇒ the unique flat pillowcase (Troyanov, v214) ⇒ curvature at the 4 marks ⇒ $\mathbb{Z}_4$-Fourier DtN (v201/v264) ⇒ $[\rho, \Lambda]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ (v198); [E] neg controls (generic $j \neq 1728$, only $\mathbb{Z}_2$; hexagonal $j{=}0$ has $\mathbb{Z}_6$, order $6 \neq 4$); [C] arithmetic rigidity ($\{1, i, -1, -i\}$ over $\mathbb{Q}$, $j{=}1728$ CM by $\mathbb{Z}[i]$, Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like c=const). Closes the reduction (axiom⇒everything), not the axiom.
v268_theta13_carrier_trace.py	FLAV.TH13.01 — the reactor angle θ_{13} *exponent* is the carrier hypercharge trace. The value $\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \mathrm{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$), complement 1/6 the neutrino ratio; [C] value 0.02311 vs PDG ∼0.0222; [E] own channel — the $\mu\tau$-breaking $\varepsilon = 3\varphi_0^{\mathrm{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting 'θ_{13} from M_ν' route); [O] CP magnitude + absolute ν scale remain open.

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Script	**What it machine-checks *(continued)***
`v269_spert_paqft_skeleton.py`	`QFT4D.SPERT.01` — the perturbative 4D S-matrix ($S_{\rm pert}$) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ($S_{\rm top}$ 2d DHR braiding v243 / $S_{\rm pert}$ 4d EG of the spectral action / $S_{\rm phys}$ LSZ on the OS Wightman functions v240), *not* a nonperturbative closure. **[E]** three distinct layers (the 2d braiding $\|M\|$=1 is statistics, not 4d cross-sections); **[E]** every spectral-action a_4 operator has mass dimension ≤ 4 in 4D (power-counting renormalizable); **[E]** a finite counterterm basis (7 marginal +3 relevant, no infinite tower); **[C]** Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g){=}T\exp(i\int g\,L_{\rm int})$ by causal factorisation + finite local extension (no path integral, no UV divergences); **[C]** adiabatic limit/gap ($\Delta{=}6\log(3/2){>}0$) $\to S_{\rm phys}$ via LSZ; **[O]** perturbative only — the ambient measure `QG.AMB.01` stays open. A contract/skeleton, explicitly perturbative.
`v270_pmns_jarlskog_assembly.py`	`FLAV.PMNS.03` — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\rm PMNS}$ (the CP *strength*), assembled from the four TFPT channels ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\phi_0}{2}$, $\sin^2\theta_{23}{=}\frac{1}{2}$, $\sin^2\theta_{13}{=}\phi_0 e^{-5/6}$, $\delta{=}4\pi/3$) into one unitary object. **[E]** the three angle channels; **[E]** $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; **[C]** $J_{\rm PMNS} = \Im(U_{e1}U_{\mu 2}U^*_{e2}U^*_{\mu 1}) = -0.02965$ (formula and full matrix agree), $J_{\rm max} = 0.03424$ — a *derived* CP strength, the leptonic analogue of the CKM J (v88); **[C]** data: $J_{\rm max}$ vs NuFIT 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative sign reproduced; **[O]** δ is the μ_6/triality phase (assembled input, *not* from M_ν/D_F) and the absolute ν-mass scale (Σm_ν, M_R) stays open. Assembles the PMNS matrix + CP strength; does not derive δ from D_F nor fix the scale.
`v271_eg_oneloop_quartic.py`	`QFT4D.SPERT.02` — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\rm pert}$ skeleton from *exists* to an actual EG number (no path integral). **[E]** $T_1{=}{:}L_{\rm int}$: needs no extension, the first extension is T_2; **[E]** the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble (two propagators) has sd=4=d so the UV singular order ω=sd$-d$=0; **[E]** $\omega{=}0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega{+}d,d){=}1$, renormalizable, no infinite tower); **[E]** the loop factor $\Omega_3/(2(2\pi)^4){=}1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action a_4 geometric quartic; **[C]** the ϕ^4 one-loop $\beta_\lambda{=}3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s,t,u channels), EG initial condition $\lambda_\Phi({\rm UV}){=}1/(16\pi^2)$; **[O]** order-2/one-loop only — the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Turns the skeleton into a concrete one-loop number.
`v272_nu_mass_scale.py`	`FLAV.NUSCALE.01` — the absolute neutrino-mass scale, typed honestly (it does *not* fabricate a Σm_ν number, which would contradict the v184 firewall). **[E]** the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\rm EW})^2/M_R$ — one irreducible UV input like $v_{\rm geo}$ (v153), while the ratios (m_2/m_3, ordering, PMNS angles v270) are pinned; **[C]** the observed m_3=0.050 eV with the TFPT PS→GUT window $M_R \in [4.2{\times}10^{13}, 2.4{\times}10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; **[O]** $M_R(y{=}1){=}6.1{\times}10^{14}$ has $\log_{c_3}(M_R/M_{\rm Pl}){=}2.57$ (non-integer), not a TFPT power, so the absolute value stays open; **[C]** the normal-ordering floor Σm_ν=0.0586 eV is consistent with cosmology ($<$ 0.072–0.12 eV); **[X]** a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1{\sim}0$ spectrum. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio), not the absolute number.
`v273_eg_gauge_running.py`	`QFT4D.SPERT.03` — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). **[E]** the one-loop gauge 2-point has scaling degree sd=4=d so $\omega{=}0 \to$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); **[E]** the one-loop b_i from the explicit carrier/SM content: $b_3{=}-7$, $b_2{=}-19/6$, $b_1{=}41/10$ (the same 41 as the carrier algebra, v159/`CAR.SM.01`); **[C]** physical RG directions $b_3{<}0$ (asymptotic freedom), $b_1{>}0$ (U(1) Landau); **[O]** the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact b_i mirrored in Wolfram.

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Script	What it machine-checks *(continued)*
`v274_scale_overdetermination.py`	`SCALE.OVERDET.01` — the single mass anchor is *over-determined*, and the absence of a derivable absolute scale is a *theorem*. **[E]** one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\rm Pl}^d$ ($\rho_\Lambda/M_{\rm Pl}^4$, $M_{\rm scal}/M_{\rm Pl}{=}c_3^{7/2}$, the spectrum); **[C]** two independent routes to $M_{\rm Pl}$ agree: gravity $(8\pi G)^{-1/2}{=}2.4353{\times}10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\rm Pl}^4{=}(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}{=}2.24$ meV) $=2.438{\times}10^{18}$ GeV, to 0.11%; **[E]** the anchor is the gravitational unit (G), the one measured dimensionful input of all physics; **[E]** theorem not gap: the seam is a conformal $c{=}8$ CFT (v157/v158), scale-invariant by construction → no intrinsic scale; **[O]**/**[E]** scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.
`v275_qgamb_roadmap.py`	`QGAMB.ROADMAP.01` — the obligation roadmap for `QG.AMB.01` (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). **[E]** Tier A gap decoupling: $\Delta_{\rm eff}{=}6\log(3/2)-2{\cdot}248\,c_3^2{=}1.648{>}0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; **[E]** Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c{=}248/31{=}8$; v77/`GATE.METRIC.03`), reduced not closed (`REDTEAM.A.01`); **[C]** the perturbative layer is in hand (v269/v271/v273); **[O]** the nonperturbative ambient measure is *not* constructed — the one genuine structural frontier; **[X]** closes IFF the seam net $=(E_8)_1$ *and* the bulk reconstruction is proved. A roadmap, not a closure.
`v276_qgeo_flat_closes_commutator.py`	`QGEO.SYM.03` — the flat $\tau{=}i$ pillowcase premise *closes* the modular-commutator chain to all orders, collapsing the bedrock `QGEO.SYM.01` to *one* geometric postulate (it does *not* prove the premise). For a quasi-free seam state `QGEO.SYM.01` $=(\omega\circ\rho{=}\omega)\Leftrightarrow[\rho,C]{=}0\Leftrightarrow[\rho,H]{=}0$. **[E]** isometry⇒commutator: if $H{=}f(\Delta_{\rm flat})$ and the order-4 deck ρ $(z{\to}iz)$ is a flat-$\tau{=}i$ isometry then $[\rho,\Delta]{=}0\Rightarrow[\rho,H]{=}0$ *exactly* (all orders), verified on the discretised square torus ($\rho^4{=}I$, $[\rho,\Delta]{=}0$, $[\rho,\sqrt{-\Delta}]{=}0$; generic non-isometry fails); **[E]** this upgrades v198 (principal symbol) + v201 (subprincipal) to the full H (flatness removes the curvature terms); **[C]** $\tau{=}i$ is forced by order-4 ($j{=}1728$; hexagonal $j{=}0$ is order-6, excluded; v214/v267); **[O]** `QGEO.SYM.01` collapses to the single statement 'the raw seam is the flat $\tau{=}i$ pillowcase quasi-free state' — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form.
`v277_seam_calderon_e8_match.py`	`QGAMB.TIERB.01` — the seam-Calderón → $(E_8)_1$ matching certificate, the concrete Tier-B step of `QG.AMB.01` (v275): it computes the full $(E_8)_1$ matching target and pins the residual to *one* bit (it does *not* close it). **[E]** $c{=}8$ (Sugawara, $248/31{=}g_{\rm car}{+}N_{\rm fam}$), but $c{=}8$ alone does not pin the net; **[E]** the discriminator is holomorphy — $\det{\rm Cartan}(E_8){=}1$ (1 primary, holomorphic) vs the $c{=}8$ counterexample $\det{\rm Cartan}(D_8{=}SO(16)){=}4$ (4 primaries, non-holomorphic), so the single discriminator is $\lvert\det K\rvert{=}1$; **[E]** the $(E_8)_1$ character $E_4/\eta^8{=}j^{1/3}{=}q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248{=}\dim E_8$ weight-1 currents (240 roots); **[E]** equivalence chain holomorphic ⇔ det Cartan 1 ⇔ unique even unimodular rank-8 (v83) ⇔ homology-sphere link ⇔ $\lvert\det K\rvert{=}1$ (v235); **[C]** seam-side match via v276 (flat $\tau{=}i$) + v260 (Kummer/K3); **[O]** residual = the single holomorphy bit — reduced to one bit, *not* closed.

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Script	What it machine-checks *(continued)*
`v278_lsz_bridge_unitarity.py`	`QFT4D.SPERT.04` — the $S_{\rm pert} \to S_{\rm phys}$ LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix ($S_{\rm phys}$ = LSZ on the OS Wightman functions, v240). **[E]** LSZ bridge: $S_{\rm phys}$ = amputate+on-shell-residue of the connected time-ordered products, and the EG T-products of $S_{\rm pert}$ *are* those correlators (OS Wick-rotation, v240) $\Rightarrow S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$; **[C]** the gap $\Delta{=}6\log(3/2){>}0$ gives Haag–Ruelle asymptotic states so LSZ is valid; **[E]** *one-loop unitarity* (optical theorem): the bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2 \operatorname{Im} M = \sum\int d\Pi\lvert M\rvert^2$ (cutting rule) holds $\Rightarrow$ matter+gauge unitary; **[E]** threshold: $\operatorname{Im} B{=}0$ below $s{=}4m^2$, turning on at the physical two-particle threshold; **[O]** gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary $\to$`QG.AMB.01`; **[C]** finite field-strength Z from the EG scheme.
`v279_qgeo_obligation_lemma.py`	`QGEO.OBLIG.01` — the `QGEO.SYM.01` bedrock written as *one* precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does *not* close it). Lemma: given the premise 'the raw seam carries the flat $\tau{=}i$ pillowcase metric $g_{\rm flat}$ (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{\rm flat}}$', then $\omega_\Sigma \circ \rho = \omega_\Sigma \to$ mark-locality $\to E_8$. **[E]** the reduction DAG from 'free RP seam' has 8 closed nodes and *exactly one* open leaf ('the raw seam state is the flat state'); **[E]** the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders, Lean `FORM.QGEO.03`), so the obligation is state-identification not geometric ambiguity; **[C]** two proof routes (RP-state uniqueness on the flat orbifold, or Troyanov + OS); **[E]** the all-orders step is a Lean theorem (`SeamDeckClosure.flat_all_orders_clock`, standard axioms, no `sorry`); **[O]** one constructive-QFT statement closes `QGEO.SYM.01`.
`v280_pillowcase_steklov.py`	`QGEO.STEKLOV.01` — a direct numerical experiment on the flat $\tau{=}i$ pillowcase orbifold (the seam geometry): the self-investigable test of the `QGEO.SYM.01` obligation (does *not* prove the premise). **[E]** pillowcase = the Z2-even sector of the flat torus, 4 cone points (2-torsion, deficit π each, sum $4\pi{=}2\pi\cdot 2$); **[E]** the order-4 deck ρ ($z{\to}iz$) descends ($[\rho, Z_2]{=}[\rho, P_{\rm even}]{=}0$) and permutes the 4 cone points as 2 fixed + 1 swapped pair; **[E]** the geometric chain is exact: $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (the v276/v279 all-orders closure on the real orbifold); **[C]** the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); **[O]** confirms the conclusion given flat geometry, the premise stays open.
`v281_holomorphy_modular_data.py`	`QGAMB.MODULAR.01` — the modular-data / anyon-condensation layer of the `QG.AMB.01` Tier-B holomorphy bit. **[E]** #anyons = $\lvert\det \mathrm{Gram}\rvert$: $E_8{\to}1$ (holomorphic), $D_8{=}SO(16){\to}4$, carrier $D_5{\oplus}A_3{\to}16$; **[E]** the anyon-condensation tower $D_5{\oplus}A_3$ (16) $\to D_8$ (4) $\to E_8$ (1), each step condensing a Lagrangian Z_2 boson, the Kitaev E_8 endpoint (v92/v235); **[E]** Gauss–Milgram recovers $c{=}8$; **[E]** E_8 is the unique holomorphic $c{=}8$ (det 1) — $SO(16)$ has the same c but 4 anyons, so holomorphy is THE discriminator (v277); **[O]** the open bit: does the seam-Calderón net condense to $\det K{=}1$? Reduced to one bit.
`v282_e8_tau_i_unification.py`	`QGAMB.UNIFY.01` — the E_8-at-$\tau{=}i$ unification: the candidate *simplification* reducing the two open obligations to *one* object (the raw seam is the $(E_8)_1$ holomorphic CFT at $\tau{=}i$). $\chi_{E_8}(\tau){=}\Theta_{E_8}/\eta^8{=}j^{1/3}$. **[E]** $\tau{=}i$ is the order-4 CM point ($j{=}1728$, the `QGEO.SYM.01` flat geometry); **[E]** $\chi_{E_8}{=}j^{1/3}$ ($\chi^3{=}q\,j$ as q-series), so $\chi_{E_8}(i){=}1728^{1/3}{=}12$ (the $(E_8)_1$ character at the SAME $\tau{=}i$, the `QG.AMB.01` holomorphy); **[E]** one object, two faces; **[C]** obligation count 2→1: prove 'the raw seam is $(E_8)_1$ at $\tau{=}i$' and *both* follow; **[O]** the unified premise stays open but is now one object.

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Script	What it machine-checks *(continued)*
`v283_live_killtest_scorecard.py`	`PRED.LIVE.01` — a live kill-test scorecard of the recent round's computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2\theta_{12}$=0.30675 (+0.31σ), $\sin^2\theta_{13}$=0.02311 (+1.45σ), θ_{23} maximal; [C] leptonic CP J_{PMNS}=−0.0297, δ=240° vs $\sim$ 232(30); [X] Σm_ν NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay τ_p=1.55×10^{35} yr for $+(15,1,1)$ (Hyper-K decisive); [E] closed readouts α^{-1}, Ω_b, β; all rows consistent $<2\sigma$, two sharp near-term kill tests. A consolidation, not new physics.
`v284_route_i_rp_uniqueness.py`	`QGEO.ROUTEI.01` — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does *not* prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $=\sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip $\text{DtN}(k)/k\to 1$, principal symbol $\lvert\xi\rvert$ with no curvature term), L4 covariance $C=(1+e^H)^{-1}$ from the DtN (v258), L5 μ_4-invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi\Rightarrow\chi_{\text{orb}}=0\Rightarrow$ the unique constant-curvature representative is flat (τ=i), so L_{open} reduces to 'the raw seam is the constant-curvature geometric state'; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step).
`v285_route_ii_seam_condensation.py`	`QGAMB.ROUTEII.01` — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma *coincides* with Route (i)'s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow\lvert\det K\rvert$=1 $\Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic c=8 $\Leftrightarrow(E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower 16→4→1 = the unique μ_4-Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share *one* open lemma — both say 'the raw seam is the $(E_8)_1$-at-τ=i object' (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at τ=i. M_{open} = `QGEO.SYM.01`.
`v286_seam_equivalence_contract.py`	`SEAM.EQUIV.01` — the Seam Equivalence contract: concentrates the whole open structure on *one* named arrow (does *not* prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the τ=i pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] **import firewall**: no Route-i module imports any Route-ii module or vice versa, so the routes converge only through `SEAM.EQUIV.01` (kills the 'E_8 into the geometry and back' circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary'.
`v287_free_rp_bulk_to_holomorphic_boundary.py`	`SEAM.EQUIV.A01` — Route A (AQFT) theorem-dependency checker for the open Gral arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary net': types the 5-lemma chain and isolates the *one* missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality ⇒ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk ⇒ single-sector boundary (μ-index 1) — the one missing standard theorem (the AQFT form of $\det K$=1); [E] L4 μ=1+c=8 ⇒ holomorphic (v234); [E] L5 holomorphic c=8 ⇒ $(E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces `SEAM.EQUIV.01` to one cited theorem.

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Script	What it machine-checks *(continued)*
`v288_full_l2_subprincipal_z4.py`	`SEAM.EQUIV.B01` — Route B (DtN): the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the 'probably provable' step and lifting v201/v284 from the finite block to L^2. **[E]** character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod 4$; **[E]** full-L^2 block-diagonality: $\Lambda = \|n\| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$, $\\|[\rho,\Lambda]\\| \sim 2{\times}10^{-15}$ on $-50..50$; **[E]** negative control: a generic curvature gives $\\|[\rho,\Lambda]\\| = 4.44 > 0$; **[E]** Lean consistency (`geom_sum_fourth_root` + `markLocal_blockDiagonal`); **[O]** the sharper residual — why is the raw seam sub-principal term mark-local?
`v289_marklocal_raw.py`	`MARKLOCAL.RAW.01` — decompose the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolate the single open analytic lemma. **[C]** L1 conic parametrix: conic Steklov DtN $= \|D_\theta\| + M_f +$ cone sources (b-calculus/Melrose, imported); **[O]** L2 **Flat-Away** — the one open analytic lemma: RP+gap+4 marks ⇒ curvature vanishes away from the marks (the flat-pillowcase realisation); **[E]** L3 orbit-source: the 4 marks form one μ_4 orbit (v195/v216); **[E]** L4 Fourier-support: μ_4-orbit curvature ⇒ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); **[E]** L5 full-operator: ⇒ $[\rho,\Lambda]{=}0$ on full L^2 (v288). 4/5 discharged; Route B reduces to exactly L2.
`v290_z4_smooth_curvature_adversary.py`	`SEAM.ADVERSARY.01` — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$: **[E]** it is $\mathbb{Z}_4$-symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$) but *not* mark-local ($f(\pi/4) = -\varepsilon \neq 0$); **[E]** it *passes* the v288 commutator $\\|[\rho,\Lambda]\\| = 0$ (same as mark-local) — so $[\rho,\Lambda]{=}0$ is necessary but *not* sufficient; **[E]** but it shifts the DtN spectrum off the flat ladder ($\max\|\Delta\lambda\| = 0.155$) and the heat trace ($\Delta = 0.019$), so the KMS weight and $(E_8)_1$ character shift; **[O]** a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input.
`v291_flataway_contract.py`	`FLATAWAY.RP.01` — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma **[O]**: given RawRPSeam with gap, chirality and the four μ_4 marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. **[E]** *spectral-rigidity*: scanning $f = \varepsilon\cos(4\theta)$, the spectrum deviation from the flat ladder is 0 only at $\varepsilon{=}0$ — flat is the unique spectral match; **[E]** *heat-kernel*: the heat-trace deviation is 0 only at $\varepsilon{=}0$; **[C]** *RP-energy*: RP+gap is a Dirichlet-energy minimisation in the Troyanov conformal class ⇒ constant-curvature minimiser (Troyanov; cf. v284). **[O]** three independent routes to one lemma; closing any one closes Route B.
`v292_flataway_heat_reduction.py`	`FLATAWAY.HEAT.01` — the Heat-Kernel route made precise: reduces Flat-Away to *one* named spectral invariant. **[E]** the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$; positive on all combinations), so $\Delta\mathrm{Tr} = 0 \Leftrightarrow f = 0$; leading invariant $\\|f\\|_{L^2}$ (Parseval). **[O]** reduction: Flat-Away ⇐ 'the seam fixes the a_2 heat coefficient to its flat value' ⇒ $\int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.
`v293_flataway_spectral_hessian.py`	`FLATAWAY.SPEC.01` — the Spectral-Rigidity route upgraded from one direction (v291) to the *full* smooth-$\mathbb{Z}_4$ space. **[E]** splitting mechanism: the degenerate flat Steklov level $\|n\|$ is split by mode $4k$ at $\|n\|{=}2k$ by $\pm\varepsilon/2$; **[E]** $S = \sum(\Delta\lambda)^2 \sim 0.50\varepsilon^2$; **[E]** the Hessian over modes $\{4,8,12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$, so the flat seam is a *strict* isolated minimum over the whole $\mathbb{Z}_4$ space; **[E]** a non-$\mathbb{Z}_4$ mode breaks the commutator outright. **[O]** pinning the seam's Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation ⇒ Flat-Away.

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Script	What it machine-checks *(continued)*
`v294_flataway_rp_energy.py`	`FLATAWAY.ENERGY.01` — the RP-energy / Troyanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: S^2 with 4 cone angles π, $\sum(2\pi-\theta)=4\pi=2\pi\chi$; [C] Troyanov/Euclidean-cone uniqueness (Troyanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f]=\int f^2$, Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: 'RP+gap realises the energy-minimiser', converging with v292/v293 on one selection principle; closing any one closes Route B.
`v295_flataway_a2_exact.py`	`FLATAWAY.A2.01` — the *exact* analytic a_2 coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \mathrm{Tr}\, e^{-t\Lambda(\varepsilon)}\|_0 = -t\hat{f}(0)\sum e^{-t\|n\|}=0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x)=e^{-tx}$ strictly convex $\Rightarrow \Lambda\mapsto \mathrm{Tr}\, e^{-t\Lambda}$ convex (Klein/Peierls), so ε=0 is the global minimum and $\Delta\mathrm{Tr}\geq 0$ for all ε, $=0$ iff $f=0$; [E] exact a_2 kernel via the divided difference of e^{-tx} (Duhamel), matched to numerics to rel.err $<10^{-6}$; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2){-}1)\geq 0$; [O] reduction now analytic: Flat-Away $\Leftarrow$ 'the seam fixes a_2'. Lean-mirrored (`FORM.FLATAWAY.01`).
`v296_flataway_a2_closed.py`	`FLATAWAY.A2.CLOSED.01` — the exact a_2 coefficient in *closed form*. [E] the operator tails telescope geometrically ($C_k+D_k e^{-4kt}=4kt(1-e^{-4kt})$), giving $W_k^{\mathrm{tail}}=t(1-e^{-4kt})/(8k(1-e^{-t}))$; [E] a finite $(4k{-}1)$-term middle (incl. the degenerate diagonal $\Phi(2k,2k)=(t^2/2)e^{-2kt}$); [E] closed form $W_k(t)=\frac{t(1-e^{-4kt})}{8k(1-e^{-t})}+\frac{1}{2}\sum_{m=1}^{4k-1}\Phi(m,4k{-}m)$ $(W_1(t)=t(2te^t+e^{3t}+3e^{2t}+e^t-1)e^{-3t}/8)$, reproducing v295 to machine precision; [E] manifestly positive; [E] small-t: $W_k=t/2+O(t^3)$ (mode-independent leading 1/2, the t^2 term cancels).
`v297_route_a_literature_stack.py`	`SEAM.EQUIV.A.LIT.01` — Route A's 'one standard import' as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic c=8 net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into 'invertible bulk $\Rightarrow (E_8)_1$', all legs established; [O] the one open hypothesis is the input of LIT-A, 'the raw quasi-free seam bulk is invertible (SRE)' = Route B's Flat-Away. [E] non-circular with Route B (the v286 firewall holds).
`v298_e8_network_attractor.py`	*E8 as a network attractor* — a computational-substrate (Wolfram-program) reading of the $(2,3,5)$/McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the E8 Kac marks $(1,2,3,4,5,6,4,2,3)$ are the *unique* attractor of the minimal lazy graph update $m\leftarrow(A+2I)/4\cdot m$ on the affine-$E_8/(2,3,5)$ McKay graph (converges from any positive start; $A\,m=2m$, top eigenvalue 2). [E] the spherical McKay tower terminates at E_8 ($2T\to E_6$, $2O\to E_7$, $2I\to E_8$, $h=30$); [E] the cascade is nested-subgraph coarse-graining $(240\to126\to72\to40\to20)$. [C] Wolfram reading: E_8 as the universal attractor of a minimal $(2,3,5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph *rewrite* whose attractor is the full TFPT structure stays open; $\varphi_0=(4/3)c_3=1/(6\pi)$ is the analytic seed, not an edge fraction.

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

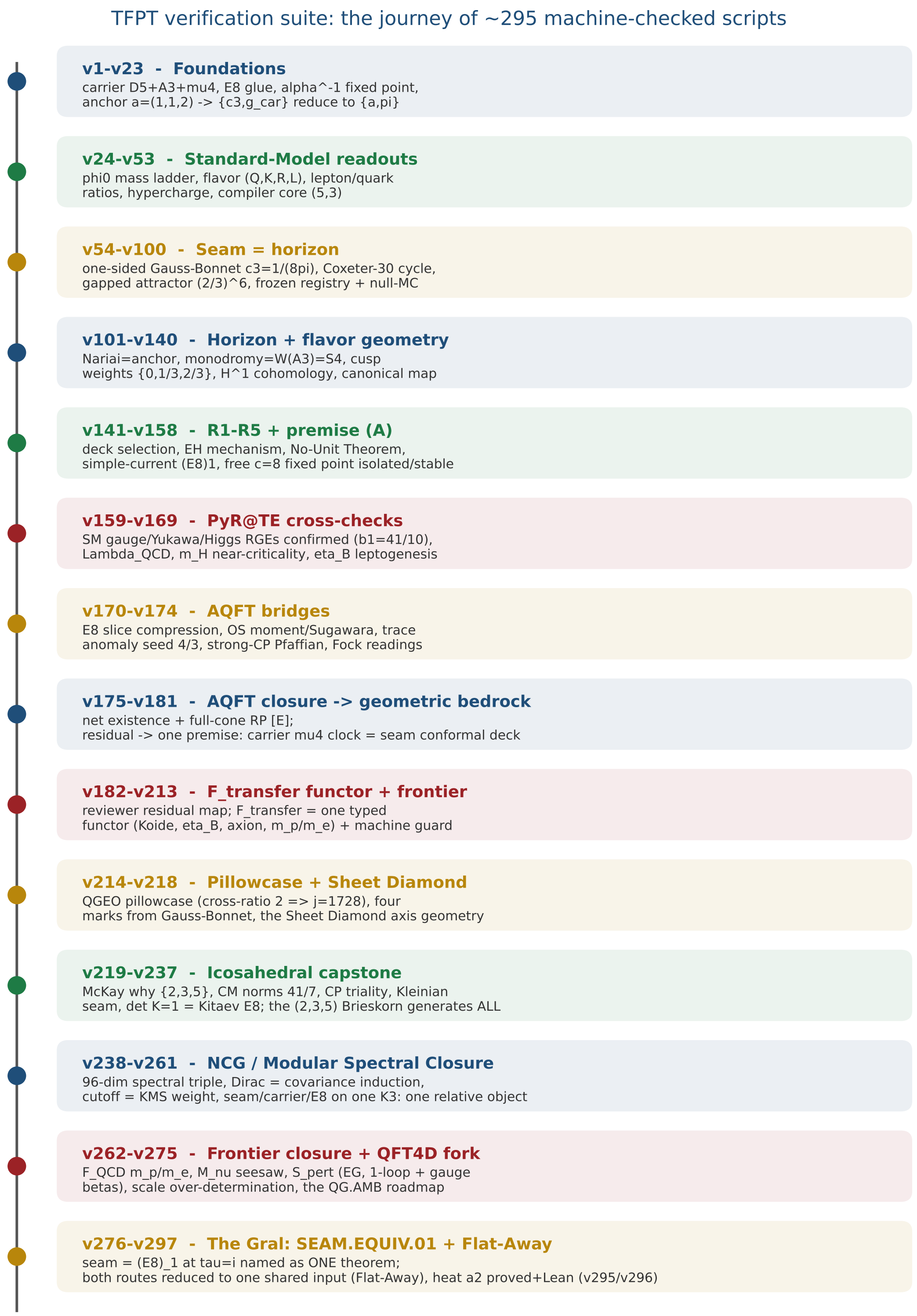

Figure 1: The development timeline of the verification suite — the eleven phases of the journey and what each did, mathematically and physically: from the foundations (carrier, E_8 glue, α^{-1}, the anchor $a = (1, 1, 2)$ to which $\{c_3, g_{\rm car}\}$ reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the $F_{\rm transfer}$ functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, to the *icosahedral capstone* — where the single $(2, 3, 5)$ Brieskorn singularity is shown to generate the whole skeleton and the closing step becomes the physical question "is the seam short-range-entangled ($\det K = 1$)?". The master script→check map follows.

Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers `vN` refer to `verification/vN_*.py`.

2026-06-18 (E8 as a network attractor — the Wolfram-program reading, v298)

- **`v298_e8_network_attractor.py`** is an honest audit of the "TFPT as a hypergraph/rewrite system" idea, on the v219 McKay bedrock. [E] the affine-E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4\, m$ on the $(2, 3, 5)$ McKay graph (converges from any positive start; $A\,m = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading "E_8 is the universal attractor of a minimal $(2, 3, 5)$ network" holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, `AUDIT OK`.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **`v296_flataway_a2_closed.py` (`FLATAWAY.A2.CLOSED.01`)** evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$) to $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k-1)$-term middle; e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small-t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.
- **`v297_route_a_literature_stack.py` (`SEAM.EQUIV.A.LIT.01`)** writes Route A's one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk ⇒ $(E_8)_1$", all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B's Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, `AUDIT OK`.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / `FORM.FLATAWAY.01`)

- **`v295_flataway_a2_exact.py` (`FLATAWAY.A2.01`)** upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\operatorname{Tr} e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since $\Lambda \mapsto \operatorname{Tr} e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon{=}0$ is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.
- **Lean `FORM.FLATAWAY.01`** (`SeamDeckClosure.lean` Part 5): the formal $\Delta{=}0 \Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route's positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. [E] build green, `AUDIT OK`.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **`v292_flataway_heat_reduction.py` (`FLATAWAY.HEAT.01`)**: the heat route, made precise. The heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact "the seam fixes the a_2 heat coefficient".

- **`v293_flataway_spectral_hessian.py` (`FLATAWAY.SPEC.01`)**: the spectral route, upgraded from one direction to the *full* smooth-$\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the "only one direction" objection is answered.
- **`v294_flataway_rp_energy.py` (`FLATAWAY.ENERGY.01`)**: the variational route. With the rigid mark divisor (4 cone angles π, Gauss-Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to "RP+gap realises the energy-minimiser". The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, `AUDIT OK`.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **`v290_z4_smooth_curvature_adversary.py` (`SEAM.ADVERSARY.01`)** is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon\cos(4\theta)$ [E] passes the v288 commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet [E] shifts the DtN spectrum ($\max|\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]{=}0$ — which is exactly why Flat-Away cannot be read off the commutator.
- **`v291_flataway_contract.py` (`FLATAWAY.RP.01`)** states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at $\varepsilon{=}0$) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, `AUDIT OK`.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **`v289_marklocal_raw.py` (`MARKLOCAL.RAW.01`)** decomposes the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks $\Rightarrow$ curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda]{=}0$ (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as 'full-L^2 QGEO proved'.
- **v287 literature matrix**: L3 ('invertible bulk $\Rightarrow$ single-sector') is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Müger/Kawahigashi–Longo–Müger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, `AUDIT OK`.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **`v286_seam_equivalence_contract.py` (`SEAM.EQUIV.01`)** names the one open theorem — *the raw RP seam state is the holomorphic $(E_8)_1$ net at $\tau{=}i$* — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an

import firewall proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no 'E_8 into the geometry and back' circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.

- **`v287_free_rp_bulk_to_holomorphic_boundary.py` (`SEAM.EQUIV.A01`)**: Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible $\Rightarrow \mu{=}1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary* (the AQFT form of $\det K{=}1$). 4/5 discharged.
- **`v288_full_l2_subprincipal_z4.py` (`SEAM.EQUIV.B01`)**: Route B (DtN) — *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \bmod 4$, so $\Lambda = |n| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$ on the full operator, $\|[\rho,\Lambda]\| \sim 2{\times}10^{-15}$; generic curvature $\to 4.44$), lifting v201/v284 to L^2. The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? **[E]** build green, `AUDIT OK`.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **`v284_route_i_rp_uniqueness.py` (`QGEO.ROUTEI.01`)** turns Route (i) (RP-state uniqueness) into a 6-lemma chain: **[E]** 5/6 lemmas are already discharged — L1 RP→DtN (v194), L2 marks→pillowcase (v195/v216/v214), L3 flat Steklov = $\sqrt{-\Delta_{\mathrm{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4-invariance (v276/Lean/v280); **[C]** Troyanov ($\chi_{\mathrm{orb}}{=}0 \Rightarrow$ unique flat $\tau{=}i$) reduces the residual to one lemma. **[O]** the one open lemma: *the raw seam is the constant-curvature geometric state.*
- **`v285_route_ii_seam_condensation.py` (`QGAMB.ROUTEII.01`)** turns Route (ii) (seam-net condensation) into a 4-lemma chain: **[E]** 3/4 discharged (no abelian sector $\Leftrightarrow \det K{=}1 \Leftrightarrow$ holomorphic $\Leftrightarrow E_8$; the tower 16→4→1), and **its open lemma coincides with Route (i)'s** — both are "the raw seam is the $(E_8)_1$-at-$\tau{=}i$ object" (v282), so proving *either* route closes *both*. **[O]** the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$. Build green, `AUDIT OK`.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at-$\tau{=}i$ unification, live scorecard, v280–v283)

- **`v280_pillowcase_steklov.py` (`QGEO.STEKLOV.01`)** runs the QGEO chain on the actual flat $\tau{=}i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ is numerically exact — route-(i) evidence for `QGEO.OBLIG.01`; the premise stays **[O]**.
- **`v281_holomorphy_modular_data.py` (`QGAMB.MODULAR.01`)**: #anyons = $|\det \mathrm{Gram}|$ (E8→1, SO(16)→4), the anyon-condensation tower $16 \to 4 \to 1$, Gauss–Milgram $\to c{=}8$; holomorphy ($\det K{=}1$) is the single Tier-B discriminator.
- **`v282_e8_tau_i_unification.py` (`QGAMB.UNIFY.01`)**: the candidate *simplification* — $\chi_{E_8}{=}j^{1/3}$ gives $\chi_{E_8}(i){=}1728^{1/3}{=}12$, so $\tau{=}i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau{=}i$), **obligation count** $2 \to 1$.
- **`v283_live_killtest_scorecard.py` (`PRED.LIVE.01`)**: the recent round's computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay $1.55{\times}10^{35}$ yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, `AUDIT OK`.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/`FORM.QGEO.03`)

- **`v279_qgeo_obligation_lemma.py` (`QGEO.OBLIG.01`) writes the bedrock as ONE precise constructive-QFT lemma.** Given the premise *the raw seam carries the flat $\tau{=}i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma{=}\sqrt{-\Delta_{\rm flat}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma$ → mark-locality → E_8. [E] the reduction DAG from 'free RP seam' has exactly *one* open leaf — 'the raw seam state is the flat state'; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Troyanov + OS).

- **`FORM.QGEO.03`: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\rm flat})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho{=}\mathrm{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full `lake build` green, 3356 jobs). The premise (the seam *is* flat) stays the [O] `QGEO.SYM.01` input. Build green, `AUDIT OK`.

2026-06-18 (4D QFT: the $S_{\rm pert} \to S_{\rm phys}$ LSZ bridge + one-loop unitarity, v278)

- **`v278_lsz_bridge_unitarity.py` (`QFT4D.SPERT.04`) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity.** [E] LSZ bridge: the EG T-products of $S_{\rm pert}$ (v269) *are* the OS Wick-rotated correlators (v240), so amputate+on-shell-residue gives $S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$; [C] the gap $\Delta{=}6\log(3/2){>}0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\mathrm{Im}\,B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\,\mathrm{Im}\,M = \sum\int d\Pi|M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector's Stelle ghost (v269/`rt_F`) is non-unitary → `QG.AMB.01`. Exact core mirrored in Wolfram (273/273). Build green, `AUDIT OK`.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón → $(E_8)_1$ match, v277)

- **`v277_seam_calderon_e8_match.py` (`QGAMB.TIERB.01`) reduces Tier-B to ONE bit.** [E] the full $(E_8)_1$ matching target is computed and matches: $c{=}8$, $\det\mathrm{Cartan}(E_8){=}1$, the holomorphic character $E_4/\eta^8{=}j^{1/3}{=}q^{-1/3}(1{+}248q{+}\dots)$ with a single primary and exactly $248{=}\dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c{=}8$ counterexample $(D_8){=}SO(16)_1$ has $\det\mathrm{Cartan}{=}4$ (non-holomorphic), so the one condition is $|\det K|{=}1$. [C] the seam side matches via v276 (flat $\tau{=}i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, `AUDIT OK`.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **`v276_qgeo_flat_closes_commutator.py` (`QGEO.SYM.03`) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]{=}0$. [E] if $H = f(\Delta_{\rm flat})$ and the order-4 deck ρ ($z{\mapsto}iz$) is a flat-$\tau{=}i$ isometry, then $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0$ *exactly*, to all orders (verified on the discretised square torus: $\rho^4{=}I$, $[\rho,\Delta]{=}0$, $[\rho,\sqrt{-\Delta}]{=}0$; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H. [C] $\tau{=}i$ is forced by order-4 ($j{=}1728$; hexagonal $j{=}0$ is order-6, excluded). [O] `QGEO.SYM.01` therefore collapses to the single statement "the raw seam IS the flat $\tau{=}i$

pillowcase quasi-free state" — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, `AUDIT OK`.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, `rt_F`)

- **`redteam/rt_F_qft4d.py` (`REDTEAM.F.01`) stress-tests the just-published round** (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν-scale, the over-determination, the `QG.AMB.01` roadmap). SURVIVES (NARROWED): two attacks land and are folded in. **(1)** the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2+M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the nonperturbative `QG.AMB.01` is the real frontier (caveat added to v269). **(2)** the 0.11% anchor over-determination is conditional on the Λ-branch readout (`LAMBDA.BRANCH.01`, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version `5.2 (rev 191)` of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI `10.5281/zenodo.20743347`. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton v269, the concrete one-loop quartic v271 and gauge β-coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute ν-mass-scale typing v272, the over-determination of the single mass anchor v274, and the `QG.AMB.01` obligation roadmap v275. Suite 273 modules (highest ID `v275`; v186/v226 skipped), Wolfram 271/271, `run_all.py` green, `AUDIT OK`, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν-mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **`v272_nu_mass_scale.py` (`FLAV.NUSCALE.01`) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ (one UV input, like v_{geo}, v153); [C] the observed m_3=0.050 eV with the TFPT PS→GUT window $M_R \in [4.2\times10^{13}, 2.4\times10^{15}]$ GeV (v249) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; [X] the NO floor Σm_ν=0.0586 eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).

- **`v273_eg_gauge_running.py` (`QFT4D.SPERT.03`) derives the SM one-loop b_i via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.

- **`v274_scale_overdetermination.py` (`SCALE.OVERDET.01`) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to $\bar{M}_{\text{Pl}}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438\times10^{18}$ GeV, to 0.11%; the seam being a conformal c=8 CFT makes "no derivable absolute scale" a theorem, not a gap. Sharpens v153/v78.

- **`v275_qgamb_roadmap.py` (`QGAMB.ROADMAP.01`) gives the open gate `QG.AMB.01` a module.** [E] Tier A gap decoupling ($\Delta_{\text{eff}} = 1.648 > 0$, v36/v76); [E]/[C] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.

- Suite 271/271 Wolfram, `run_all` green, `AUDIT OK`, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **`v271_eg_oneloop_quartic.py` (`QFT4D.SPERT.02`) takes the v269 $S_{\rm pert}$ skeleton from "exists" to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product T_2; the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble has sd=4=d, so ω=sd$-d$=0 $\Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d, d)$=1, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry 3 = s, t, u channels), EG initial condition $\lambda_\Phi(\mathrm{UV}) = 1/(16\pi^2)$. [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Exact core mirrored in Wolfram (270/270). [E] build green, `AUDIT OK`.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **`v270_pmns_jarlskog_assembly.py` (`FLAV.PMNS.03`) assembles the four TFPT channels** ($\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$, $\sin^2\theta_{23}=\frac{1}{2}$, $\sin^2\theta_{13}=\varphi_0^{\rm ret}e^{-5/6}$, $\delta=4\pi/3$) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength.** [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\rm PMNS} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ ($J_{\max} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: $J_{\max}$ vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative CP-violating sign reproduced. [O] δ is the μ_6/triality phase (assembled, not from M_ν/D_F) and the absolute ν-mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, `AUDIT OK`.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **`v269_spert_paqft_skeleton.py` (`QFT4D.SPERT.01`) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: $S_{\rm top}$ (2d DHR braiding $|M|$=1, statistics, v243), $S_{\rm pert}$ (4d Epstein–Glaser/BV S-matrix of the spectral action), $S_{\rm phys}$ (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators dim ≤ 4; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit ($S_{\rm pert} \to S_{\rm phys}$ via LSZ). [O] perturbative only — the ambient measure `QG.AMB.01` stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, `AUDIT OK`.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **`v268_theta13_carrier_trace.py` (`FLAV.TH13.01`) closes the θ_{13} exponent origin.** The value $\sin^2\theta_{13} = \varphi_0^{\rm ret}e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = \mathrm{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). [E] own channel: the $\mu\tau$-breaking $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) "θ_{13} from M_ν" route flagged by the new script atlas.

[O] the CP magnitude and the absolute ν-mass scale remain the open PMNS items. Exact core mirrored in Wolfram. **[E]** build green, `AUDIT OK`.

2026-06-18 (Infrastructure: a generated causal "script atlas" so the suite is navigable at a glance)

- **`verification/make_script_atlas.py` generates `verification/script_atlas.md`** — a single grouped, dependency-aware view fusing the existing single sources (`status_ledger.csv` for claim→status→dependencies→supersedes, `script_registry.csv`/`script_clusters.csv` for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. **[E]** build green, `AUDIT OK`.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for `QGEO.SYM.01`, v267)

- **`v267_qgeo_rigidity_minimal_axiom.py` (`QGEO.SYM.02`) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). **[E]** the *single* bare statement "the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks" ($=\omega\circ\rho=\omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a; cross-ratio 2 $\Leftrightarrow$ j=1728 (only $\lambda\in\{-1,\frac{1}{2},2\}$) $\Leftrightarrow$ the order-4 CM automorphism on y^2=x^3−x; and the unique flat pillowcase metric, mark-locality, $[\rho,\Lambda]$=0 and $\omega\circ\rho$=ω all follow (v214/v201/v264/v198). **[E]** neg controls: generic config $j\neq 1728$ ($\mathbb{Z}_2$ only), hexagonal j=0 ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. **[C]** second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. **[O]** the bare order-4 symmetry stays the one foundational postulate (like c=const) — the *reduction* is closed (axiom ⇒ everything), the axiom is not. **[E]** build green, `AUDIT OK`.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, +(15, 1, 1) survives, Hyper-K decisive, v266)

- **`v266_ps_threshold_proton.py` (`PS.PROTON.02`/`PS.THRESHOLD.02`) decides branch B of the v265 fork.** The explicit two-step Pati–Salam thresholds give d=6 $\tau_p(p\to e^+\pi^0)$ (writes `ps_threshold_proton.json`): **[X]** the minimal 16-Higgs content (M_{GUT}∼2.4×10^{15}) gives τ_p∼4.4×10^{33} yr < Super-K and is *killed* even at the high hadronic band; **[C]** the E_8-allowed $(15, 1, 1)$ [the single 45] raises M_{GUT}∼5.9×10^{15} ⇒ τ_p∼1.6×10^{35} yr, proton-safe *now*; **[X]** but that sits at the Hyper-K reach (∼1.4×10^{35} yr) — a *dated* kill test decisive within the decade. So branch B is promoted from "well-mannered candidate" to *content-selected (must be* +(15, 1, 1)*), Hyper-K-decisive UV branch*; **[O]** structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. **[E]** build green, `AUDIT OK`.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **`v265_qft4d_fork_freeze.py` (`QFT4D.FORK.01`) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree** (`qft4d_fork_freeze.json`), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). **[X]** SM-only 4D-GUT unification is killed (1-loop: at the α_1=α_2 scale

α_3^{-1} off by $\sim$5.6, spread $\sim$8.8 at 2×10^{16} GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W$=3/8 (v248), the $O(1)$ κ=M_{PS}/M_s band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake τ_p window; carrier gauging stays the contract). [E] a prose guard (`QFT4D.NO_OVERCLAIM.01`) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows `QFT4D.BOUNDARY.DEFAULT.01`, `QFT4D.SMONLYGUT.01`, `PS.UVBRANCH.01`, `PS.THRESHOLD.01`, `PS.PROTON.KILL.01`. A fork-policy box added to `tfpt_research_contracts`; [E] build green, `AUDIT OK`.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F, and the raw-seam⇒mark-locality reduction, v262–v264)

- **`v262_fqcd_mp_me.py` (`FR.MPME.02`) implements the one contract-only F_{transfer} pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z)$=0.1179 with n_f thresholds (carrier $b_3 = -7$, [E]) to $\alpha_s(2\,\text{GeV})$$\sim$0.30 and $\Lambda^{(3)}_{\overline{\text{MS}}}$$\sim$0.33 GeV; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e$$\sim$1824 vs the measured 1836.15 within the $\sim$ 7% external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]).
- **`v263_mnu_from_df.py` (`FLAV.PMNS.02`) makes M_ν a seesaw operator of D_F.** [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$-symmetric inputs) $\mu\tau$-symmetric; [C] it reproduces θ_{23}=45° and $\sin^2\theta_{12}$=$\frac{1}{3}$−$\frac{\varphi_0^{\text{ret}}}{2}$ (v9) under the seam misalignment; [O] θ_{13}=0 in the leading texture, the CP *magnitude* (240°) is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **`v264_raw_seam_marklocal.py` (`QGEO.ENERGY.03`) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$-invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]$=0 ⇒ $\omega \circ \rho$=ω on this metric. So the whole bedrock collapses to the single metric statement "the raw RP seam carries the uniformising flat pillowcase metric" (=`QGEO.SYM.01`). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] `AUDIT OK`, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit "what is *not* closed" line)

- **Five paper sections that predated the `v238`–`v261` round are brought up to date** (additive, no status marker moved). The `introduction` closure-ledger table and the "five questions" item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ closed modulo `QGEO.SYM.01` (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` "ambient closure" section now states that the cutoff of its S_{rel} *is* the seam KMS weight (f_2/f_0=1, v259) and D_{rel} the covariance induction (v258); the `tfpt_2` "(4) Constructive QFT closure" section adds the relative-object pointer; `tfpt_4_frontier`'s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and `origin_theory`'s honest-status keybox notes that the whole boundary QFT collapses onto the same `QGEO.SYM.01` postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (`10 ok, 0 failed`), `AUDIT OK`.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The `v258`–`v261` closure is now reflected on every "what is open" surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary G_{net} entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise `QGEO.SYM.01` via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card ("the structural residual is one condition") carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, `AUDIT OK`.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, `v258`–`v261`)

- **`v258_dirac_covariance_induction.py` (`PS.DIRAC.03`) closes the "posited Yukawa vs derived operator" gap of v250/v252.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, β=1) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ.
- **`v259_modular_cutoff_kappa.py` (`PS.SPECACT.02`) fixes the last open spectral-action input.** The cutoff f is the seam's own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (v239) and $2\pi = 1/(4c_3)$ (v58) fixes β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa\approx 1.3$ pins $c_{PS}/c_{\text{grav}}\approx 1.7$).
- **`v260_k3_kummer_unification.py` (`ARCH.K3.01`) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z\mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(\text{K3},\mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1, even, signature $(3,19)$); [E] the seam is $T^2/(z\mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]|=2^4=16$ Kummer nodes $= \dim S^+$ (v197), $16=4\times 4$; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 "glue $D_5\oplus A_3+\mu_4\Rightarrow E_8$" is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.
- **`v261_modular_spectral_closure.py` (`QFT.MSC.01`) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6\log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP

holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from `changelog.tex` and audit-gated)

- **The canonical changelog now also drives a public `/changelog` website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf`/`\emph`/`\texttt`/`\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] build (`npm run build` green, `/changelog` prerendered static) and `bash build.sh audit` (`AUDIT OK`).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **`v257_quasiorient_modpd.py` (`PS.NCG.ORIENT.02`) corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: [E] strict orientability fails ($\gamma \notin \mathrm{span}\{\lambda(a)\rho(b)\}$); [E] *quasi-orientability* holds (the left/right A-irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\mathrm{LR}}(\mathcal{H}^{\mathrm{even}}, \mathcal{H}^{\mathrm{odd}}) = \{0\}$); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal-L/R-neutrino case); [C] modified Poincaré duality holds (CCM), Stephan's alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **`v254_inner_fluctuations.py` (`PS.NCG.FLUCT.01`) closes v250 #5 + the no-junk obligation.** The inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1,2)_{1/2}$ — no junk, no triplet, no $(10,1,3)/\mathbf{126}$. [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E] σ is an SM singlet $(1,1,0)$ with $B-L = 2$. [C] σ = scalaron.
- **`v255_spectral_action_expansion.py` (`PS.SPECACT.01`) closes the spectral-action expansion (v252 #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: [E] structure ($a_0$ cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N = 96$); [E] $b/a^2 = 1/3$ (top-dominated); [E] $\sin^2\theta_W = 3/8$; [E] the R^2 spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic $\to \sim$125 GeV with σ; [C] $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\mathrm{grav}}}$ explicit; [O] exact decimal needs the cutoff moments.

- **`v256_orientability_poincare.py` (`PS.NCG.ORIENT.01`) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\mathrm{Tr}(\gamma\,\pi(p)J\pi(q)J^{-1})$ to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null $\sim(1,1,-1)$ on the SM K-theory $\mathbb{Z}^3$; [E] the orientability local condition $[\gamma_F,\pi(a)]=0$. [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, `v252`–`v253`)

- **`v252_full_finite_triple.py` (`PS.DIRAC.02`) advances the Dirac contract `v250` from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ($J^2{=}+1$, $JD{=}DJ$), grading ($\gamma^2{=}1$, $[\gamma,a]{=}0$, $J\gamma{=}-\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}}{=}m_D^2/M_R$. [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.
- **`v253_kappa_scalaron_ps.py` (`PS.KAPPA.01`) discharges residual 6(b) of `v249`.** κ in $M_{PS}=\kappa\,M_s$ ($M_s=c_3^{7/2}\bar{M}\approx 3.06{\times}10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors $\times$ the two E_8-allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa=\sqrt{(f_2/f_0)\,c_{PS}/c_{\text{grav}}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/σ — the CCvS Pati–Salam mechanism, `v251`)

- **`v251_first_order_sigma.py` (`PS.DIRAC.01`) advances the Dirac contract `v250` from a contract to an exhibited mechanism.** On a faithful lepton-block model ($H=H_F\oplus H_F^c$, $H_F=\mathbb{C}^3$), the real even structure is checked ($J^2{=}+1$, $\gamma^2{=}1$, $D\gamma{=}-\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R\nu_R^c=\sigma$) block — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ. Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] $\sigma=$ scalaron (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, `v247`–`v250`)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** `v247_e8_branching_no126.py` (`PS.E8BRANCH.01`): the E_8 hull branches under $SO(10){\times}SU(4)$ as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the $SO(10)$ content is exactly $\{1,10,16,45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and 10+16+45 is E_8-supplied (only one 45 $\Rightarrow$ proton-marginal). `v248_ps_algebra.py` (`PS.ALGEBRA.01`): $A_F=\mathbb{H}_L\oplus\mathbb{H}_R\oplus M_4(\mathbb{C})\to SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$, exact hypercharges $Y=T_{3R}+(B{-}L)/2$, matching $\alpha_1^{-1}=\frac{3}{5}\alpha_{2R}^{-1}+\frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W=3/8$ (colour $SU(4)_c\subset D_5$, distinct from the family A_3). `v249_ps_unification.py` (`PS.RGTEST.01`): the two-step PS$\to SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0$–1.7. `v250_dirac_triple_contract.py` (`CONTRACT.QFT4D.DIRAC.01`): the spectral-triple contract ($H_F = 3\times16$, the seven NCG axioms, σ =scalaron) that would turn "carrier is gauged" from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **`v246_unification_data_crosscheck.py` (`QFT4D.RGTEST.01`) — the first hard data confrontation of the spectral-action prediction.** TFPT's gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}$–10^{17} GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed [X]/[O]: with "no new state" there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- **`v245_spectral_matter_content.py` (`QFT4D.MATTER.01`) discharges the computable core of the 4d contract v244.** The carrier half-spinor (the $SO(10)$ **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10}+\bar{\mathbf{5}}+\mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions $+\,\nu_R$), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$), the spectral-action normalisation gives $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\,g_1$), and the Higgs is the $(1,2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of `CONTRACT.QFT4D.01` is now [E]; only the seam→Dirac realisation and the run-down couplings stay [O]. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238).** `v239_kms_thermal_time.py` (`DYN.KMS.01`): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. `v240_gns_os_reconstruction.py` (`DYN.GNS.01`): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{\mathrm{OS}} = -\log T = -L$ is positive with gap Δ (the v64 mass gap). `v241_dhr_sectors.py` (`DHR.SECTORS.01`): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \to 4 \to 1$ (v235/v237). `v242_spin_statistics_modular.py` (`DHR.MODULAR.01`): spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. `v243_haag_ruelle_braiding.py` (`DHR.SMATRIX.01`): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (`CONTRACT.QFT4D.01`): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [O] contract. Net: the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one* *contract* (`CONTRACT.QFT4D.01`). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New `v238_modular_lindblad_dynamics.py` (`DYN.SEMIGROUP.01`, cluster registry).** A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel's generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time). [E] keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real≤ 0 spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra = `QGEO.SYM.01` (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (`experiments/`, all `data_limited`, not load-bearing).** From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior ($w_{\rm in}$=−1) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\rm in} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai $Q_{\rm geom}$=3/8 equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}$=3/8 (shared de Sitter endpoint 1/2); since 1/3<3/8<4/9<1/2 this is a light-trapping ECO, yielding an echo template (round-trip delay ∼0.7 ms at 62 $M_\odot$, amplitude $\leq (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\rm geom}$ map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES ∼3.3σ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).

- **η_B — full BDP Boltzmann ODE solve confirms the route at the frozen scale (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\rm CP} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10}, with *no* free M_R dial. This moves the leptogenesis route from "solve pending" to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.

- **The Petz recovery map and the rank-one baby universe, realised numerically (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected λ=1 (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the

Petz identification v221 deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.

- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts search-targets`).** The Stage-1 machinery (Kerr-ringdown subtraction → matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.

- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors `figures/`)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, infrastructure).** The Python red-team surface still emitted the historical *three-residual* "A reduced, not closed" form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement "the seam–Calderón boundary net is holomorphic with $c = 8$" ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \to (E8)_1$ glue realised at Lie level (v143) and its odd sectors twisted/Ramond ($E_8 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$, v148), while the $q(A_3)$ normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ($120{+}128 = 248$; glue index $= |\mu_4| = N_{\mathrm{fam}}{+}1 = 4$); status stays [C]/[O] (reduced, not closed).

- **Shadow export now mirrors `figures/` but not `website/` (`scripts/export-shadow.sh`, `verification/make_script_index.py`, infrastructure).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh notes` runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (`v231_cp_mu6_phases`, `tfpt_5_redteam`, [E]/[C]).** Sharper than v220/v225 (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the $j{=}0$ Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\mathrm{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM,lead}} + \pi =$ (CM unit) $\times$ ($\mathbb{Z}_2$ sheet, $\rho^3{=}{-1}$). The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the v88 misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays [C] (`CP.MU6.01`).

- **The seam as the E_8 Kleinian singularity (`v232_e8_kleinian_seam`, `origin_theory`, [E]/[C]).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise `QGEO.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine-E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}}+N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2)-curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1{=}0$, $\pi_1{=}2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8; `TOPO.E8.01`, bedrock stays [O]).

- **CP is the universal family/triality phase, only sheet-split (`v233_cp_triality_phase`, `tfpt_5_redteam`, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3(\text{family/triality})\times\mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (`CP.TRIALITY.01`).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (`v234_seam_holomorphy_selection`, `tfpt_research_contracts`, [E]/[O]).** The whole structural residual is *one* condition — "the seam carries no nontrivial abelian sector" — with three provably-equivalent faces, all forcing E_8: (i) holomorphy (μ-index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal #(mark-1) $= |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8; and holomorphic $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8. So P2, G_{net}, Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is "free bulk $\Rightarrow$ holomorphic boundary" (`GATE.HOLO.01`).
- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow$ $\det K{=}1$ (`v235_seam_chern_simons`, `tfpt_research_contracts`, [E]/[O]).** A genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K-matrix with #anyons $= |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic) — all rank 8, $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ — so holomorphic $= E_8 =$ the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order-$|\mu_4|$ Lagrangian glue (det 16$\to$1, vs. a partial order-2 isotropic $\to 4{=}D_8$). The residual narrows to one integer condition "the free RP seam condenses the Lagrangian glue (det =1)" = the sheet/QGEO selection, now with holomorphy$\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`).
- **The $(2,3,5)$ Brieskorn singularity is the one generator of the skeleton (`v236_brieskorn_capstone`, `origin_theory`, [E]).** Capstone of the icosahedral round: the singularity $x^2{+}y^3{+}z^5$, whose exponents are the atoms $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$, has Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\text{rank}\,E_8$ (a *fifth* origin of the "8"), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8, link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h\rangle{=}\mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (`TOPO.BRIESKORN.01`).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow$ $\det K{=}1$ $\Leftrightarrow$ the seam is SRE (`v237_seam_sre_closure`, `tfpt_research_contracts`, [E]/[O]).** Sharpens `GATE.HOLO.02`'s residual from definitional to falsifiable: the K-matrix ground-state degeneracy on a genus-g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so "no topological degeneracy on any closed surface" $\Leftrightarrow$ $\det K{=}1$; among rank-8 even positive-definite K ($c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement "the free RP seam is SRE (no topological

order)" (`GATE.HOLO.03`).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2,3,5\}$ (`v219_icosahedral_mckay`, `origin_theory`, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1,2,3,4,5,6,10\}$ — it contains the $2,3,5$ rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1,2,2,3,3,4,4,5,6\}$, sum $30{=}h(E_8)$, sum of squares $120{=}|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8, not a P2 proof (`MCKAY.E8.01`).
- **CM-norm duality: 41 (square) and 7 (hex) (`v222_cm_norm_duality`, `origin_theory`, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j{=}1728$) gives $N(g_{\mathrm{car}}{+}i|\mu_4|){=}5^2{+}4^2{=}41{=}10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3{+}2i){=}13{=}\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j{=}0$) gives $N(3{+}2\omega){=}7{=}$ scalaron. The $(3,2)$ split generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (`CMNORM.DUAL.01`).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (`v223_coxeter_totative_clock`, `origin_theory`, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7\rangle{=}\{1,7,13,19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h){=}$rank and $h{=}2{\cdot}3{\cdot}5$ (`COX.CLOCK.01`).
- **$248{=}120{+}128$ as a magnitude/phase channel typing (`v227_degree_exponent_channel_split`, `tfpt_5_redteam`, [E]).** Exponents sum $120{=}|R^+(E_8)|$ (magnitude channel), degrees sum $128{=}\mathrm{rank}\cdot\dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; `E8.CHAN.01`).
- **P2 as the mode space of a degree-4 seam divisor (`v228_rr_index_gate`, `origin_theory`, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D{=}4{=}|\mu_4|$ on $\mathbb{P}^1$ forces $h^0{=}5{=}g_{\mathrm{car}}$, rank $H_1{=}3{=}N_{\mathrm{fam}}$, $h^0{+}H_1{=}8{=}\mathrm{rank}\,E_8$, $\Lambda^{\mathrm{even}}(\mathbb{C}^5){=}16{=}\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (`QGEO.RR.01`).
- **The seam as a finite recoverability code (`v221_seam_qecc`, `origin_theory`, [E]/[C]).** Code dimension $\dim S^+{=}16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1,(2/3)^6,(1/3)^6\}$, recovery bound $\|T^n\delta\|{\le}(2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (`QEC.SEAM.01`).
- **CP lives in the hexagonal phase fiber (`v220_cp_hexagonal_modulus`, `tfpt_5_redteam`, [E]/[C]).** The seam deck is the square modulus $j{=}1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j{=}0$ ($\mathbb{Z}/6$, $\arg\rho{=}\pi/3$). The frozen $\delta{=}\pi/3{+}3\lambda_C^2{=}68.654°$ has leading term $\pi/3{=}\arg\rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (`CP.MOD.01`).
- **The dual normal frame (d,n) with CP as orientation (`v225_dual_normal_frame`, `tfpt_5_redteam`, [E]/[C]).** $d{=}a^\top R^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1){=}-\frac{1}{2}$ Nariai; $n{=}(5,-9,6)$ reads det on the first-generation column; $\det(\mathbf{1},d,n){=}21{=}3{\cdot}7$; $\mathrm{Im}\det(\mathbf{1},d,\rho n){=}21\sin(\pi/3)$ — CP as orientation (`DUAL.FRAME.01`).
- **F_{transfer} as one path on the Sheet Diamond (`v224_diamond_ftransfer_path`, `tfpt_2_standard_model`, [E]/[C]).** $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$: K,C,F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1,8,10),(1,8,16)$; the Koide source→pole is the 1-D projection (`FTR.PATH.01`).

- **The charged leptons as an étale Frobenius algebra (`v229_lepton_frobenius_algebra`, `tfpt_2_standard_model`, [E]/[C]).** $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$, product $\frac{32}{9}$; $A = \mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (`LEP.FROB.01`).
- **The center budget $(7, 11, 13)$ as three local norms (`v230_center_budget_norms`, `tfpt_2_standard_model`, [E]).** $C = R + Q\,\mathrm{diag}(1, 0, 0)$ row sums $(7, 11, 13)$: $7 = N_{\mathbb{Z}[\omega]}(3+2\omega)$ (hex), $13 = N_{\mathbb{Z}[i]}(3+2i)$ (square), $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (`CENTER.NORM.01`).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (`v218_diamond_axis_geometry`, `tfpt_2_standard_model`, [E]).** Sharpens `v94`/`v95` with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C + xU) = 14 + 6x$ (slope $6 = |R^+(A_3)|$) and quadratic on the sheet axis $\det(C + yV) = 14 + 14y + 4y^2$ (2nd difference $8 = \mathrm{rank}\, E_8$); the anchor block has $\det B(C + xU) = 2 \det(C + xU)$ and sheet curvature $6 = |R^+(A_3)|$ — two curvatures $(8, 6) = (\mathrm{rank}\, E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\mathrm{Pl}(K) \to \mathrm{Pl}(C) \to \mathrm{Pl}(F)$ lifts in steps $(1, 8, 10) = (N_\Phi, \mathrm{rank}\, E_8, A_\Lambda)$ then $(1, 8, 16) = (N_\Phi, \mathrm{rank}\, E_8, \dim S^+)$ — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2\,\mathrm{Disc}(q)$ with squares $(1, 3, 4, 6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3{\cdot}5{\cdot}7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28 = 2 \dim G_2$, $52 = \dim F_4$ stay audit-only, not spine, as in `v94`/`v95`); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger `DIAMOND.AXIS.01`, `DIAMOND.PLUCKER.01`, `DIAMOND.SPECTRAL.01`; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger `QGEO.MARKS.03`).** The K1 lever of the `v215` kill-test (“exactly four marks”) is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\mathrm{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, $(2, 2, 2, 2)$, and three independent criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), $\mathrm{rank}\, H^1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2, 2, 2, 2)$. The numerical sibling `v217_marks_emergence_scan` ([C], ledger `QGEO.EMERGE.01`) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. `v216` exact (Wolfram extension → 248/248); `v217` numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **`QGEO.SYM.01` as a kill-test, not a proof-promise (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.KILL.01`).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198`/`v199`/`v201`) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4-deck-invariant). Levers: **K1** exactly four marks ($b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio $2 \Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [E] (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214/v54/v56`). Suite → 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of `QGEO.SYM.01` (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.PILLOW.01`).** The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2,2,2,2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the `v201` "flat away from the marks". **(2)** New link: cross-ratio$(\mu_4) = 2$ (`v168`) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$; all six harmonic cross-ratios give 1728) ⇒ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: "seam = uniformising flat pillowcase metric" implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. [O] It does *not* close `QGEO.SYM.01`: the choice "RP seam metric = the constant-curvature representative" stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/j-invariant/CM arithmetic); Klein-J values mpmath-numerical. Suite → 214 modules.

2026-06-15 (the F_transfer functor contract CONTRACT.F.01)

- **F_{transfer} formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]: **(1)** μ_4-deck equivariance (the Koide multiplier $\lambda_2{=}(2/3)^6{=}64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53{=}a^T(R{+}Q)\mathbf{1}$, $\text{Pl}(F){=}(1, 22, 30)$, $\|\text{Pl}(K)\|_1{=}11$); **(3)** positivity/stochasticity ($\text{spec}\, T$ positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit ($b_3{=}{-}7$ a carrier output). The four instances are F_{pole} (`v183`), $F_{\text{Boltzmann}}$ (`v212`), F_{relic} (`v211`), F_{QCD} (`v164`). `CONTRACT.F.01` is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the `v187` typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger `CONTRACT.F.01`.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (`v211_axion_spine_angle`, `tfpt_4_frontier`, [C]).** A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170°$ (`v185`, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108°$ exactly (no fit), the finite-T solver reaches Ω_{DM} at ≈106°, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); [X] $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.

- **Leptogenesis scalaron-decuple branch (`v212_leptogenesis_decuple`, `tfpt_4_frontier`, [C]).** Both Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\rm scal}, \varphi_0^{\rm ret}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than `v184`'s $M_R\varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10}. The coupling mechanism is posited not derived, so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the route, not the theory (`FR.ETAB.04`; Route A independent).

2026-06-15 (Lean: QGEO.COHOM.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (`CohomologyGrading.lean`, ledger `FORM.QGEO.03`, [O]). A new module machine-proves the COHOM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry/admit`, axioms `propext, Classical.choice, Quot.sound` only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4 - 1)$ (k=1, 2, 3) have pullback character $\rho^*\omega_k = i^k\omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since i^4=1), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3)$ = the A_3 exponents = Spec(Q_+), rank 3 = $N_{\rm fam}$. The MODULE *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`Mobius Uniformisation.lean`, ledger `FORM.QGEO.02`, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry/admit`, axioms `propext, Classical.choice, Quot.sound` only): the clock $\rho : z \mapsto iz$ has ρ^4=id and $\rho^2 \neq$id (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma\rho\sigma = \rho^{-1}$ (so $\langle\rho, \sigma\rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes $1, -1$, swaps $i, -i$); and the multiplier classification "$z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow$ $\zeta = \pm i$". Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [O], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201/v210` (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext, Classical.choice, Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j<4} \zeta^j$ = (if ζ=1 then 4 else 0) for ζ^4=1 (so a μ_4-mark sum is supported only on modes $\equiv 0$ mod 4); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]).** A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of `v175`). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via `v201`).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\rm top}^2$ projected through the seam-loop phase gives $a_\mu^{\rm seam} = \delta_2/(2\pi) = 45/(524288\,\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger `FR.MUONG2.01`; Wolfram-mirrored.

- **An independent gravitational $3/4$ (`v205_xi_threequarter`, `tfpt_4_frontier`, [E]/[C]).** $\xi = c_3/\varphi_{\rm tree} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\rm ret}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (`v152`), an over-determination. Ledger `GRAV.XI.01`; Wolfram-mirrored.

- **Quantitative H_0 from the Λ branch (`v206_h0_lambda_branch`, `tfpt_4_frontier`, [C]).** $\delta_\Sigma = 48\, e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5$–67.1 km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger `GRAV.H0.01`.

- **Asymptotic Safety as a second QG route (`v207_asymptotic_safety`, `tfpt_4_frontier`, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close $G_{\rm metric}$. Ledger `GRAV.ASYMP.01`.

- **Black-hole thermodynamics from the seam** (`v208_bh_thermodynamics`, `tfpt_horizon_readouts`, [E]/[C]). Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to `v198/v199`) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from `v28`). Ledger `HOR.BHTHERMO.01`; Wolfram-mirrored.

- **The black hole as a seam fixed point** (`v209_bh_regular_core`, `tfpt_horizon_readouts`, [C]). No singularity ($\varphi \to \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger `HOR.BHCORE.01`.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (`tfpt_3`, [O]).** Added a paragraph to the "Möbius origin of π" section: $\text{Möb}(\hat{\mathbb{C}}) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\mathrm{PSL}(2, \mathbb{C})$ rotation (v180) and the RP$\to$OS causal cone is the same Lorentz structure — so "Möbius origin of π" and "one Lorentzian cone" are two faces of one boundary $\mathrm{PSL}(2,\mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to c_3 (`tfpt_4_frontier`, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\rm rad} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1-\varphi_{\rm seam}) = 170.4^\circ$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **`HOR.EHT.01` (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\rm BH}(r) = 16c_3^4\, Q_e^{\rm eff} Q_m^{\rm eff}/r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\rm top}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\rm top} = 48c_3^4$ that fixes the α-kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\rm rad} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude Q_eQ_m is an MHD/GR weight, never a pixel prediction. **Dated kill test** [X]: the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \to \pi\nu\bar{\nu}$ — v202)

- **`FR.RAREKAON.01` (v202, [C]).** The closed TFPT CKM point ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}|V_{cb}|$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test** [X]: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new `FR.RAREKAON.` family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **`QGEO.SUBPRIN.01` (v201, [E]/[O]).** The v199 bedrock residual — "the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements" — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4-character-block-diagonal iff f is $\mathbb{Z}_4$-invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4-mark sum $f = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g, hence is automatically $\mathbb{Z}_4$-invariant ([E]). So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks

($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to "the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)", structurally definitional — the narrowest form yet of `QGEO.SYM.01`, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]).** The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170°$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6$–2.3 (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120$–$130°$ or extra dilution) — this *refines* v185's optimistic "can be all-DM" toward an over-production tension. The near-hilltop regime is O(1)-uncertain, so a full finite-T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.
- **Full finite-T solve confirms it (`axion_relic/full_finiteT_solve.py`).** A converged nonlinear misalignment solve ($\theta'' + (3 + \mathrm{d}\ln H/\mathrm{d}N)\theta' + (m_a(T)/H)^2 \sin\theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i{=}1 \to \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170°$ — $\sim$5.5$\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106°$. The over-production is thus *confirmed* (not the optimistic "all-DM"); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **`v199_seam_state_invariance.py` [E]/[O] (claim `QGEO.STATE.01`, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1+\operatorname{sgn} H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to "the bounded sub-principal symbol has no off-character matrix elements" — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **`v200_seam_variational_scan.py` [C]/[O] (claim `QGEO.VARI.02`, work package B).** The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, `experiments/` only, not the ledger): the F_{transfer} four-pipeline scaffold (`experiments/ftransfer/`) with one CONTRACT per interface, and confirmation that the FRB preregistration (`frb_tfpt_v1.yaml`) and its status semantics already exist.* Suite now 199 modules, highest ID v200; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **`v198_modular_commutator_reduction.py` [O] (claim `QGEO.MODULAR.01`).** The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \mathrm{diag}(|n|)$ and the clock $\rho{:}z \mapsto iz = \mathrm{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4-invariant (v177). So the entire residual collapses to the single *state-invariance* "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of `QGEO.SYM.01`, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 197 modules, highest ID `v198`; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **`v195_marks_lefschetz_character.py` [E]/[O] (claim `QGEO.MARKS.02`).** The four seam marks are *forced* (not posited): $\rho{:}z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and $\operatorname{rank} H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So Marks reduces from the geometric posit "D_4 marks" to the milder premise "H^1 carries the A_3-exponent rep" — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **`v196_seam_energy_functional.py` [E]/[O] (claim `QGEO.VARI.01`).** The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the `QGEO.SYM.01` conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4-breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **`v197_rr_carrier_clifford_d5.py` [E]/[C] (claim `ARCH.RRCAR.02`).** Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \to g_{\text{car}} \to D_5$. [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop "$\pi - 3\phi_0$" proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID `v197`; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **`v194_raw_seam_dtn_rp.py` [O] (claim `QGEO.DTN.01`).** The next hard lever — "is the raw RP-seam DtN definable from RP data alone?" (sub-claim 2 of `QGEO.ENERGY.02`) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \backslash \mu_4$. (3) [O] the residual relocates: the full-L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So `QGEO.SYM.01` reduces from a definitional postulate to "intrinsic BW geometric modular covariance" — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **`v193_qgeo_energy_commutator.py` [O] (claim `QGEO.ENERGY.02`).** The non-circular sharpening of `QGEO.ENERGY.01`: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1, 2, 3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, `QGEO.SYM.01` becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to "Constructive QFT closure of the admissible physical sector", and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **`v189_riemann_roch_carrier.py` [E]/[C] (claim `ARCH.RRCAR.01`).** The pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already `QGEO.COHOM.01`) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by $\operatorname{rank}(D_5) = 5$. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **`v190_nariai_entropy_bound.py` [E] (claim `HOR.NARIAI.02`).** A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3} S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v191_universal_branch_line.py` [C] (claim `HOR.BRANCHLINE.01`).** The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed [C]* — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known 2/3 ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v192_qgeo_energy_clock.py` [O] (claim `QGEO.ENERGY.01`).** An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed:* it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (v186 skipped).
- *All four are validated external-review proposals (`problem_b`); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version `5.1 (rev 120)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20687564`, DOI `10.5281/zenodo.20687564`. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-`QGEO.SYM.01` two-layer clarification. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **`v184_etaB_anchored_boltzmann.py` [C] (claim `FR.ETAB.03`).** An honest test of the external-review η_B proposal ($M_1 = M_R\varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (c_3-/φ_0-powers off ~75%/~40%), so $M_1 = M_R\varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- **`v185_axion_relic_solver.py` [C] (claim `FR.DM.03`).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\rm scal}/128 \approx 2.39\times10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) CAN be all dark matter, but the harmonic misalignment under-produces (~21% at $\theta \sim O(1)$), so the $\theta_i \approx 170°$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **`v187_ftransfer_laws.py` [E] (claim `FR.TRANSFER.01`).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B, axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **`v188_frontier_wording_guard.py` [E] (claim `AUDIT.WORDING.01`).** A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording → v184 [C]; the old "open relic scale" dark-matter line → v185; the old "near, not exact" Koide line → v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-`QGEO.SYM.01` two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID `v188` (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, $v_{\rm geo}$ as unit gauge, $G_{\rm net}$ a corollary of `QGEO.SYM.01`, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **`v183_koide_f_corner_transfer.py` [E]/[C] (claim `FR.KOIDE.07`).** An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78{+}8{+}2{\cdot}27{\cdot}3$) and $a^T(R+Q)\mathbf{1} = 53 = 52{+}1$, where $F = R{+}Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades `FR.KOIDE.03` from "near miss + conjecture" to "operator-sourced source→pole transfer": the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an F_{transfer} special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, "one clock, two geometries") is exactly `HOR.TRISECT.01`/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **`v182_reviewer_residual_map.py` [E]/[C] (claim `REVIEW.MAP.01`).** A careful external review raised four concerns (A mapping $2D \to 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{$`QGEO.SYM.01`$, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B (≡ `QGEO.SYM.01`), D (Möbius forced by $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2,\mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a "Rosetta" translation lexicon** (website `RosettaLexicon`): TFPT's compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer}, v_{geo}, `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \le 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** `QGEO.SYM.01` is now framed confidently as TFPT's *one fundamental postulate* (the role "c = const" plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website "what is open" card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (`AX.P1.01`, `AX.P2.01`, `QGEO.ISO.01`) had *unquoted commas* in a status/external field (e.g. "a=(1,1,2)", "PSL(2,C)"), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.

- G_{net} **typed in three explicit levels** (website "What is open" card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (`QGEO.SYM.01`). The target object's mathematics is closed; only the physical coupling of the raw seam is open.
- **`QGEO.SYM.01` now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than `CONF/ISO`, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing "*given* `QGEO.SYM.01`, the closure follows" rather than "the theory derives the seam geometry".
- **Claim-language detox.** "no free fundamental number" softened to "no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)" (`origin_theory`, `introduction`, website); the site title softened from "the Standard Model Derived" to "the Standard-Model *Skeleton* Derived".
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version `5.1 (rev 114)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20686284`, DOI `10.5281/zenodo.20686284`. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise `QGEO.SYM.01`), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The `README` and the Zenodo description were refreshed (181 modules), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from "build a quantum-gravity measure" to the single definitional bedrock `QGEO.SYM.01`; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ∼181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1,1,2)$ plus π (with $g_{\text{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01`/v23, `ARCH.CORE.01`/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare "declared inputs". Fixed: the ledger rows `AX.P1.01`/`AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\text{car}} = e_2(a)$, bootstrap-forced) rather than "declared axiom". No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **`v181_clock_is_conformal_symmetry.py` [E]/[O] (claim `QGEO.SYM.01`).** The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: "the seam's conformal structure *is* the μ_4-deck structure of the carrier clock". This is bedrock — a physical/definitional statement tying P2 to the seam's conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE`(v175)→ $\cdots$ →`SYM.01`(v181) ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` "latest reductions" prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A $\times 2$, G_{net}), the `glossary` G_{net} entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **`v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`).** Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) *isometry⇒ conformal⇒Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i, free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **`v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). Marks's residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the same $\chi = 2$ that gives the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). Kernel then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: "the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism". The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **`v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`).** An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so Marks reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So Kernel reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- **`v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`).** A second external residual-class audit pointed out that the v176 statement takes the marks/D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as Realize = Marks · Uniform · Cohom · Module · Kernel · Net and closes *four* nodes *constructively* (all validated symbolically before adoption): Uniform [E] — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening v168's cross-ratio to the full uniformisation); Cohom [E] — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; Module [E] — a *unique* μ_4-equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); Net [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4-equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- **`v176_seam_collar_realisation.py` [E]/[O] (claim `QGEO.REALIZE.02`).** Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible c=8 data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4-equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (v154/v175). The whole structural residual thus reduces to the *one* open premise `QGEO.REALIZE.01`: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1} v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \to \frac{53}{54} u$ source→pole transfer is already typed [C] (`FR.KOIDE.03`); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version `5.1 (rev 105)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20681451`, DOI `10.5281/zenodo.20681451`. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The `README` and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **`v175_net_existence_full_cone.py` [E]/[O] (claim `QFT.NETRP.01`).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m*: the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\operatorname{spec}(t)$, so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) A_2 *assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation `QGEO.REALIZE.01` (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **`v174_seam_fock_readings.py` [E] (claim `QFT.FOCK.01`).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind `v161`. (2) *Witten index = the Plücker norm* 11: the four μ_4-corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL (deg ≤ 2) keeps $\sum_{k\le 2}\binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5+3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the `v157`/`v158` rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **`v170_diamond_slice_compression.py` [E] (claim `E8.SLICE.01`).** E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3,4,6,10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3,4,8,14,20,32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7,11,13) + s(3,3,3) + t(1,2,3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **`v171_os_moment_cluster.py` [E]/[C] (claim `QFT.OSMOMENT.01`).** Atomic OS Moment Lemma: $\mathrm{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster 729/665, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6\log\frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).
- **`v172_trace_anomaly_seed.py` [E] (claim `SEED.ANOMALY.01`).** The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7$ = scalaron exponent $48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2\cdot 7^2/\dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).
- **`v173_pfaffian_cp_car.py` [E] (claim `FLAV.STRONGCP.02`).** Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5-Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version `5.1 (rev 97)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20678568`, DOI `10.5281/zenodo.20678568`. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **`v168_qgeo_rigidity.py` [E]/[C] (claims `QGEO.RIGID.01`, `QGEO.REALIZE.01`).** Hardens the *finite* half of the one remaining Tier-1 hinge `GATE.QGEO` (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4-1)$ $(k = 1, 2, 3)$ are μ_4-eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = \text{A}_3$ exponents $= \text{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \backslash \mu_4$ — the QGEO Rigidity Theorem (`QGEO.RIGID.01`, [E]). The seam-collar realisation stays the one premise (`QGEO.REALIZE.01`, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **`v169_etaB_boltzmann_interface.py` [C] (claim `FR.ETAB.02`).** Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240°$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10}, untuned) — the route is *viable*. But M_1/washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **`v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`).** A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral c=8 fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim$0.022 at one loop to $\sim$0.002). The double condition predicts $m_H \sim 129$–134 GeV; the same condition at the scalaron scale gives $\sim$107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ, reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **`v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`).** Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2-suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2} Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2/(16\pi^2)\ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta \sin^2\theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2-driven quark mass ratios $(m_t/m_b \sim 41)$ by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **`v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`).** The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-

loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest "[O] — not forced" frontier ratio. Cited in `tfpt_4_frontier`.

- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_transfer + irreducible core, v167)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q-realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology b_1=N_{fam}=3); the EH step (R3) folds into v_{geo} (v152) and the ambient measure into G_{net} (v76). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer}, one functor with four typed conditional interfaces (Koide, η_B, axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor. A_2 **(net existence) closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–

Mack–Todorov), the holomorphic c=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)$=16/16=1, 248=120+128, character $E_4/\eta^8=j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $|k|$ (v156/v157). R_1 **(seam clock) reduces to** `GATE.QGEO` **[C]**: a μ_4-equivariant flat gapped generator is *unique* (= $A_0^\star$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification "seam boundary = $\mathbb{P}\setminus\mu_4$" = `GATE.QGEO` (cohomology established, v137, b_1=3=$N_{\rm fam}$). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\rm geo}\}$; premise (A) = (A_2 assembly) + (R_1=`GATE.QGEO`) + the $\{\pi, v_{\rm geo}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module** `v162_seam_transport_identification.py` **[E]/[O].** Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content.* The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4-equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4|\,\mathrm{diag}\,X$ for U=diag$(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights spec$(A_0^\star)$=$\{0, \frac{1}{3}, \frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6\log\frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of ω_k=$|k|$ (v113/v156). **Closure:** with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, v161)

- **New module** `v161_one_particle_reduction.py` **[E]/[O].** *Discharges* the scope premise of v160 ("the transport is gapped on the *full* Schwinger cone") via Bogoliubov second quantisation. A quasi-free transport is T_Σ=$\Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) spec$(\Gamma(t)|_{\deg m})$ = products of m one-particle eigenvalues (compound-matrix theorem, all m=0..6); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation ⇔ one vacuum); (4) quasi-freeness is *forced* — the 120= dim $\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160's three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger `GATE.METRIC.17`). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) ("the seam bulk is a reflection-positive free field") as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 — verified by the *signed* set-partition/Pfaffian moment↔cumulant inversion on a concrete pure Majorana covariance (the v113 "one kernel = net" property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray ⇒ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0, the value $(2/3)^6$ only sets the rate. **The honest residual:** v56's gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement "the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56", which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.
- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection` × `{YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as `2026-06-07 / 2026-06-08` sort by the later date). A manual prepend had left five `2026-06-12` entries above seven `2026-06-13` blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral c=8 fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s "reaches-the-fixed-point" residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the h=1 currents are chiral (v157, not $(1,1)$-marginal), and the lowest interaction (quartic, h=2) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the b_1=41/10 cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee–Uhlmann — order-1 ΨDO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic c=8 theory is *rigid* — every field has $\bar{h}$=0, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}). With $(E_8)_1$ unique (v83), premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=$6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point" — freeness then forced by rigidity, not fine-tuned (ledger `GATE.METRIC.14`).

- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** "$B \cong (E_8)_1$" is no longer an assertion: the chiral seam net is built explicitly and the identification *checked.* (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5{+}3$; (c) the weight-1 currents are $248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q-series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN $= |k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger `GATE.METRIC.13`).
- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN $= |k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the 248=120+128 split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- **G_{net}'s last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) "the Calderón involution is sheet-odd" (the double-cover deck, structural) and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from "the seam bulk is a reflection-positive free field", the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by G_{net}, the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger `GATE.METRIC.12`).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- **v_{geo} re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\mathrm{point}} \sim v_{\mathrm{geo}}$, $1/G \sim v_{\mathrm{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay $\{$one unit, $\pi\}$ (ledger `ANCHOR.VGEO.02`).

- G_{net} **stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has $[B{:}A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ ([E]/[C]), and F_{transfer} as one functor with four typed interfaces — "not structural gaps". The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\text{geo}}\}$; one seam-net theorem; frontier = F_{transfer}) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series "Paper N" cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series "Paper N" attributions and removed them: four literal "Paper 6" references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old "Paper 4" (OS/admissibility) and "Paper 5" (APS/Hodge) references in `tfpt_2`, the old "Paper 3" transport-kernel attributions in `tfpt_2`, and the internal "Paper 1/2/4" cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts $\leftrightarrow$ papers $\leftrightarrow$ ledger $\leftrightarrow$ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d{=}2$ that ratio *is* the induced Newton constant $1/G$ (v68). So v_{geo} (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π}, and `SEAM.THEOREM.01`'s residual reduces to the kernel-identification premise alone. *Audit (not promoted):* the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/`origin_theory`/`next.txt`/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI `10.5281/zenodo.`

20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (`ZENODO_TOKEN` secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ∼270 files / ∼3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts ("Why the interpretation is internally consistent", "Dependency chain of the birefringence test", "The precise reformulation", "The three theses honestly graded", "The whole story", "One-line summary") become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, . . .) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card ("five short documents", missing `tfpt_5`) *and* the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box "clock" wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon "Scope/typing" marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document's `.aux`/`.toc`/`.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\rm cone}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the "Calderón version" demanded by v150 needs no separate derivation.

`SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger `SEAM.EHMODEL.02`).

- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/`origin_theory/next.txt` moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4′: both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6,3,5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2,8,121)$ frame data and the w_0-lift exact. R4′ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on `SEAM.THEOREM.01`.** The conical heat-trace deficit is exact and t-independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ-regularised replica variation is *finite and cutoff-independent*, $\Delta \log\det' = 2C(\gamma)\ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi)\int\sqrt{g}R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger `SEAM.EHMODEL.01`).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4′: the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4|e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\rm car} = N^2_{\rm fam}$ (4+5=9, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2| g_{\rm car} \|\mathrm{Pl}(K)\|_1$. R4′ = derive *one* map (ledger `GATE.UWALL.13`).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z\mapsto iz,\ z\mapsto 1/z\rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity +1 on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger `FLAV.QGEO.07`).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is det′-clean (v147).** Variance ratio $(1-\alpha)$ (forced by norm = area) gives the Born-squared ratio $(1-\alpha)^{p_2}$, the ln series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger `HOR.CLOCK.13`).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\rm NS} + 128_{\rm R}$); the one remaining net statement is irreducibly twisted-sector (ledger `GATE.METRIC.10`). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + `GATE.WOLFRAM.02` + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\rm Pl}$ vs $M_{\rm Pl}$ — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** "How each of the seven papers simplifies" → "How the legacy layers simplify (bridge map)" with the lead-in "(Paper 6)" reference replaced by the scalaron-reheating chain (v86); "(Paper~6/7)" in the `tfpt_3` bridge table reworded; the `tfpt_1` "single remaining geometric input (H2)" passage and the `tfpt_2` "single remaining input (U)" passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).
- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes **[E]**/**[C]**/**[O]** instead of the legacy [*A*]/[*I*]/[*L*]/[*B*]/[*P*]/[*R*] badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + c=8 ⇔ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never "Paper 5–7"); the marker system on every website surface migrated to [*E*]/[*C*]/[*O*]/[*X*] (fine types named in prose; the script index keeps quoting the scripts' internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net}: Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review's narrative reframings (anchor-algebra trilogy, "one seam clock" framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A\Sigma$ (v97/v98) pairs the Q_+=1 cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \to -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading $= Q_+ \circ \rho$. `GATE.QGEO` keeps only its realisation premise — no discrete freedom left (ledger `FLAV.QGEO.06`).
- **R4′ narrowed: three pairings → two (v142, frame integrality).** Integer covectors in the $(\mathbf{1}, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \bmod 11$; the index *is* $\|\mathrm{Pl}(K)\|_1$); with $(x, y) = (|\mathbb{Z}_2|, \mathrm{rank}\, E_8)$ the σ-pairing is forced $\equiv 0 \bmod 11$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).

- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$-graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1's det-ratio step derived in-family (v144).** e_2-rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1-\Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays [C] with the $6 \to \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (`SEAM.THEOREM.01`); the gate keeps its [O] typing untouched.
- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the `vN` scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri`/`\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the `vN` scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` ("Three views of one spine": anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers

the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): tfpt_3 from 39 to 27 boxes (e.g. "Purpose and discipline", the five "(i)–(v)" audit-lemma boxes, "Net", "The connection in one sentence", "The unification", "The question and the honest answer", "Net: two axioms"); tfpt_4 from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item "Honest status of . . . " keyboxes are kept — tfpt_4 is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in tfpt_3 and tfpt_4 (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in tfpt_4 (Koide flow-time) replaced by the 4-class [C]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [E]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): tfpt_1 from 21 to 14 loud boxes (+3 `keyeq`); tfpt_2 from 52 to 43 (e.g. "In one sentence", "The one-paragraph version", the four "Sharpening" notes, "What the simplification means net", "The disciplined main statement", "What improves. . . ", "Scope and honesty"). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout tfpt_1 and tfpt_2 (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** tfpt_2's inflation-pivot table gains the v86 scalaron-reheating point ($N_\star$=51.4 ⇒ n_s=0.9611, r=0.0045, A_s-disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ...\tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L]` → [E], `[N]/[P]` → [E]/[C]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.

- **Status tables reconciled against the ledger.** The introduction's inflation/$N_\star$ rows are made consistent with v86/`COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star$=51.4 (n_s=0.9611, r=0.0045), replacing the stale point values $N_\star$=55 (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [E] → [C](conditional on M_{Star}=M_{scal} + reheating), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today` + version + revision). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry author, date and `v5.2 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the "Reduction in one line" and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to `tex-artefacts/`.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh notes` deliverable. The `make_manifest.py` `TEX` list and the workflow rule were updated to the new paths.

- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long vN→check table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction's eleven `keybox`es are reduced to the single headline "In one sentence" card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winbox`es become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winbox`es, one `gapbox`) plus the two shared orientation cards.
- **Document-count consistency.** The introduction's document map gained the missing `tfpt_5_redteam` row; "seven/eight companion documents" and the ledger "six documents" wording are corrected to the true count (nine active documents); `README.md` ("10 ok", 140 scripts) and the workflow rule ("9 docs") are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox`/`audbox`/`intbox`/`resbox`/`roadbox`) across tfpt_1, tfpt_2, tfpt_3 and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer's prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the 2/3 *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document's layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in tfpt_1, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), [C] (readout / standard physics in TFPT units) to the existing [C] (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction's one-sentence claim is softened from "derives all constants" to "constructs a discrete compiler; physical readouts run through named, status-typed bridges"; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says "stays [P]" — ratios are **[E]** closed on the derived stratum, only the absolute scale stays **[O]**, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36c_3^4$ (leading c_3, fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO's first 59.1-day result (0.3092 ± 0.0087, compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. "Complete solutions of the five open questions" becomes "Closure status of the five former open questions".

- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\to v_{\text{geo}} \to G_{\text{net}}$); F_{transfer} is named as one missing functor with Koide/η_B/axion/m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv$ `tfpt_N`). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four tcolorbox families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` ("TFPT status at a glance") gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \to 12 \to 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4′ selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from "find" to "identify".
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights (spec $A_0^\star$); $\det(\mathbf{1} - A_0^\star) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law → zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).

- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1,1,-2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s,t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\mathrm{rowsum}(V) = \mathrm{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the $A_0^\star$ zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1,2,3) = \mathrm{Spec}(Q_+) =$ the A_3 exponents and cusp weights $= \mathrm{spec}\, A_0^\star$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the R4′ residue is three pairings; the $H^1 \to$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded "single remaining input" claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ∼110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s,t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass ("The Sheet Diamond and the Winding Line" in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\mathrm{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\mathrm{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank $= c$) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- **U_{wall} made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue $A_0^\star$ (diag $= a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff$

$a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.

- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.